MATH4043GU

IZ TECÚM

ABSTRACT. Lecture notes for MATH4043GU.

CONTENTS

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[1]Based on lectures by Gyujin Oh [2].

[2]Based on lectures by Austin Lei, others [1].

Part 1. **Algebraic Number Theory**[3]

1. Number Fields

This section fixes the integral, linear-algebraic, and discriminant invariants of a finite extension of $\mathbb{Q}$.

1.1. Algebraic Integers, Trace, and Discriminant.

Definition 1.1.1 (Number field and ring of integers)**.** A number field is a finite extension $K/\mathbb{Q}$ of degree $n = [K : \mathbb{Q}]$. An element $\alpha \in K$ is integral over $\mathbb{Z}$ when it satisfies a monic polynomial in $\mathbb{Z}[X]$, and

$$\mathscr{O}_K := \{\alpha \in K : \alpha^r + a_{r-1}\alpha^{r-1} + \cdots + a_0 = 0 \text{ for some } a_0, \dots, a_{r-1} \in \mathbb{Z}\}.$$

For $\alpha \in K$, its minimal polynomial $f_\alpha(X) \in \mathbb{Q}[X]$ is the unique monic irreducible polynomial satisfying $f_\alpha(\alpha) = 0$. An element $\theta \in K$ is primitive when $K = \mathbb{Q}(\theta)$.

Proposition 1.1.2 (Integral-element criteria)**.** *Let* $\alpha, \alpha_1, \dots, \alpha_s \in K$.

(1)

$$\alpha \in \mathscr{O}_K \iff \mathbb{Z}[\alpha] \textit{ is finitely generated over } \mathbb{Z} \iff \exists 0 \neq M \subset K : M \textit{ is finite over } \mathbb{Z} \wedge \alpha M \subset M.$$

Proof. Use a monic relation and the determinant trick.

$$\alpha^r = -a_{r-1}\alpha^{r-1} - \cdots - a_0 \implies \mathbb{Z}[\alpha] = \mathbb{Z} + \mathbb{Z}\alpha + \cdots + \mathbb{Z}\alpha^{r-1}.$$

$$\mathbb{Z}[\alpha] \text{ finite over } \mathbb{Z} \implies M := \mathbb{Z}[\alpha] \neq 0, \qquad \alpha M \subset M.$$

$$M = \mathbb{Z}\omega_1 \oplus \cdots \oplus \mathbb{Z}\omega_t, \qquad \omega := \begin{bmatrix} \omega_1 \\ \vdots \\ \omega_t \end{bmatrix}.$$

$$A := \begin{bmatrix} a_{11} & \cdots & a_{1t} \\ \vdots & \ddots & \vdots \\ a_{t1} & \cdots & a_{tt} \end{bmatrix}, \qquad A \in \mathrm{Mat}_t(\mathbb{Z}), \qquad \alpha \cdot \omega = A \cdot \omega.$$

$$\big(\alpha \cdot I_t - A\big) \cdot \omega = 0 \implies \det(\alpha \cdot I_t - A) = 0, \qquad \det(X \cdot I_t - A) \in \mathbb{Z}[X] \text{ monic}.$$

The three conditions are equivalent. □

(2)

$$\alpha_1, \dots, \alpha_s \in \mathscr{O}_K \iff \mathbb{Z}[\alpha_1, \dots, \alpha_s] \textit{ is finitely generated over } \mathbb{Z}.$$

Proof. Reduce every exponent by a monic relation.

$$f_i(\alpha_i) = 0, \qquad f_i(X) \in \mathbb{Z}[X] \text{ monic}, \qquad d_i := \deg(f_i).$$

$$\mathbb{Z}[\alpha_1, \dots, \alpha_s] = \sum_{0 \le e_i < d_i} \mathbb{Z}\alpha_1^{e_1} \cdots \alpha_s^{e_s} \implies \mathbb{Z}[\alpha_1, \dots, \alpha_s] \text{ is finite over } \mathbb{Z}.$$

$$\mathbb{Z}[\alpha_1, \dots, \alpha_s] \text{ finite over } \mathbb{Z} \implies \alpha_i \mathbb{Z}[\alpha_1, \dots, \alpha_s] \subset \mathbb{Z}[\alpha_1, \dots, \alpha_s] \overset{1}{\implies} \alpha_i \in \mathscr{O}_K.$$

This proves the finite-family criterion. □

(3)

$$\alpha, \beta \in \mathscr{O}_K \implies \alpha \pm \beta \in \mathscr{O}_K, \qquad \alpha\beta \in \mathscr{O}_K, \qquad \mathscr{O}_K \subset K \textit{ is a subring}.$$

[3]Based on lectures by Gyujin Oh [2].

Proof. Work in the common finite algebra.

$$\alpha,\beta\in\mathcal{O}_K \overset{2}{\Longrightarrow} B:=\mathbb{Z}[\alpha,\beta] \text{ is finite over } \mathbb{Z}.$$
$$\gamma\in\{\alpha+\beta,\alpha-\beta,\alpha\beta\} \implies \gamma\in B, \qquad \gamma B\subset B.$$
$$\gamma B\subset B \overset{1}{\Longrightarrow} \gamma\in\mathcal{O}_K, \qquad 0,1\in\mathcal{O}_K.$$

Thus $\mathcal{O}_K$ is a ring. ☐

Proposition 1.1.3 (Primitive and minimal polynomials, [2])**.** *Let $K/\mathbb{Q}$ be a number field and $\alpha\in K$.*

(1)
$$\exists\,\theta\in\mathcal{O}_K : K=\mathbb{Q}(\theta), \qquad \deg(f_\theta)=[K:\mathbb{Q}].$$

Proof. Scale a generator supplied by the primitive-element theorem.

$$K=\mathbb{Q}(\eta), \qquad f_\eta(X)=X^n+a_{n-1}X^{n-1}+\cdots+a_0\in\mathbb{Q}[X].$$
$$0\neq c\in\mathbb{Z}, \qquad c^j a_{n-j}\in\mathbb{Z} \quad (1\le j\le n), \qquad \theta:=c\eta.$$
$$c^n f_\eta(X/c)=X^n+ca_{n-1}X^{n-1}+\cdots+c^n a_0\in\mathbb{Z}[X].$$
$$c^n f_\eta(\theta/c)=0 \implies \theta\in\mathcal{O}_K, \qquad \mathbb{Q}(\theta)=\mathbb{Q}(\eta)=K.$$

Hence K admits an integral primitive element. ☐

(2)
$$\alpha\in\mathcal{O}_K \iff f_\alpha(X)\in\mathbb{Z}[X].$$

Proof. Compare the minimal polynomial with all conjugates of α.

$$g(\alpha)=0, \quad g(X)\in\mathbb{Z}[X] \text{ monic} \implies g(\sigma(\alpha))=0 \implies \sigma(\alpha)\in\overline{\mathbb{Z}}.$$
$$f_\alpha(X)=\prod_{i=1}^{d}(X-\alpha_i)=X^d+b_{d-1}X^{d-1}+\cdots+b_0, \qquad b_j\in\mathbb{Q}\cap\overline{\mathbb{Z}}.$$
$$\mathbb{Q}\cap\overline{\mathbb{Z}}=\mathbb{Z} \implies b_j\in\mathbb{Z} \implies f_\alpha(X)\in\mathbb{Z}[X].$$
$$f_\alpha(X)\in\mathbb{Z}[X] \text{ monic} \implies \alpha\in\mathcal{O}_K.$$

This proves the minimal-polynomial criterion. ☐

Definition 1.1.4 (Norm and trace)**.** Let $m_\alpha\colon K\to K$ be multiplication by α, and let $\sigma_1,\dots,\sigma_n\colon K\hookrightarrow\mathbb{C}$ be the embeddings of K. Set

$$m_\alpha(x):=\alpha x, \qquad \Sigma_K(x\otimes c):=\begin{bmatrix} c\sigma_1(x)\\ \vdots\\ c\sigma_n(x)\end{bmatrix}, \qquad D_\alpha:=\begin{bmatrix}\sigma_1(\alpha) & 0 & \cdots & 0\\ 0 & \sigma_2(\alpha) & \ddots & \vdots\\ \vdots & \ddots & \ddots & 0\\ 0 & \cdots & 0 & \sigma_n(\alpha)\end{bmatrix}.$$

$$\begin{tikzcd} K\otimes_{\mathbb{Q}}\overline{\mathbb{Q}} \arrow[r, "m_\alpha\otimes\mathrm{id}_{\overline{\mathbb{Q}}}"] \arrow[d, "\Sigma_K"', "\sim"] & K\otimes_{\mathbb{Q}}\overline{\mathbb{Q}} \arrow[d, "\sim"', "\Sigma_K"] \\ \overline{\mathbb{Q}}^n \arrow[r, "D_\alpha"'] & \overline{\mathbb{Q}}^n \end{tikzcd} \qquad \Sigma_K\circ(m_\alpha\otimes\mathrm{id}_{\overline{\mathbb{Q}}})=D_\alpha\cdot\Sigma_K.$$

$$\mathrm{N}_{K/\mathbb{Q}}(\alpha):=\det(m_\alpha)=\prod_{i=1}^{n}\sigma_i(\alpha), \qquad \mathrm{Tr}_{K/\mathbb{Q}}(\alpha):=\mathrm{tr}(m_\alpha)=\sum_{i=1}^{n}\sigma_i(\alpha). \tag{1.1.1}$$

Proposition 1.1.5 (Norm and trace identities)**.** *Let $\alpha,\beta\in K$, and let $f_\alpha(X)=X^d+a_{d-1}X^{d-1}+\cdots+a_0$.*

(1) *For* $e = [K : \mathbb{Q}(\alpha)]$,

$$\mathrm{N}_{K/\mathbb{Q}}(\alpha) = (-1)^n a_0^e, \qquad \mathrm{Tr}_{K/\mathbb{Q}}(\alpha) = -e \cdot a_{d-1}, \qquad n = de.$$

Proof. Restrict the embeddings of K to $\mathbb{Q}(\alpha)$.

$$\{\sigma_i(\alpha) : 1 \le i \le n\} = \{\alpha_1, \ldots, \alpha_d\} \text{ with every } \alpha_j \text{ repeated } e \text{ times.}$$

$$(1.1.1) \implies \mathrm{N}_{K/\mathbb{Q}}(\alpha) = \left(\prod_{j=1}^{d} \alpha_j\right)^e, \qquad \mathrm{Tr}_{K/\mathbb{Q}}(\alpha) = e \cdot \sum_{j=1}^{d} \alpha_j.$$

$$\prod_{j=1}^{d} \alpha_j = (-1)^d a_0, \qquad \sum_{j=1}^{d} \alpha_j = -a_{d-1}, \qquad de = n.$$

$$\mathrm{N}_{K/\mathbb{Q}}(\alpha) = (-1)^{de} a_0^e = (-1)^n a_0^e, \qquad \mathrm{Tr}_{K/\mathbb{Q}}(\alpha) = -e \cdot a_{d-1}.$$

The formulas follow from Viète's identities. □

(2) *For* $q \in \mathbb{Q}$,

$$\mathrm{N}_{K/\mathbb{Q}}(\alpha\beta) = \mathrm{N}_{K/\mathbb{Q}}(\alpha) \cdot \mathrm{N}_{K/\mathbb{Q}}(\beta), \qquad \mathrm{Tr}_{K/\mathbb{Q}}(\alpha + \beta) = \mathrm{Tr}_{K/\mathbb{Q}}(\alpha) + \mathrm{Tr}_{K/\mathbb{Q}}(\beta),$$
$$\mathrm{N}_{K/\mathbb{Q}}(q) = q^n, \qquad \mathrm{Tr}_{K/\mathbb{Q}}(q) = n \cdot q.$$

Proof. Use the multiplication endomorphisms.

$$m_{\alpha\beta} = m_\alpha \circ m_\beta, \qquad m_{\alpha+\beta} = m_\alpha + m_\beta, \qquad m_q = q \cdot I_K.$$
$$\det(m_\alpha \circ m_\beta) = \det(m_\alpha) \cdot \det(m_\beta), \qquad \mathrm{tr}(m_\alpha + m_\beta) = \mathrm{tr}(m_\alpha) + \mathrm{tr}(m_\beta).$$
$$\det(q \cdot I_K) = q^n, \qquad \mathrm{tr}(q \cdot I_K) = n \cdot q.$$

These are the asserted identities. □

(3)

$$\alpha \in \mathscr{O}_K \implies \mathrm{N}_{K/\mathbb{Q}}(\alpha), \mathrm{Tr}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}.$$

Proof. Apply the integral minimal-polynomial criterion.

$$\alpha \in \mathscr{O}_K \overset{2}{\implies} f_\alpha(X) \in \mathbb{Z}[X] \implies a_0, a_{d-1} \in \mathbb{Z}.$$
$$\mathrm{N}_{K/\mathbb{Q}}(\alpha) = (-1)^n a_0^e, \qquad \mathrm{Tr}_{K/\mathbb{Q}}(\alpha) = -e \cdot a_{d-1}.$$
$$a_0, a_{d-1}, e \in \mathbb{Z} \implies \mathrm{N}_{K/\mathbb{Q}}(\alpha), \mathrm{Tr}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}.$$

This proves integrality of norm and trace. □

Theorem 1.1.6 (Trace pairing)**.** *The trace pairing is nondegenerate and induces the displayed isomorphism,*

$$B_K(x, y) := \mathrm{Tr}_{K/\mathbb{Q}}(xy), \qquad K \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{Q}), \qquad x \longmapsto \big(y \longmapsto B_K(x, y)\big).$$

Proof. Test a nonzero element against its inverse.

$$x \neq 0, \qquad y := x^{-1}, \qquad B_K(x, y) = \mathrm{Tr}_{K/\mathbb{Q}}(1) = n \neq 0.$$
$$B_K(x, K) = 0 \implies x = 0 \implies \ker(K \to \mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{Q})) = 0.$$
$$\dim_{\mathbb{Q}}(K) = n = \dim_{\mathbb{Q}}(\mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{Q})) \implies K \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{Q}).$$

Thus the trace form is perfect over $\mathbb{Q}$. □

Theorem 1.1.7 (Integral bases)**.** *For every number field* $K/\mathbb{Q}$ *of degree* n,

$$\mathscr{O}_K \cong \mathbb{Z}^n, \qquad \mathrm{rank}_{\mathbb{Z}}(\mathscr{O}_K) = n.$$

Proof. Bound $\mathscr{O}_K$ between two lattices using the trace pairing.

$$K = \mathbb{Q}(\theta), \quad \theta \in \mathscr{O}_K, \qquad \omega := \begin{bmatrix} 1 & \theta & \cdots & \theta^{n-1} \end{bmatrix}, \qquad L := \mathbb{Z} + \mathbb{Z}\theta + \cdots + \mathbb{Z}\theta^{n-1} \subset \mathscr{O}_K.$$

$$G_\omega := \begin{bmatrix} \mathrm{Tr}_{K/\mathbb{Q}}(1) & \mathrm{Tr}_{K/\mathbb{Q}}(\theta) & \cdots & \mathrm{Tr}_{K/\mathbb{Q}}(\theta^{n-1}) \\ \mathrm{Tr}_{K/\mathbb{Q}}(\theta) & \mathrm{Tr}_{K/\mathbb{Q}}(\theta^2) & \cdots & \mathrm{Tr}_{K/\mathbb{Q}}(\theta^n) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Tr}_{K/\mathbb{Q}}(\theta^{n-1}) & \mathrm{Tr}_{K/\mathbb{Q}}(\theta^n) & \cdots & \mathrm{Tr}_{K/\mathbb{Q}}(\theta^{2n-2}) \end{bmatrix} \in \mathrm{Mat}_n(\mathbb{Z}), \qquad D := \det(G_\omega) \neq 0.$$

$$x = \sum_{j=1}^n c_j \omega_j \in \mathscr{O}_K \implies G_\omega \cdot \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \mathrm{Tr}_{K/\mathbb{Q}}(x\omega_1) \\ \vdots \\ \mathrm{Tr}_{K/\mathbb{Q}}(x\omega_n) \end{bmatrix} \in \mathbb{Z}^n.$$

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \frac{1}{D} \cdot \mathrm{adj}(G_\omega) \cdot \begin{bmatrix} \mathrm{Tr}_{K/\mathbb{Q}}(x\omega_1) \\ \vdots \\ \mathrm{Tr}_{K/\mathbb{Q}}(x\omega_n) \end{bmatrix} \in \frac{1}{D} \cdot \mathbb{Z}^n, \qquad L \subset \mathscr{O}_K \subset \frac{1}{D} \cdot L.$$

$$\mathscr{O}_K \text{ is finite and torsion-free over } \mathbb{Z}, \qquad \mathbb{Q} \otimes_{\mathbb{Z}} \mathscr{O}_K = K \implies \mathscr{O}_K \cong \mathbb{Z}^n.$$

This gives the asserted free module. □

Definition 1.1.8 (Integral bases and discriminants)**.** An integral basis is a $\mathbb{Z}$-basis $\omega = (\omega_1, \dots, \omega_n)$ of $\mathscr{O}_K$. For any $\mathbb{Q}$-basis $\alpha = (\alpha_1, \dots, \alpha_n)$ of K, set

$$S_\alpha := \begin{bmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_1(\alpha_n) \\ \vdots & \ddots & \vdots \\ \sigma_n(\alpha_1) & \cdots & \sigma_n(\alpha_n) \end{bmatrix}, \qquad G_\alpha := \begin{bmatrix} \mathrm{Tr}_{K/\mathbb{Q}}(\alpha_1\alpha_1) & \cdots & \mathrm{Tr}_{K/\mathbb{Q}}(\alpha_1\alpha_n) \\ \vdots & \ddots & \vdots \\ \mathrm{Tr}_{K/\mathbb{Q}}(\alpha_n\alpha_1) & \cdots & \mathrm{Tr}_{K/\mathbb{Q}}(\alpha_n\alpha_n) \end{bmatrix},$$

$$\mathrm{disc}(\alpha) := \det(G_\alpha).$$

Proposition 1.1.9 (Trace matrices and lattice index)**.** *Let α and β be $\mathbb{Q}$-bases of K.*

(1)

(1.1.2) $$G_\alpha = S_\alpha^{\mathsf{T}} \cdot S_\alpha, \qquad \mathrm{disc}(\alpha) = \det(S_\alpha)^2 = \det(G_\alpha) \neq 0.$$

Proof. Compute the entries through the embeddings.

$$\left(S_\alpha^{\mathsf{T}} \cdot S_\alpha\right)_{ij} = \sum_{k=1}^n \sigma_k(\alpha_i)\sigma_k(\alpha_j) = \sum_{k=1}^n \sigma_k(\alpha_i\alpha_j).$$

$$\sum_{k=1}^n \sigma_k(\alpha_i\alpha_j) \overset{1.1.1}{\iff} \mathrm{Tr}_{K/\mathbb{Q}}(\alpha_i\alpha_j) \implies G_\alpha = S_\alpha^{\mathsf{T}} \cdot S_\alpha.$$

$$\det(G_\alpha) = \det(S_\alpha^{\mathsf{T}}) \cdot \det(S_\alpha) = \det(S_\alpha)^2.$$

$$\det(G_\alpha) = 0 \iff B_K \text{ is degenerate} \overset{1.1.6}{\iff} \perp.$$

This proves the trace-matrix identity and nonvanishing. □

(2) *Suppose*

$$\begin{bmatrix} \beta_1 & \cdots & \beta_n \end{bmatrix} = \begin{bmatrix} \alpha_1 & \cdots & \alpha_n \end{bmatrix} \cdot M, \qquad M \in \mathrm{GL}_n(\mathbb{Q}).$$

If $\alpha = \omega$ is integral, $M \in \mathrm{Mat}_n(\mathbb{Z})$, and $L := \mathbb{Z}\beta_1 + \cdots + \mathbb{Z}\beta_n \subset \mathscr{O}_K$, then

(1.1.3) $$\mathrm{disc}(\beta) = \det(M)^2 \cdot \mathrm{disc}(\alpha), \qquad [\mathscr{O}_K : L] = |\det(M)|, \qquad \mathrm{disc}(\beta) = [\mathscr{O}_K : L]^2 \cdot \mathrm{disc}(\omega).$$

Proof. Apply each embedding to the coordinate identity.

$$S_\beta = S_\alpha \cdot M, \qquad G_\beta = M^\mathsf{T} \cdot G_\alpha \cdot M.$$

$$\operatorname{disc}(\beta) = \det(M^\mathsf{T}) \cdot \det(G_\alpha) \cdot \det(M) = \det(M)^2 \cdot \operatorname{disc}(\alpha).$$

$$\begin{tikzcd} \mathbb{Z}^n \arrow[r, "M"] \arrow[d, "\sim"'] & \mathbb{Z}^n \arrow[d, "\sim"] \\ L \arrow[r, hook] & \mathscr{O}_K \end{tikzcd} \qquad \mathscr{O}_K/L \cong \mathbb{Z}^n/(M\cdot\mathbb{Z}^n).$$

$$[\mathscr{O}_K : L] = \#(\mathbb{Z}^n/(M\cdot\mathbb{Z}^n)) = |\det(M)| \implies \operatorname{disc}(\beta) = [\mathscr{O}_K : L]^2 \cdot \operatorname{disc}(\omega).$$

This proves both transformation laws. □

Definition 1.1.10 (Field discriminant)**.** For any integral basis ω of $\mathscr{O}_K$,

$$\operatorname{disc}(K/\mathbb{Q}) := \operatorname{disc}(\omega) \in \mathbb{Z}\setminus\{0\}.$$

Equation (1.1.3) makes this independent of ω.

Corollary 1.1.11 (Principal-ideal index)**.** *For every* $0 \neq \alpha \in \mathscr{O}_K$,

$$[\mathscr{O}_K : \alpha\mathscr{O}_K] = |\mathrm{N}_{K/\mathbb{Q}}(\alpha)|.$$

Proof. Write multiplication by α in an integral basis.

$$A_\alpha := \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \in \operatorname{Mat}_n(\mathbb{Z}), \qquad \alpha \cdot \begin{bmatrix} \omega_1 & \cdots & \omega_n \end{bmatrix} = \begin{bmatrix} \omega_1 & \cdots & \omega_n \end{bmatrix} \cdot A_\alpha.$$

$$\alpha\mathscr{O}_K = \mathbb{Z}(\alpha\omega_1) + \cdots + \mathbb{Z}(\alpha\omega_n), \qquad [\mathscr{O}_K : \alpha\mathscr{O}_K] = |\det(A_\alpha)|.$$

$$\det(A_\alpha) = \det(m_\alpha) = \mathrm{N}_{K/\mathbb{Q}}(\alpha) \implies [\mathscr{O}_K : \alpha\mathscr{O}_K] = |\mathrm{N}_{K/\mathbb{Q}}(\alpha)|.$$

This proves the index formula. □

Theorem 1.1.12 (Quadratic integers)**.** *Let* $d \in \mathbb{Z}\setminus\{0,1\}$ *be squarefree and* $K = \mathbb{Q}(\sqrt{d})$. *The two cases are*

$$(1.1.4) \quad \mathscr{O}_K = \begin{cases} \mathbb{Z}[\sqrt{d}], & d \equiv 2,3 \pmod 4, \\ \mathbb{Z}\left[\dfrac{1+\sqrt{d}}{2}\right], & d \equiv 1 \pmod 4, \end{cases} \qquad \operatorname{disc}(K/\mathbb{Q}) = \begin{cases} 4d, & d \equiv 2,3 \pmod 4, \\ d, & d \equiv 1 \pmod 4. \end{cases}$$

(1) *For* $\alpha = a + b\sqrt{d} \in K$,

$$\alpha \in \mathscr{O}_K \iff \alpha = \frac{r+s\sqrt{d}}{2}, \quad r,s\in\mathbb{Z}, \quad r^2 \equiv ds^2 \pmod 4.$$

Proof. Use the trace, norm, and squarefreeness of d.

$$\alpha \in \mathscr{O}_K \overset{3}{\implies} r := \operatorname{Tr}_{K/\mathbb{Q}}(\alpha) = 2a \in \mathbb{Z}, \qquad 4db^2 = r^2 - 4\mathrm{N}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}.$$

$$s := 2b = u/v \in \mathbb{Q}, \quad (u,v) = 1, \quad ds^2 \in \mathbb{Z} \implies v^2 \mid d \implies v = 1 \implies s \in \mathbb{Z}.$$

$$\mathrm{N}_{K/\mathbb{Q}}(\alpha) = \frac{r^2 - ds^2}{4} \in \mathbb{Z} \iff r^2 \equiv ds^2 \pmod 4.$$

$$r,s \in \mathbb{Z}, \quad r^2 \equiv ds^2 \pmod 4 \implies \alpha^2 - r\alpha + \frac{r^2-ds^2}{4} = 0 \implies \alpha \in \mathscr{O}_K.$$

This proves the quadratic integrality criterion. □

(2) *If* $d \equiv 2,3 \pmod 4$*, the first lines of Equation* (1.1.4) *hold.*

Proof. Reduce the criterion modulo 4.

$$r^2 \equiv ds^2 \pmod 4, \quad d \equiv 2,3 \pmod 4 \implies r \equiv s \equiv 0 \pmod 2.$$
$$\overset{1}{\implies} \alpha \in \mathbb{Z}[\sqrt{d}], \qquad (\sqrt{d})^2 - d = 0 \implies \mathscr{O}_K = \mathbb{Z}[\sqrt{d}].$$
$$G_{(1,\sqrt{d})} = \begin{bmatrix} \mathrm{Tr}_{K/\mathbb{Q}}(1) & \mathrm{Tr}_{K/\mathbb{Q}}(\sqrt{d}) \\ \mathrm{Tr}_{K/\mathbb{Q}}(\sqrt{d}) & \mathrm{Tr}_{K/\mathbb{Q}}(d) \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2d \end{bmatrix}.$$
$$\mathrm{disc}(K/\mathbb{Q}) = \det(G_{(1,\sqrt{d})}) = 2 \cdot 2d - 0 \cdot 0 = 4d.$$

This proves the first case. □

(3) *If $d \equiv 1 \pmod 4$, the second lines of Equation* (1.1.4) *hold.*

Proof. Set $\omega = (1+\sqrt{d})/2$ and reduce the criterion modulo 4.

$$r^2 \equiv s^2 \pmod 4 \iff r \equiv s \pmod 2, \qquad \alpha = \frac{r-s}{2} + s\omega \in \mathbb{Z}[\omega].$$
$$\omega^2 - \omega + \frac{1-d}{4} = 0, \qquad \sqrt{d} = 2\omega - 1 \implies \mathscr{O}_K = \mathbb{Z}[\omega].$$
$$\mathrm{Tr}_{K/\mathbb{Q}}(\omega) = 1, \qquad \mathrm{Tr}_{K/\mathbb{Q}}(\omega^2) = \frac{1+d}{2}, \qquad G_{(1,\omega)} = \begin{bmatrix} 2 & 1 \\ 1 & \dfrac{1+d}{2} \end{bmatrix}.$$
$$\mathrm{disc}(K/\mathbb{Q}) = \det(G_{(1,\omega)}) = 2 \cdot \frac{1+d}{2} - 1 \cdot 1 = d.$$

This proves the second case. □

Theorem 1.1.13 (Cyclotomic integers, [2])**.** *Let $m \geq 1$, let ζ_m be a primitive m-th root of unity, and let $\Phi_m(X)$ be the m-th cyclotomic polynomial. Then*

$$\begin{tikzcd} \mathbb{Z}[X] \arrow[r, two heads, "X\mapsto\zeta_m"] \arrow[dr, two heads, "\mathrm{mod}(\Phi_m(X))"'] & \mathbb{Z}[\zeta_m] \\ & \mathbb{Z}[X]/(\Phi_m(X)) \arrow[u, "\sim"'] \end{tikzcd} \qquad \mathscr{O}_{\mathbb{Q}(\zeta_m)} = \mathbb{Z}[\zeta_m] = \bigoplus_{j=0}^{\varphi(m)-1} \mathbb{Z}\zeta_m^j.$$

Definition 1.1.14 (Dedekind domain)**.** An integral domain A is Dedekind when

A is Noetherian, $\qquad$ A is integrally closed in $\mathrm{Frac}(A)$, $\qquad$ $0 \neq \mathfrak{p} \subsetneq A$ prime $\implies \mathfrak{p}$ is maximal.

Theorem 1.1.15 (Dedekind structure)**.** *Let $K/\mathbb{Q}$ be a number field.*

(1) *$\mathscr{O}_K$ is Noetherian.*

Proof. Regard every ideal as a subgroup of the integral lattice.

$$\mathscr{O}_K \cong \mathbb{Z}^n \overset{1.1.7}{\implies} \mathfrak{a} \subset \mathscr{O}_K \implies \mathfrak{a} \cong \mathbb{Z}^r \quad (0 \leq r \leq n).$$
$$\mathfrak{a} = \mathbb{Z}\alpha_1 + \cdots + \mathbb{Z}\alpha_r, \qquad \alpha_i \in \mathfrak{a}.$$
$$\mathfrak{a} \subset \mathscr{O}_K\alpha_1 + \cdots + \mathscr{O}_K\alpha_r \subset \mathfrak{a} \implies \mathfrak{a} = \mathscr{O}_K\alpha_1 + \cdots + \mathscr{O}_K\alpha_r.$$

Thus every ideal is finitely generated. □

(2) *$\mathscr{O}_K$ is integrally closed in K.*

Proof. Apply transitivity through a finite coefficient algebra.

$$x^r + a_{r-1}x^{r-1} + \cdots + a_0 = 0, \qquad a_0, \ldots, a_{r-1} \in \mathscr{O}_K.$$

$$B := \mathbb{Z}[a_0, \ldots, a_{r-1}] \overset{2}{\Longrightarrow} B \text{ is finite over } \mathbb{Z}.$$

$$B[x] = B + Bx + \cdots + Bx^{r-1} \implies B[x] \text{ is finite over } B \implies B[x] \text{ is finite over } \mathbb{Z}.$$

$$xB[x] \subset B[x] \overset{1}{\Longrightarrow} x \in \mathscr{O}_K.$$

Hence $\mathscr{O}_K$ is integrally closed. □

(3) *Every nonzero prime ideal $\mathfrak{p} \subset \mathscr{O}_K$ is maximal.*

Proof. Reduce modulo a nonzero principal ideal inside $\mathfrak{p}$.

$$0 \neq \alpha \in \mathfrak{p}, \qquad \alpha\mathscr{O}_K \subset \mathfrak{p}, \qquad \#(\mathscr{O}_K/\alpha\mathscr{O}_K) \overset{1.1.11}{\Longrightarrow} |\mathrm{N}_{K/\mathbb{Q}}(\alpha)| < \infty.$$

$$\mathscr{O}_K/\alpha\mathscr{O}_K \twoheadrightarrow \mathscr{O}_K/\mathfrak{p} \qquad \#(\mathscr{O}_K/\mathfrak{p}) < \infty.$$

$$\mathfrak{p} \text{ prime} \implies \mathscr{O}_K/\mathfrak{p} \text{ is a finite integral domain} \implies \mathscr{O}_K/\mathfrak{p} \text{ is a field.}$$

$$\mathscr{O}_K/\mathfrak{p} \text{ is a field} \iff \mathfrak{p} \text{ is maximal.}$$

This proves maximality. □

Consequently, $\mathscr{O}_K$ is a Dedekind domain.

2. Prime Ideals

Prime-ideal factorization replaces element factorization and records splitting through ideal valuations, residue fields, and Frobenius classes.

2.1. Factorization, Splitting, and Frobenius.

Definition 2.1.1 (Fractional ideals)**.** A nonzero fractional ideal of K is a nonzero $\mathscr{O}_K$-submodule $\mathfrak{a} \subset K$ satisfying

$$\exists 0 \neq d \in \mathscr{O}_K : d\mathfrak{a} \subset \mathscr{O}_K.$$

The ideal product, inverse, and ideal group are

$$\mathfrak{a}\mathfrak{b} := \left\{ \sum_{i=1}^{r} a_i b_i : a_i \in \mathfrak{a},\ b_i \in \mathfrak{b} \right\}, \qquad \mathfrak{a}^{-1} := \{x \in K : x\mathfrak{a} \subset \mathscr{O}_K\}, \qquad J_K := \{\mathfrak{a} : \mathfrak{a} \text{ is fractional}\}.$$

Theorem 2.1.2 (Unique prime-ideal factorization, [2])**.** *Every $\mathfrak{a} \in J_K$ has unique finite-support integers $v_\mathfrak{p}(\mathfrak{a})$ satisfying*

$$\mathfrak{a} = \prod_{\substack{0 \neq \mathfrak{p} \subset \mathscr{O}_K \\ \mathfrak{p} \text{ prime}}} \mathfrak{p}^{v_\mathfrak{p}(\mathfrak{a})}, \qquad v_\mathfrak{p}(\mathfrak{a}\mathfrak{b}) = v_\mathfrak{p}(\mathfrak{a}) + v_\mathfrak{p}(\mathfrak{b}), \tag{2.1.1}$$

$$v_\mathfrak{p}(\mathfrak{a} + \mathfrak{b}) = \min\{v_\mathfrak{p}(\mathfrak{a}), v_\mathfrak{p}(\mathfrak{b})\}, \qquad v_\mathfrak{p}(\mathfrak{a} \cap \mathfrak{b}) = \max\{v_\mathfrak{p}(\mathfrak{a}), v_\mathfrak{p}(\mathfrak{b})\}.$$

For nonzero integral ideals,

$$\mathfrak{a} \mid \mathfrak{b} \iff \mathfrak{a} \supseteq \mathfrak{b}.$$

Moreover,

$$\mathfrak{a}^{-1} = \prod_{\mathfrak{p}} \mathfrak{p}^{-v_\mathfrak{p}(\mathfrak{a})}, \qquad \mathfrak{a}\mathfrak{a}^{-1} = \mathscr{O}_K, \qquad (J_K, \cdot) \textit{ is an abelian group.}$$

Definition 2.1.3 (Coprime ideals)**.** For nonzero integral ideals $\mathfrak{a}, \mathfrak{b} \subset \mathscr{O}_K$,

$$\gcd(\mathfrak{a}, \mathfrak{b}) := \mathfrak{a} + \mathfrak{b}, \qquad \operatorname{lcm}(\mathfrak{a}, \mathfrak{b}) := \mathfrak{a} \cap \mathfrak{b},$$

$$\mathfrak{a} + \mathfrak{b} = \mathscr{O}_K \iff \min\{v_\mathfrak{p}(\mathfrak{a}), v_\mathfrak{p}(\mathfrak{b})\} = 0 \quad \text{for every } \mathfrak{p}.$$

Theorem 2.1.4 (Chinese remainder theorem)**.** *Let* $\mathfrak{a}_1,\dots,\mathfrak{a}_s \subset \mathscr{O}_K$ *be pairwise coprime. Then*

$$\begin{tikzcd} \mathscr{O}_K \arrow[r, "x\mapsto (x+\mathfrak{a}_i)_{i=1}^s"] \arrow[d, two heads] & \prod_{i=1}^s \mathscr{O}_K/\mathfrak{a}_i \\ \mathscr{O}_K/\prod_{i=1}^s \mathfrak{a}_i \arrow[ur, "\sim"'] & \end{tikzcd} \qquad \bigcap_{i=1}^s \mathfrak{a}_i = \prod_{i=1}^s \mathfrak{a}_i.$$

Proof. First treat two coprime ideals.

$$\mathfrak{a}+\mathfrak{b} = \mathscr{O}_K \implies \exists u \in \mathfrak{a},\ v \in \mathfrak{b} : u+v = 1.$$

$$x \in \mathfrak{a}\cap\mathfrak{b} \implies x = xu+xv \in \mathfrak{a}\mathfrak{b} \implies \mathfrak{a}\cap\mathfrak{b} = \mathfrak{a}\mathfrak{b}.$$

$$\mathfrak{b}_i := \prod_{j\neq i}\mathfrak{a}_j, \qquad \mathfrak{a}_i+\mathfrak{b}_i = \mathscr{O}_K \implies \exists e_i \in \mathfrak{b}_i : e_i \equiv 1 \pmod{\mathfrak{a}_i}.$$

$$(x_1+\mathfrak{a}_1,\dots,x_s+\mathfrak{a}_s) = \left(\sum_{i=1}^s x_ie_i+\mathfrak{a}_1,\dots,\sum_{i=1}^s x_ie_i+\mathfrak{a}_s\right), \qquad \ker\left(\mathscr{O}_K \to \prod_i \mathscr{O}_K/\mathfrak{a}_i\right) = \prod_i \mathfrak{a}_i.$$

This proves the quotient isomorphism. □

Theorem 2.1.5 (Two generators)**.** *For every nonzero integral ideal* $\mathfrak{a}\subset\mathscr{O}_K$ *and every* $0\neq\alpha\in\mathfrak{a}$*,*

$$\exists \beta \in \mathfrak{a} : \mathfrak{a} = \alpha\mathscr{O}_K+\beta\mathscr{O}_K.$$

Proof. Use the module form of the Chinese remainder theorem.

$$S := \{\mathfrak{p} : v_{\mathfrak{p}}(\alpha\mathscr{O}_K) > v_{\mathfrak{p}}(\mathfrak{a})\}, \qquad \#(S)<\infty, \qquad S=\varnothing \implies \mathfrak{a} = \alpha\mathscr{O}_K,\ \beta := \alpha.$$

$$S\neq\varnothing, \qquad \mathfrak{a}\mathfrak{p}\subsetneq\mathfrak{a} \implies \exists x_{\mathfrak{p}} \in \mathfrak{a}\setminus\mathfrak{a}\mathfrak{p} \quad (\mathfrak{p}\in S).$$

$$\mathfrak{a} \longrightarrow \prod_{\mathfrak{p}\in S}\mathfrak{a}/\mathfrak{a}\mathfrak{p} \text{ is surjective} \implies \exists\beta\in\mathfrak{a} : \beta \equiv x_{\mathfrak{p}} \pmod{\mathfrak{a}\mathfrak{p}}.$$

$$v_{\mathfrak{p}}(\alpha\mathscr{O}_K+\beta\mathscr{O}_K) = \min\{v_{\mathfrak{p}}(\alpha\mathscr{O}_K), v_{\mathfrak{p}}(\beta\mathscr{O}_K)\} = v_{\mathfrak{p}}(\mathfrak{a}) \quad \text{for every } \mathfrak{p}.$$

Thus $\mathfrak{a} = (\alpha,\beta)$. □

Definition 2.1.6 (Ideal norm)**.** For every nonzero integral ideal $\mathfrak{a}\subset\mathscr{O}_K$,

$$\mathrm{N}(\mathfrak{a}) := \#(\mathscr{O}_K/\mathfrak{a}).$$

Proposition 2.1.7 (Ideal norms)**.** *Let* $\mathfrak{a},\mathfrak{b}\subset\mathscr{O}_K$ *be nonzero integral ideals.*

(1)

$$\mathrm{N}(\mathfrak{a}\mathfrak{b}) = \mathrm{N}(\mathfrak{a})\cdot\mathrm{N}(\mathfrak{b}).$$

Proof. Reduce by Equation (2.1.1) to one prime factor.

$$\mathfrak{a}\mathfrak{p}\subset\mathfrak{a}, \qquad 0 \longrightarrow \mathfrak{a}/\mathfrak{a}\mathfrak{p} \longrightarrow \mathscr{O}_K/\mathfrak{a}\mathfrak{p} \longrightarrow \mathscr{O}_K/\mathfrak{a} \longrightarrow 0$$

$$\mathfrak{p}\cdot(\mathfrak{a}/\mathfrak{a}\mathfrak{p}) = 0, \qquad \nexists\mathfrak{c} : \mathfrak{a}\mathfrak{p}\subsetneq\mathfrak{c}\subsetneq\mathfrak{a} \quad \text{by Equation (2.1.1).}$$

$$\dim_{\mathscr{O}_K/\mathfrak{p}}(\mathfrak{a}/\mathfrak{a}\mathfrak{p}) = 1 \implies \#(\mathfrak{a}/\mathfrak{a}\mathfrak{p}) = \mathrm{N}(\mathfrak{p}) \implies \mathrm{N}(\mathfrak{a}\mathfrak{p}) = \mathrm{N}(\mathfrak{a})\cdot\mathrm{N}(\mathfrak{p}).$$

$$\mathfrak{b} = \prod_i \mathfrak{p}_i^{e_i} \implies \mathrm{N}(\mathfrak{a}\mathfrak{b}) = \mathrm{N}(\mathfrak{a})\cdot\prod_i \mathrm{N}(\mathfrak{p}_i)^{e_i} = \mathrm{N}(\mathfrak{a})\cdot\mathrm{N}(\mathfrak{b}).$$

This proves multiplicativity. □

(2) *Every nonzero prime ideal* $\mathfrak{p}\subset\mathscr{O}_K$ *lies over a unique rational prime* p*, and*

$$\mathfrak{p}\cap\mathbb{Z} = p\mathbb{Z}, \qquad \mathscr{O}_K/\mathfrak{p}\cong\mathbb{F}_{p^f}, \qquad \mathrm{N}(\mathfrak{p}) = p^f \quad \textit{for some } f\geq 1.$$

Proof. Contract $\mathfrak{p}$ to $\mathbb{Z}$.

$$0 \neq \alpha \in \mathfrak{p} \implies 0 \neq \mathrm{N}_{K/\mathbb{Q}}(\alpha) \in \alpha\mathscr{O}_K \subset \mathfrak{p} \implies \mathfrak{p} \cap \mathbb{Z} \neq 0.$$

$$\mathfrak{p} \cap \mathbb{Z} \subset \mathbb{Z} \text{ is nonzero and prime} \implies \mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z} \quad \text{for a unique rational prime } p.$$

$$\mathbb{F}_p \hookrightarrow \mathscr{O}_K/\mathfrak{p}, \qquad \mathscr{O}_K/\mathfrak{p} \text{ is a finite field} \implies \mathscr{O}_K/\mathfrak{p} \cong \mathbb{F}_{p^f}.$$

$$\mathrm{N}(\mathfrak{p}) = \#(\mathscr{O}_K/\mathfrak{p}) = p^f.$$

This proves the prime-ideal formula. □

Definition 2.1.8 (Prime decomposition)**.** For a rational prime p, write

$$n := [K : \mathbb{Q}], \qquad p\mathscr{O}_K = \prod_{i=1}^{g} \mathfrak{p}_i^{e_i}, \qquad k_i := \mathscr{O}_K/\mathfrak{p}_i \cong \mathbb{F}_{p^{f_i}}, \qquad f_i := [k_i : \mathbb{F}_p].$$

Here $e_i = e(\mathfrak{p}_i \mid p)$ and $f_i = f(\mathfrak{p}_i \mid p)$. Moreover,

behavior of p	numerical condition
split completely	$e_i = f_i = 1$ $(1 \le i \le g)$, $\quad g = n$
inert	$g = 1, \quad e_1 = 1, \quad f_1 = n$
totally ramified	$g = 1, \quad e_1 = n, \quad f_1 = 1$
ramified	$e_i > 1$ for some i

Theorem 2.1.9 (The efg relation)**.** *With Definition 2.1.8,*

$$\sum_{i=1}^{g} e_i f_i = [K : \mathbb{Q}]. \tag{2.1.2}$$

Proof. Take ideal norms of the prime factorization of $p\mathscr{O}_K$.

$$p \in \mathscr{O}_K \setminus \{0\} \overset{1.1.11}{\implies} \mathrm{N}(p\mathscr{O}_K) = [\mathscr{O}_K : p\mathscr{O}_K] = |\mathrm{N}_{K/\mathbb{Q}}(p)| = p^n.$$

$$p\mathscr{O}_K = \prod_{i=1}^{g} \mathfrak{p}_i^{e_i} \overset{1}{\implies} \mathrm{N}(p\mathscr{O}_K) = \prod_{i=1}^{g} \mathrm{N}(\mathfrak{p}_i)^{e_i} = \prod_{i=1}^{g} p^{e_i f_i}.$$

$$p^n = p^{\sum_i e_i f_i} \implies n = \sum_{i=1}^{g} e_i f_i.$$

This proves the degree relation. □

Theorem 2.1.10 (Discriminant and ramification, [2])**.** *For every rational prime p,*

$$p \mid \operatorname{disc}(K/\mathbb{Q}) \iff p \text{ ramifies in } K \iff \mathscr{O}_K/p\mathscr{O}_K \text{ is not reduced}.$$

Theorem 2.1.11 (Dedekind–Kummer)**.** *Let $K = \mathbb{Q}(\alpha)$, where $\alpha \in \mathscr{O}_K$ has minimal polynomial $f(X) \in \mathbb{Z}[X]$, put $n := [K : \mathbb{Q}] = \deg(f)$, and let*

$$p \nmid [\mathscr{O}_K : \mathbb{Z}[\alpha]], \qquad \overline{f}(X) = \prod_{i=1}^{g} \overline{g}_i(X)^{e_i} \in \mathbb{F}_p[X]$$

with distinct monic irreducible $\overline{g}_i(X)$ and monic lifts $g_i(X) \in \mathbb{Z}[X]$. Then

$$p\mathscr{O}_K = \prod_{i=1}^{g} \mathfrak{p}_i^{e_i}, \qquad \mathfrak{p}_i = (p, g_i(\alpha)), \qquad f(\mathfrak{p}_i \mid p) = \deg(g_i).$$

Proof. Reduce the monogenic order modulo p.

$$m := [\mathscr{O}_K : \mathbb{Z}[\alpha]], \quad am + bp = 1, \quad x \in \mathscr{O}_K \implies x \equiv amx \pmod{p\mathscr{O}_K}, \qquad mx \in \mathbb{Z}[\alpha].$$

$$\begin{tikzcd}
\mathbb{Z}[X] \arrow[r, two heads, "X\mapsto\alpha"] \arrow[d, two heads] & \mathbb{Z}[\alpha] \arrow[r, hook] \arrow[d, two heads] & \mathscr{O}_K \arrow[d, two heads] \\
\mathbb{F}_p[X] \arrow[r, two heads] & \mathbb{F}_p[X]/(\overline{f}) \arrow[r, "\sim", "\#=p^n"'] & \mathscr{O}_K/p\mathscr{O}_K
\end{tikzcd}$$

$$(\overline{g}_i^{e_i}) + (\overline{g}_j^{e_j}) = \mathbb{F}_p[X] \quad (i \neq j) \implies \mathbb{F}_p[X]/(\overline{f}) \xrightarrow{\sim} \prod_{i=1}^{g} \mathbb{F}_p[X]/(\overline{g}_i^{e_i}).$$

$$\mathfrak{p}_i = (p, g_i(\alpha)), \qquad \mathscr{O}_K/\mathfrak{p}_i \cong \mathbb{F}_p[X]/(\overline{g}_i), \qquad \mathrm{N}(\mathfrak{p}_i) = p^{\deg(g_i)}.$$

$$\prod_i \mathfrak{p}_i^{e_i} \subseteq p\mathscr{O}_K, \qquad \mathrm{N}\left(\prod_i \mathfrak{p}_i^{e_i}\right) = p^{\sum_i e_i \deg(g_i)} = p^{\deg(f)} = \mathrm{N}(p\mathscr{O}_K) \implies p\mathscr{O}_K = \prod_i \mathfrak{p}_i^{e_i}.$$

This proves the factorization criterion. □

Theorem 2.1.12 (Quadratic splitting, [2])**.** *Let $K/\mathbb{Q}$ be quadratic, let $D = \mathrm{disc}(K/\mathbb{Q})$, and let $\chi_D(p) = \left(\frac{D}{p}\right)$ be the Kronecker symbol. For an odd rational prime p,*

$$\chi_D(p) = \begin{cases} 1, & p\mathscr{O}_K = \mathfrak{p}\mathfrak{p}', \quad \mathfrak{p} \neq \mathfrak{p}', \quad (e,f,g) = (1,1,2), \\ -1, & p\mathscr{O}_K \textit{ is prime}, \quad (e,f,g) = (1,2,1), \\ 0, & p\mathscr{O}_K = \mathfrak{p}^2, \quad (e,f,g) = (2,1,1). \end{cases}$$

For $p = 2$,

$$D \equiv 1 \pmod 8 \implies 2 \textit{ splits}, \qquad D \equiv 5 \pmod 8 \implies 2 \textit{ is inert}, \qquad 2 \mid D \implies 2 \textit{ ramifies}.$$

Example 2.1.13 (The field $\mathbb{Q}(\sqrt{-5})$)**.** Let $K = \mathbb{Q}(\sqrt{-5})$. Then

$$2\mathscr{O}_K = \mathfrak{p}^2, \quad \mathfrak{p} = (2, 1+\sqrt{-5}), \qquad 3\mathscr{O}_K = \mathfrak{q}\mathfrak{q}', \quad \mathfrak{q} = (3, 1+\sqrt{-5}), \quad \mathfrak{q}' = (3, 1-\sqrt{-5}),$$
$$(1+\sqrt{-5})\mathscr{O}_K = \mathfrak{p}\mathfrak{q}, \qquad (1-\sqrt{-5})\mathscr{O}_K = \mathfrak{p}\mathfrak{q}',$$
$$6 = 2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5}), \qquad 6\mathscr{O}_K = \mathfrak{p}^2\mathfrak{q}\mathfrak{q}' = (\mathfrak{p}\mathfrak{q})(\mathfrak{p}\mathfrak{q}').$$

Definition 2.1.14 (Decomposition and inertia)**.** Let $K/\mathbb{Q}$ be Galois with $G := \mathrm{Gal}(K/\mathbb{Q})$, and let $\mathfrak{P} \mid p$. Set

$$D(\mathfrak{P} \mid p) := \{\sigma \in G : \sigma(\mathfrak{P}) = \mathfrak{P}\},$$
$$I(\mathfrak{P} \mid p) := \{\sigma \in D(\mathfrak{P} \mid p) : \sigma(x) \equiv x \pmod{\mathfrak{P}} \text{ for every } x \in \mathscr{O}_K\}.$$

Theorem 2.1.15 (Galois transitivity)**.** *Let $K/\mathbb{Q}$ be Galois, let $G = \mathrm{Gal}(K/\mathbb{Q})$, and let p be rational prime. For any $\mathfrak{P} \mid p$,*

$$G/D(\mathfrak{P} \mid p) \xrightarrow{\sim} \{\mathfrak{Q} \subset \mathscr{O}_K : \mathfrak{Q} \mid p\}, \qquad \sigma D(\mathfrak{P} \mid p) \longmapsto \sigma(\mathfrak{P}).$$

Consequently, all ramification indices equal e, all residue degrees equal f, and $efg = |G|$.

Proof. Exclude two distinct orbits by simultaneous approximation.

$$\mathscr{P}_1 \neq \mathscr{P}_2, \quad \mathfrak{P} \in \mathscr{P}_1, \quad \mathfrak{Q} \in \mathscr{P}_2 \overset{2.1.4}{\implies} \exists \alpha \in \mathscr{O}_K : \alpha \in \mathfrak{P}, \quad \alpha \equiv 1 \pmod{\mathfrak{R}} \ (\mathfrak{R} \in \mathscr{P}_2).$$
$$\mathrm{N}_{K/\mathbb{Q}}(\alpha) = \prod_{\sigma \in G} \sigma(\alpha) \in \mathfrak{P} \cap \mathbb{Z} = p\mathbb{Z} \subset \mathfrak{Q}.$$
$$\sigma(\alpha) \in \mathfrak{Q} \iff \alpha \in \sigma^{-1}(\mathfrak{Q}), \qquad \sigma^{-1}(\mathfrak{Q}) \in \mathscr{P}_2 \implies \sigma(\alpha) \notin \mathfrak{Q}.$$
$$\prod_{\sigma \in G} \sigma(\alpha) \in \mathfrak{Q} \quad \wedge \quad \sigma(\alpha) \notin \mathfrak{Q} \text{ for every } \sigma \implies \bot.$$

Thus the action is transitive, and Equation (2.1.2) gives $efg = |G|$. $\square$

Theorem 2.1.16 (Decomposition–inertia sequence, [2])**.** *Let $k_{\mathfrak{P}} := \mathscr{O}_K/\mathfrak{P} \cong \mathbb{F}_{p^f}$. Reduction induces*

$$1 \longrightarrow I(\mathfrak{P} \mid p) \hookrightarrow D(\mathfrak{P} \mid p) \xrightarrow{\sigma \mapsto (\bar{x} \mapsto \overline{\sigma(x)})} \mathrm{Gal}(k_{\mathfrak{P}}/\mathbb{F}_p) \longrightarrow 1 \tag{2.1.3}$$

and

$$|D(\mathfrak{P} \mid p)| = ef, \qquad |I(\mathfrak{P} \mid p)| = e, \qquad D(\mathfrak{P} \mid p)/I(\mathfrak{P} \mid p) \cong \mathrm{Gal}(\mathbb{F}_{p^f}/\mathbb{F}_p).$$

Writing $K^I := K^{I(\mathfrak{P}|p)}$ and $K^D := K^{D(\mathfrak{P}|p)}$,

$$\begin{tikzcd}
K \arrow[d, dash, "e"'] & 1 \arrow[d, hook] \\
K^I \arrow[d, dash, "f"'] & I(\mathfrak{P} \mid p) \arrow[d, hook] \\
K^D \arrow[d, dash, "g"'] & D(\mathfrak{P} \mid p) \arrow[d, hook] \\
\mathbb{Q} & G
\end{tikzcd} \qquad [K : K^I] = e, \quad [K^I : K^D] = f, \quad [K^D : \mathbb{Q}] = g.$$

For $\sigma \in G$,

$$D(\sigma(\mathfrak{P}) \mid p) = \sigma D(\mathfrak{P} \mid p)\sigma^{-1}, \qquad I(\sigma(\mathfrak{P}) \mid p) = \sigma I(\mathfrak{P} \mid p)\sigma^{-1}.$$

Definition 2.1.17 (Frobenius)**.** Assume p is unramified in the Galois extension $K/\mathbb{Q}$. The Frobenius element $\mathrm{Frob}(\mathfrak{P} \mid p) \in D(\mathfrak{P} \mid p)$ is the unique lift in Equation (2.1.3) satisfying

$$\mathrm{Frob}(\mathfrak{P} \mid p)(x) \equiv x^p \pmod{\mathfrak{P}} \quad \text{for every } x \in \mathscr{O}_K.$$

Its Artin symbol is

$$\left(\frac{K/\mathbb{Q}}{\mathfrak{P}}\right) := \mathrm{Frob}(\mathfrak{P} \mid p).$$

Proposition 2.1.18 (Frobenius properties)**.** *Let p be unramified in the Galois extension $K/\mathbb{Q}$, and set $G = \mathrm{Gal}(K/\mathbb{Q})$.*

(1) *For every $\sigma \in G$,*

$$\mathrm{Frob}(\sigma(\mathfrak{P}) \mid p) = \sigma \mathrm{Frob}(\mathfrak{P} \mid p)\sigma^{-1}, \qquad \mathrm{ord}(\mathrm{Frob}(\mathfrak{P} \mid p)) = f,$$

and the elements $\mathrm{Frob}(\mathfrak{P} \mid p)$ *for* $\mathfrak{P} \mid p$ *form a conjugacy class* $\mathrm{Frob}_p \subset G$*.*

Proof. Conjugate the defining congruence.

$$\sigma \mathrm{Frob}(\mathfrak{P} \mid p)\sigma^{-1}(x) \equiv \sigma\big((\sigma^{-1}(x))^p\big) = x^p \pmod{\sigma(\mathfrak{P})}.$$

$$\sigma \mathrm{Frob}(\mathfrak{P} \mid p)\sigma^{-1}(x) \equiv x^p \pmod{\sigma(\mathfrak{P})} \overset{2.1.17}{\Longrightarrow} \sigma \mathrm{Frob}(\mathfrak{P} \mid p)\sigma^{-1} = \mathrm{Frob}(\sigma(\mathfrak{P}) \mid p).$$

$$\{\mathrm{Frob}(\mathfrak{Q} \mid p) : \mathfrak{Q} \mid p\} = \{\sigma \mathrm{Frob}(\mathfrak{P} \mid p)\sigma^{-1} : \sigma \in G\} =: \mathrm{Frob}_p \quad \text{by Theorem 2.1.15.}$$

$$p \text{ unramified} \implies I(\mathfrak{P} \mid p) = 1 \overset{2.1.3}{\Longrightarrow} D(\mathfrak{P} \mid p) \cong \mathrm{Gal}(\mathbb{F}_{p^f}/\mathbb{F}_p) = \langle \bar{x} \mapsto \bar{x}^p \rangle \implies \mathrm{ord}(\mathrm{Frob}(\mathfrak{P} \mid p)) = f.$$

This proves conjugacy and the order formula. $\square$

(2)

$$\mathrm{Frob}(\mathfrak{P} \mid p) = 1 \iff \textit{p splits completely in } K.$$

Proof. Use the order and degree relations.

$$\mathrm{Frob}(\mathfrak{P} \mid p) = 1 \overset{1}{\iff} f = 1.$$
$$p \text{ unramified} \implies e = 1, \qquad efg = |G| = [K:\mathbb{Q}].$$
$$e = f = 1 \iff g = [K:\mathbb{Q}] \iff p \text{ splits completely}.$$

This proves the splitting criterion. □

Theorem 2.1.19 (Frobenius orbits, [2]). *Let $H \le G$, let $M = K^H$, and let p be unramified in K. For $\mathfrak{P} \mid p$, put $F = \mathrm{Frob}(\mathfrak{P} \mid p)$. Then*

$$\begin{tikzcd}
H\backslash G \arrow[r, two heads] \arrow[dr, bend right, "H\sigma\mapsto\sigma(\mathfrak{P})\cap\mathscr{O}_M"'] & (H\backslash G)/\langle F\rangle \arrow[d, "\sim"] \\
 & \{\mathfrak{q} \subset \mathscr{O}_M : \mathfrak{q} \mid p\}
\end{tikzcd}$$

where $\langle F\rangle$ acts on the right. The residue degree of the prime corresponding to an orbit is its orbit length.

Theorem 2.1.20 (Chebotarev, [2]). *Let $K/\mathbb{Q}$ be Galois with group G, and let $C \subset G$ be a conjugacy class. Then*

$$\lim_{x\to\infty} \frac{\#\{p \le x : p \textit{ unramified}, \mathrm{Frob}_p = C\}}{\#\{p \le x : p \textit{ prime}\}} = \frac{|C|}{|G|}. \tag{2.1.4}$$

Theorem 2.1.21 (Cyclotomic arithmetic, [2]). *Let $K_m = \mathbb{Q}(\zeta_m)$, and identify its ring of integers by Theorem 1.1.13. Then*

$$\mathrm{Gal}(K_m/\mathbb{Q}) \xrightarrow{\sim} (\mathbb{Z}/m\mathbb{Z})^\times, \qquad \sigma_a \longmapsto a, \qquad \sigma_a(\zeta_m) = \zeta_m^a.$$

For $m > 2$,

$$\mathrm{disc}(K_m/\mathbb{Q}) = (-1)^{\varphi(m)/2} \frac{m^{\varphi(m)}}{\prod_{\ell \mid m} \ell^{\varphi(m)/(\ell-1)}}, \qquad \ell \textit{ ranges over rational primes}.$$

For a rational prime p, write $m = p^a m_0$, where $(p, m_0) = 1$, and set $\mathrm{ord}_1(p) = 1$. Then

$$p\mathscr{O}_{K_m} = \prod_{i=1}^{g} \mathfrak{P}_i^e, \qquad e = \varphi(p^a), \qquad f = \mathrm{ord}_{m_0}(p), \qquad g = \frac{\varphi(m_0)}{f}.$$

Hence

$$p \nmid m \implies \mathrm{Frob}(\mathfrak{P}_i \mid p) = \sigma_p, \quad f = \mathrm{ord}_m(p), \quad p \textit{ splits completely} \iff p \equiv 1 \pmod{m},$$
$$p \textit{ ramifies in } K_m \iff p^a > 2 \iff p \mid m^\flat, \qquad m^\flat := \begin{cases} m/2, & m \equiv 2 \pmod 4, \\ m, & m \not\equiv 2 \pmod 4. \end{cases}$$

Corollary 2.1.22 (Dirichlet). *Let $(a, m) = 1$. The rational primes $p \equiv a \pmod m$ have density $1/\varphi(m)$, hence are infinite.*

Proof. Apply Chebotarev to the cyclotomic extension.

$$\mathrm{Gal}(K_m/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times \implies \{\sigma_a\} \text{ is a conjugacy class of size } 1.$$
$$p \nmid m, \qquad \mathrm{Frob}_p = \sigma_p = \sigma_a \iff p \equiv a \pmod m.$$
$$\mathrm{Frob}_p = \{\sigma_a\} \iff p \equiv a \pmod m \overset{2.1.4}{\implies} \lim_{x\to\infty} \frac{\#\{p \le x : p \equiv a \pmod m\}}{\#\{p \le x : p \text{ prime}\}} = \frac{1}{|(\mathbb{Z}/m\mathbb{Z})^\times|} = \frac{1}{\varphi(m)} > 0.$$

This proves Dirichlet's theorem. □

Lemma 2.1.23 (Quadratic Gauss sum, [2])**.** *For an odd rational prime p and $b \in (\mathbb{Z}/p\mathbb{Z})^\times$, put*

$$\tau_p := \sum_{a=1}^{p-1} \left(\frac{a}{p}\right) \zeta_p^a, \qquad p^* := (-1)^{(p-1)/2} p.$$

Then

$$\sigma_b(\tau_p) = \left(\frac{b}{p}\right) \tau_p, \qquad \tau_p^2 = p^*.$$

Theorem 2.1.24 (Quadratic reciprocity)**.** *For distinct odd rational primes p, q,*

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

Proof. Use the quadratic Gauss sum in $\mathbb{Q}(\zeta_p)$.

$$\sigma_b(\tau_p) = \left(\frac{b}{p}\right) \tau_p, \qquad \tau_p^2 = p^* \quad \text{by Lemma 2.1.23.}$$

$$\mathbb{Q}(\sqrt{p^*}) = \mathbb{Q}(\tau_p) \subset \mathbb{Q}(\zeta_p), \qquad \sigma_q|_{\mathbb{Q}(\tau_p)} = \mathrm{id}_{\mathbb{Q}(\tau_p)} \iff \left(\frac{q}{p}\right) = 1.$$

$$\left(\frac{p^*}{q}\right) = 1 \overset{2.1.12}{\iff} q \text{ splits in } \mathbb{Q}(\sqrt{p^*}) \iff \sigma_q|_{\mathbb{Q}(\tau_p)} = \mathrm{id}_{\mathbb{Q}(\tau_p)}.$$

$$\left(\frac{p^*}{q}\right) = \left(\frac{q}{p}\right), \qquad \left(\frac{p^*}{q}\right) = \left(\frac{-1}{q}\right)^{(p-1)/2} \left(\frac{p}{q}\right).$$

$$\left(\frac{-1}{q}\right) = (-1)^{(q-1)/2} \implies \left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

This proves quadratic reciprocity. □

3. Class Groups, Units, and Quadratic Forms

The Minkowski embedding converts ideal classes and units into lattice problems, while binary quadratic forms make the imaginary-quadratic class group explicit.

3.1. **Geometry of Numbers and Quadratic Arithmetic.**

Definition 3.1.1 (Ideal class group)**.** Let

$$P_K := \{\alpha \mathscr{O}_K : 0 \neq \alpha \in K\} \subset J_K, \qquad \mathrm{Cl}(K) := J_K/P_K, \qquad h_K := \#(\mathrm{Cl}(K)).$$

Thus

$$\mathrm{Cl}(K) = 1 \iff \mathscr{O}_K \text{ is a principal ideal domain} \iff \mathscr{O}_K \text{ is a unique factorization domain.}$$

The defining maps form the exact sequence

$$1 \longrightarrow \mathscr{O}_K^\times \hookrightarrow K^\times \xrightarrow{\alpha \mapsto \alpha \mathscr{O}_K} J_K \twoheadrightarrow \mathrm{Cl}(K) \longrightarrow 1.$$

Definition 3.1.2 (Lattices)**.** A lattice in $\mathbb{R}^n$ is

$$\Lambda = \mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_n, \qquad \det\big(\begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}\big) \neq 0.$$

Its fundamental parallelotope and covolume are

$$\mathscr{P}_\Lambda := \left\{ \sum_{i=1}^n t_i v_i : 0 \le t_i < 1 \right\}, \qquad \mathrm{vol}(\Lambda) := \mathrm{vol}(\mathscr{P}_\Lambda) = \left|\det\big(\begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}\big)\right|.$$

Theorem 3.1.3 (Minkowski convex-body theorem). *Let $\Lambda \subset \mathbb{R}^n$ be a lattice and let $S \subset \mathbb{R}^n$ be measurable, convex, and centrally symmetric. Then*

$$\mathrm{vol}(S) > 2^n \mathrm{vol}(\Lambda) \implies (S \cap \Lambda) \setminus \{0\} \neq \varnothing.$$

Proof. Apply the quotient map to the half-body.

$$\begin{tikzcd} \frac{1}{2}S \arrow[r, hook, "\iota"] \arrow[dr, "q|_{\frac{1}{2}S}"'] & \mathbb{R}^n \arrow[d, two heads, "q"] \\ & \mathbb{R}^n/\Lambda \end{tikzcd} \qquad q(x) = x + \Lambda.$$

$$q|_{\frac{1}{2}S} \text{ injective} \implies 2^{-n}\mathrm{vol}(S) = \mathrm{vol}\big(\tfrac{1}{2}S\big) \leq \mathrm{vol}(\Lambda) \implies \perp.$$
$$\exists x \neq y \in \tfrac{1}{2}S : x - y \in \Lambda, \qquad \tfrac{1}{2}S - \tfrac{1}{2}S \subset S \implies 0 \neq x - y \in S \cap \Lambda.$$

This proves the lattice-point assertion. ☐

Definition 3.1.4 (Minkowski embedding). Let K have real embeddings $\sigma_1, \ldots, \sigma_{r_1}$ and one embedding τ_j from each complex-conjugate pair, where $r_1 + 2r_2 = n = [K : \mathbb{Q}]$. The unscaled Minkowski embedding is

$$\iota_K \colon K \hookrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \cong \mathbb{R}^n, \qquad x \longmapsto (\sigma_1(x), \ldots, \sigma_{r_1}(x), \tau_1(x), \ldots, \tau_{r_2}(x)).$$

Proposition 3.1.5 (Arithmetic covolume). *For every nonzero integral ideal $\mathfrak{a} \subset \mathscr{O}_K$,*

$$\mathrm{vol}(\iota_K(\mathscr{O}_K)) = 2^{-r_2}\sqrt{|\mathrm{disc}(K/\mathbb{Q})|}, \qquad \mathrm{vol}(\iota_K(\mathfrak{a})) = 2^{-r_2}\mathrm{N}(\mathfrak{a})\sqrt{|\mathrm{disc}(K/\mathbb{Q})|}. \tag{3.1.1}$$

Proof. Fix an integral basis $\omega_1, \ldots, \omega_n$, list all embeddings as $\rho_1, \ldots, \rho_n$, and pass from conjugate coordinates to real coordinates.

$$(\rho_1, \ldots, \rho_n) := (\sigma_1, \ldots, \sigma_{r_1}, \tau_1, \overline{\tau}_1, \ldots, \tau_{r_2}, \overline{\tau}_{r_2}).$$
$$A := \begin{bmatrix} \rho_1(\omega_1) & \cdots & \rho_1(\omega_n) \\ \vdots & & \vdots \\ \rho_n(\omega_1) & \cdots & \rho_n(\omega_n) \end{bmatrix}, \qquad \det(A)^2 = \mathrm{disc}(K/\mathbb{Q}).$$
$$\begin{bmatrix} \Re(z) \\ \Im(z) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2i} & -\frac{1}{2i} \end{bmatrix} \cdot \begin{bmatrix} z \\ \overline{z} \end{bmatrix}, \qquad \left|\det\left(\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2i} & -\frac{1}{2i} \end{bmatrix}\right)\right| = \frac{1}{2}.$$
$$\mathrm{vol}(\iota_K(\mathscr{O}_K)) = 2^{-r_2}|\det(A)| = 2^{-r_2}\sqrt{|\mathrm{disc}(K/\mathbb{Q})|}.$$
$$[\mathscr{O}_K : \mathfrak{a}] = \mathrm{N}(\mathfrak{a}) \implies \mathrm{vol}(\iota_K(\mathfrak{a})) = \mathrm{N}(\mathfrak{a}) \cdot \mathrm{vol}(\iota_K(\mathscr{O}_K)).$$

This proves both covolume formulas. ☐

Theorem 3.1.6 (Minkowski ideal-class bound). *Every class $C \in \mathrm{Cl}(K)$ contains an integral ideal $\mathfrak{a}$ satisfying*

$$\mathrm{N}(\mathfrak{a}) \leq M_K, \qquad M_K := \frac{n!}{n^n}\left(\frac{4}{\pi}\right)^{r_2}\sqrt{|\mathrm{disc}(K/\mathbb{Q})|}. \tag{3.1.2}$$

Proof. Choose an integral ideal $\mathfrak{b}$ representing C^{-1} and use a weighted cross-polytope.

$$S_t := \left\{ (x,z) \in \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} : \sum_{i=1}^{r_1} |x_i| + 2\sum_{j=1}^{r_2} |z_j| \leq t \right\}, \qquad \mathrm{vol}(S_t) = 2^{r_1}\left(\frac{\pi}{2}\right)^{r_2}\frac{t^n}{n!}.$$
$$T_{\mathfrak{b}} := n!\left(\frac{4}{\pi}\right)^{r_2}\mathrm{N}(\mathfrak{b})\sqrt{|\mathrm{disc}(K/\mathbb{Q})|}, \qquad t^n > T_{\mathfrak{b}} \overset{3.1.1}{\implies} \mathrm{vol}(S_t) > 2^n\mathrm{vol}(\iota_K(\mathfrak{b})).$$

$$\mathrm{vol}(S_t) > 2^n \mathrm{vol}(\iota_K(\mathfrak{b})) \overset{3.1.3}{\Longrightarrow} \exists 0 \neq \alpha \in \mathfrak{b} : \iota_K(\alpha) \in S_t,$$

$$|\mathrm{N}_{K/\mathbb{Q}}(\alpha)| = \prod_{i=1}^{r_1} |\sigma_i(\alpha)| \cdot \prod_{j=1}^{r_2} |\tau_j(\alpha)|^2,$$

$$|\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \leq \left(\frac{\sum_{i=1}^{r_1} |\sigma_i(\alpha)| + 2\sum_{j=1}^{r_2} |\tau_j(\alpha)|}{n} \right)^n \leq \left(\frac{t}{n}\right)^n.$$

$$\mathfrak{a} := \alpha\mathfrak{b}^{-1} \subset \mathscr{O}_K, \qquad [\mathfrak{a}] = C, \qquad \mathrm{N}(\mathfrak{a}) = \frac{|\mathrm{N}_{K/\mathbb{Q}}(\alpha)|}{\mathrm{N}(\mathfrak{b})} \leq \frac{t^n}{n^n \mathrm{N}(\mathfrak{b})}.$$

$$M_K < \lfloor M_K \rfloor + 1 \implies \exists t > 0 : T_{\mathfrak{b}} < t^n < n^n \mathrm{N}(\mathfrak{b})(\lfloor M_K \rfloor + 1) \implies \mathrm{N}(\mathfrak{a}) < \lfloor M_K \rfloor + 1 \implies \mathrm{N}(\mathfrak{a}) \leq M_K.$$

This proves Equation (3.1.2). □

Proposition 3.1.7 (Class-group consequences)**.** *The Minkowski bound has the following consequences.*

(1) *The group* $\mathrm{Cl}(K)$ *is finite and*

$$\mathrm{Cl}(K) = \langle [\mathfrak{p}] : \mathfrak{p} \subset \mathscr{O}_K \text{ prime}, \ \mathrm{N}(\mathfrak{p}) \leq M_K \rangle.$$

Proof. Factor the bounded representatives from Theorem 3.1.6.

$$C = [\mathfrak{a}], \quad \mathrm{N}(\mathfrak{a}) \leq M_K, \quad \mathfrak{a} = \prod_i \mathfrak{p}_i^{e_i} \overset{2.1.7}{\Longrightarrow} \mathrm{N}(\mathfrak{p}_i) \leq M_K.$$

$$\mathrm{N}(\mathfrak{p}) = p^{f(\mathfrak{p}|p)} \leq M_K \implies p \leq M_K, \qquad \#\{\mathfrak{p} : \mathrm{N}(\mathfrak{p}) \leq M_K\} < \infty.$$

$$\mathrm{N}(\mathfrak{a}) = \prod_i \mathrm{N}(\mathfrak{p}_i)^{e_i} \leq M_K \implies 0 \leq e_i \leq \frac{\log(M_K)}{\log(\mathrm{N}(\mathfrak{p}_i))}.$$

Thus only finitely many classes occur, and the displayed prime classes generate them. □

(2) *For every nonzero prime ideal* $\mathfrak{p} \subset \mathscr{O}_K$*,*

$$\mathfrak{p} \text{ is principal} \iff \exists 0 \neq \alpha \in \mathfrak{p} : |\mathrm{N}_{K/\mathbb{Q}}(\alpha)| = \mathrm{N}(\mathfrak{p}).$$

Proof. Compare containment and index.

$$\mathfrak{p} \text{ is principal} \implies \exists 0 \neq \alpha \in \mathfrak{p} : \mathfrak{p} = \alpha\mathscr{O}_K, \quad \mathrm{N}(\mathfrak{p}) = [\mathscr{O}_K : \alpha\mathscr{O}_K] \overset{1.1.11}{\Longrightarrow} \mathrm{N}(\mathfrak{p}) = |\mathrm{N}_{K/\mathbb{Q}}(\alpha)|.$$

$$\alpha \in \mathfrak{p} \implies \alpha\mathscr{O}_K \subseteq \mathfrak{p}, \quad [\mathscr{O}_K : \alpha\mathscr{O}_K] = [\mathscr{O}_K : \mathfrak{p}] \cdot [\mathfrak{p} : \alpha\mathscr{O}_K].$$

$$\alpha \in \mathfrak{p} \quad \wedge \quad |\mathrm{N}_{K/\mathbb{Q}}(\alpha)| = \mathrm{N}(\mathfrak{p}) \overset{1.1.11}{\Longrightarrow} [\mathscr{O}_K : \alpha\mathscr{O}_K] = [\mathscr{O}_K : \mathfrak{p}] \implies [\mathfrak{p} : \alpha\mathscr{O}_K] = 1.$$

$$[\mathfrak{p} : \alpha\mathscr{O}_K] = 1 \implies \mathfrak{p} = \alpha\mathscr{O}_K.$$

This proves the principal-prime test. □

(3) *If* $K \neq \mathbb{Q}$*, then*

$$|\mathrm{disc}(K/\mathbb{Q})| \geq \left(\frac{\pi}{4}\right)^{2r_2} \frac{n^{2n}}{(n!)^2} > 1, \qquad \exists p : p \text{ ramifies in } K.$$

Proof. Apply Equation (3.1.2) to the principal class.

$$[\mathfrak{a}] = [\mathscr{O}_K] \implies \exists 0 \neq \alpha \in \mathscr{O}_K : \mathfrak{a} = \alpha\mathscr{O}_K \implies \mathrm{N}(\mathfrak{a}) = |\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \in \mathbb{Z}_{\geq 1}.$$

$$1 \leq M_K \iff |\mathrm{disc}(K/\mathbb{Q})| \geq \left(\frac{\pi}{4}\right)^{2r_2} \frac{n^{2n}}{(n!)^2} > 1 \quad (n > 1).$$

$$|\mathrm{disc}(K/\mathbb{Q})| > 1 \implies \exists p : p \mid \mathrm{disc}(K/\mathbb{Q}) \overset{2.1.10}{\Longrightarrow} p \text{ ramifies in } K.$$

This proves both assertions. □

Theorem 3.1.8 (Hermite, [2])**.** *For every $B > 0$,*

$$\#\{K/\mathbb{Q} \text{ up to isomorphism} : |\mathrm{disc}(K/\mathbb{Q})| \leq B\} < \infty.$$

Definition 3.1.9 (Logarithmic embedding and regulator)**.** Put $s = r_1 + r_2$, and define

$$\lambda_K \colon K^\times \longrightarrow \mathbb{R}^s, \qquad \lambda_K(x) := \big(\log(|\sigma_1(x)|), \dots, \log(|\sigma_{r_1}(x)|), 2\log(|\tau_1(x)|), \dots, 2\log(|\tau_{r_2}(x)|)\big),$$

$$H_K := \left\{(y_1, \dots, y_s) \in \mathbb{R}^s : \sum_{i=1}^{s} y_i = 0\right\}, \qquad \mu_K := \{\zeta \in K^\times : \zeta^m = 1 \text{ for some } m \geq 1\}.$$

For units $\varepsilon_1, \dots, \varepsilon_{s-1}$ whose classes form a $\mathbb{Z}$-basis of $\mathscr{O}_K^\times/\mu_K$, let

$$L_K := \begin{bmatrix} \log(|\sigma_1(\varepsilon_1)|) & \cdots & \log(|\sigma_1(\varepsilon_{s-1})|) \\ \vdots & & \vdots \\ \log(|\sigma_{r_1}(\varepsilon_1)|) & \cdots & \log(|\sigma_{r_1}(\varepsilon_{s-1})|) \\ 2\log(|\tau_1(\varepsilon_1)|) & \cdots & 2\log(|\tau_1(\varepsilon_{s-1})|) \\ \vdots & & \vdots \\ 2\log(|\tau_{r_2}(\varepsilon_1)|) & \cdots & 2\log(|\tau_{r_2}(\varepsilon_{s-1})|) \end{bmatrix}.$$

The regulator $\mathrm{Reg}(K)$ is the absolute determinant of any matrix obtained from L_K by deleting one row.

Theorem 3.1.10 (Dirichlet unit theorem, [2])**.** *The image $\Lambda_K := \lambda_K(\mathscr{O}_K^\times)$ is a full lattice in H_K, and*

$$\begin{tikzcd} 1 \arrow[r] & \mu_K \arrow[r, hook] & \mathscr{O}_K^\times \arrow[r, two heads, "\lambda_K"] & \Lambda_K \arrow[r] \arrow[d, hook] & 0 \\ & & & H_K \arrow[d, hook] & \\ & & & \mathbb{R}^{r_1+r_2} & \end{tikzcd} \qquad \mathscr{O}_K^\times \cong \mu_K \times \mathbb{Z}^{r_1+r_2-1}. \tag{3.1.3}$$

Moreover, $\mathrm{Reg}(K) > 0$ *unless* $r_1 + r_2 - 1 = 0$*, when* $\mathrm{Reg}(K) := 1$.

Corollary 3.1.11 (Units in cyclotomic fields)**.** *Let $m > 2$ and $K_m = \mathbb{Q}(\zeta_m)$. Then*

$$\mathscr{O}_{K_m}^\times \cong \mu_{K_m} \times \mathbb{Z}^{\varphi(m)/2-1}, \qquad \mu_{K_m} = \begin{cases} \mu_m, & 2 \mid m, \\ \mu_{2m}, & 2 \nmid m. \end{cases}$$

Proof. Compute the signature and torsion subgroup.

$$m > 2 \implies r_1(K_m) = 0, \qquad r_2(K_m) = \frac{\varphi(m)}{2}.$$

$$2 \mid m \implies \mu_{K_m} = \langle \zeta_m \rangle = \mu_m, \qquad 2 \nmid m \implies K_m = K_{2m}, \quad \mu_{K_m} = \mu_{2m}.$$

$$r_1(K_m) = 0 \quad \wedge \quad r_2(K_m) = \frac{\varphi(m)}{2} \overset{3.1.3}{\implies} \mathrm{rank}(\mathscr{O}_{K_m}^\times/\mu_{K_m}) = r_1 + r_2 - 1 = \frac{\varphi(m)}{2} - 1.$$

This proves the stated decomposition. □

Definition 3.1.12 (Continued fractions)**.** For $a_0 \in \mathbb{Z}$ and $a_i \in \mathbb{Z}_{>0}$ for $i \geq 1$, set

$$[a_0; a_1, a_2, \dots] := a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \ddots}}.$$

Its convergents $p_k/q_k = [a_0;\dots,a_k]$ satisfy

$$p_{-2}=0,\quad p_{-1}=1,\quad q_{-2}=1,\quad q_{-1}=0,\qquad p_k=a_kp_{k-1}+p_{k-2},\quad q_k=a_kq_{k-1}+q_{k-2},$$
$$\begin{bmatrix} p_k & p_{k-1}\\ q_k & q_{k-1}\end{bmatrix} = \begin{bmatrix} a_0 & 1\\ 1 & 0\end{bmatrix}\cdot\begin{bmatrix} a_1 & 1\\ 1 & 0\end{bmatrix}\cdot\dots\cdot\begin{bmatrix} a_k & 1\\ 1 & 0\end{bmatrix},\qquad p_kq_{k-1}-p_{k-1}q_k=(-1)^{k+1}.$$

Theorem 3.1.13 (Lagrange and Legendre, [2])**.** *For $\xi\in\mathbb{R}\setminus\mathbb{Q}$,*

$$[\mathbb{Q}(\xi):\mathbb{Q}]=2 \iff \xi \textit{ has an eventually periodic simple continued fraction}.$$

Moreover,

$$\left|\xi-\frac{a}{b}\right|<\frac{1}{2b^2},\qquad (a,b)=1,\qquad b>0 \implies \frac{a}{b}=\frac{p_k}{q_k} \textit{ for some } k.$$

Theorem 3.1.14 (Pell equations and fundamental units, [2])**.** *Let $d>1$ be squarefree and*

$$\sqrt{d}=[a_0;\overline{a_1,\dots,a_{\ell-1},2a_0}].$$

Assume that ℓ is the least period. Then

$$p_{\ell-1}^2-dq_{\ell-1}^2=(-1)^\ell,\qquad x^2-dy^2=-1 \textit{ is solvable in } x,y\in\mathbb{Z} \iff 2\nmid\ell,$$
$$\min\{x+y\sqrt{d}>1 : x,y\in\mathbb{Z},\ x^2-dy^2=1\}=\begin{cases} p_{\ell-1}+q_{\ell-1}\sqrt{d}, & 2\mid\ell,\\ p_{2\ell-1}+q_{2\ell-1}\sqrt{d}, & 2\nmid\ell.\end{cases}$$

The order $\mathbb{Z}[\sqrt{d}]$ satisfies

$$\mathbb{Z}[\sqrt{d}]^\times=\{\pm\varepsilon_d^k : k\in\mathbb{Z}\},\qquad \varepsilon_d:=p_{\ell-1}+q_{\ell-1}\sqrt{d}.$$

If $d\equiv 2,3 \pmod 4$, this is $\mathscr{O}_K^\times$. If $d\equiv 1 \pmod 4$, then

$$u\in\mathscr{O}_K^\times \iff u=\frac{x+y\sqrt{d}}{2},\quad x,y\in\mathbb{Z},\quad x\equiv y \pmod 2,\quad x^2-dy^2=\pm4,$$

and the least such $u>1$ is a fundamental unit of $\mathscr{O}_K$.

Definition 3.1.15 (Binary quadratic forms)**.** Let $D<0$ with $D\equiv 0,1 \pmod 4$. A primitive positive-definite binary quadratic form of discriminant D is

$$Q(x,y)=ax^2+bxy+cy^2,\qquad a,b,c\in\mathbb{Z},\qquad (a,b,c)=1,\qquad a>0,\qquad b^2-4ac=D.$$

Proper equivalence is the $\mathrm{SL}_2(\mathbb{Z})$-action determined by

$$M:=\begin{bmatrix} p & q\\ r & s\end{bmatrix}\in\mathrm{SL}_2(\mathbb{Z}),\qquad M\cdot\begin{bmatrix} x\\ y\end{bmatrix}=\begin{bmatrix} x'\\ y'\end{bmatrix},\qquad Q'(x,y):=Q(x',y').$$

The form $[a,b,c]$ is reduced when

$$|b|\le a\le c,\qquad b\ge 0 \text{ if } |b|=a \text{ or } a=c.$$

Write $\mathscr{F}(D)$ for the proper-equivalence classes and $h(D):=\#(\mathscr{F}(D))$.

Theorem 3.1.16 (Reduction, [2])**.** *Every class in $\mathscr{F}(D)$ contains exactly one reduced form. For a reduced form $[a,b,c]$,*

$$3a^2\le|D|,\qquad 1\le a\le\sqrt{|D|/3},\qquad |b|\le a,$$

so $h(D)<\infty$.

Theorem 3.1.17 (Gauss composition and ideal classes, [2])**.** *Let $D<0$ be a fundamental discriminant and $K=\mathbb{Q}(\sqrt{D})$. Gauss composition makes $\mathscr{F}(D)$ an abelian group, and*

$$\begin{tikzcd} \mathscr{F}(D)\times\mathscr{F}(D) \arrow[r, "*"] \arrow[d, "\Phi_D\times\Phi_D"'] & \mathscr{F}(D) \arrow[d, "\Phi_D", "\sim"'] \\ \mathrm{Cl}(K)\times\mathrm{Cl}(K) \arrow[r, "{([\mathfrak{a}],[\mathfrak{b}])\mapsto[\mathfrak{a}\mathfrak{b}]}"'] & \mathrm{Cl}(K) \end{tikzcd} \qquad \Phi_D([a,b,c])=\left[a\mathbb{Z}+\frac{-b+\sqrt{D}}{2}\mathbb{Z}\right]. \tag{3.1.4}$$

The identity, inverse, and principal form are

$$[a,b,c]^{-1}=[a,-b,c],\qquad Q_D(x,y)=\begin{cases}x^2-\dfrac{D}{4}y^2, & D\equiv 0 \pmod 4,\\ x^2+xy+\dfrac{1-D}{4}y^2, & D\equiv 1 \pmod 4.\end{cases}$$

In particular,

$$h(D)=h_K=\#(\mathrm{Cl}(K)).$$

Corollary 3.1.18 (Prime representation)**.** *Let $p\nmid D$ be an odd rational prime. Then*

$$p \text{ is represented by a primitive positive-definite form of discriminant } D \iff \left(\frac{D}{p}\right)=1.$$

Proof. Translate splitting into an ideal of norm p.

$$\left(\frac{D}{p}\right)=1 \overset{2.1.12}{\iff} p\mathscr{O}_K=\mathfrak{p}\overline{\mathfrak{p}},\qquad \mathrm{N}(\mathfrak{p})=p.$$

$$\exists b\in\mathbb{Z}: b^2\equiv D \pmod{4p},\quad \mathfrak{p}=p\mathbb{Z}+\frac{-b+\sqrt{D}}{2}\mathbb{Z}.$$

$$Q_{\mathfrak{p}}:=\left[p,b,\frac{b^2-D}{4p}\right],\qquad \Phi_D([Q_{\mathfrak{p}}])=[\mathfrak{p}],\qquad Q_{\mathfrak{p}}(1,0)=p\implies p \text{ is represented.}$$

$$Q(x,y)=p\implies (x,y)=1\implies \exists r,s\in\mathbb{Z}: M:=\begin{bmatrix}x & r\\ y & s\end{bmatrix}\in\mathrm{SL}_2(\mathbb{Z}),\qquad \det(M)=xs-ry=1.$$

$$M\cdot\begin{bmatrix}u\\ v\end{bmatrix}=\begin{bmatrix}xu+rv\\ yu+sv\end{bmatrix},$$

$$Q'(u,v):=Q(xu+rv,yu+sv)=pu^2+b'uv+c'v^2,$$

$$D\equiv(b')^2 \pmod p \implies \left(\frac{D}{p}\right)=1.$$

This proves the representation criterion. $\square$

Corollary 3.1.19 (Principal representation)**.** *For an odd rational prime $p\nmid D$,*

$$p=Q_D(x,y) \text{ for some } x,y\in\mathbb{Z} \iff p\mathscr{O}_K=\mathfrak{p}\overline{\mathfrak{p}} \text{ with } [\mathfrak{p}]=1 \text{ in } \mathrm{Cl}(K).$$

Proof. Use the identity under Equation (3.1.4).

$$\Phi_D([Q_D])=[\mathscr{O}_K],\qquad p=Q_D(x,y)\implies \exists\alpha\in\mathscr{O}_K: |\mathrm{N}_{K/\mathbb{Q}}(\alpha)|=p.$$

$$0\neq\alpha\in\mathscr{O}_K \quad\wedge\quad |\mathrm{N}_{K/\mathbb{Q}}(\alpha)|=p \overset{1.1.11}{\implies} \mathrm{N}(\alpha\mathscr{O}_K)=[\mathscr{O}_K:\alpha\mathscr{O}_K]=|\mathrm{N}_{K/\mathbb{Q}}(\alpha)|=p.$$

$$\mathfrak{p}:=\alpha\mathscr{O}_K,\qquad p\mathscr{O}_K=(\alpha\overline{\alpha})\mathscr{O}_K=\mathfrak{p}\overline{\mathfrak{p}}.$$

$$p\mathscr{O}_K=\mathfrak{p}\overline{\mathfrak{p}},\quad \mathfrak{p}=\alpha\mathscr{O}_K\implies p=|\mathrm{N}_{K/\mathbb{Q}}(\alpha)|.$$

$$\omega_D := \begin{cases} \sqrt{D}/2, & D \equiv 0 \pmod 4, \\ (1+\sqrt{D})/2, & D \equiv 1 \pmod 4, \end{cases} \qquad \alpha = x + y\omega_D \implies \mathrm{N}_{K/\mathbb{Q}}(\alpha) = Q_D(x,y) = p.$$

This proves the principal representation criterion. $\square$

Example 3.1.20 (Two squares)**.** For $D = -4$, one has $K = \mathbb{Q}(i)$, $\mathrm{Cl}(K) = 1$, and $Q_D(x,y) = x^2 + y^2$. Hence

$$p = x^2 + y^2 \text{ for some } x, y \in \mathbb{Z} \iff p = 2 \text{ or } p \equiv 1 \pmod 4.$$

Theorem 3.1.21 (Genus theory, [2])**.** *Factor the fundamental discriminant as $D = q_1^* \cdots q_t^*$ into prime discriminants. If Q represents m with $(m, D) = 1$, set*

$$\chi_i([Q]) := \left(\frac{q_i^*}{m}\right), \qquad \mathscr{G}_D := \left\{ (\varepsilon_1, \dots, \varepsilon_t) \in \{\pm 1\}^t : \prod_{i=1}^{t} \varepsilon_i = 1 \right\}.$$

The characters are well-defined, two classes have the same genus exactly when all χ_i agree, and

$$\begin{tikzcd} 1 \arrow[r] & \mathscr{F}(D)^2 \arrow[r, hook] \arrow[d, "\Phi_D", "\sim"'] & \mathscr{F}(D) \arrow[r, two heads, "(\chi_1,\dots,\chi_t)"] \arrow[d, "\Phi_D", "\sim"'] & \mathscr{G}_D \arrow[r] \arrow[d, equal] & 1 \\ 1 \arrow[r] & \mathrm{Cl}(K)^2 \arrow[r, hook] & \mathrm{Cl}(K) \arrow[r, two heads] & \mathscr{G}_D \arrow[r] & 1. \end{tikzcd}$$

Consequently,

$$\mathrm{Cl}(K)/\mathrm{Cl}(K)^2 \cong (\mathbb{Z}/2\mathbb{Z})^{t-1}, \qquad t > 1 \implies 2^{t-1} \mid h_K.$$

Example 3.1.22 (The classes of $\mathbb{Q}(\sqrt{-5})$ and $\mathbb{Q}(\sqrt{-23})$)**.** For $K = \mathbb{Q}(\sqrt{-5})$,

$$\mathrm{disc}(K/\mathbb{Q}) = -20, \qquad M_K = \frac{2}{\pi}\sqrt{20} < 3.$$

Let $\mathfrak{p}$ be the prime above 2 from Example 2.1.13. Then

$$\mathrm{N}(\mathfrak{p}) = 2 \quad \wedge \quad \nexists a, b \in \mathbb{Z} : \mathrm{N}_{K/\mathbb{Q}}(a + b\sqrt{-5}) = a^2 + 5b^2 = 2 \overset{2}{\implies} \mathfrak{p} \text{ is nonprincipal.}$$

$$\mathfrak{p}^2 = 2\mathscr{O}_K \quad \text{by Example 2.1.13} \implies [\mathfrak{p}]^2 = 1.$$

$$M_K < 3 \quad \wedge \quad 2\mathscr{O}_K = \mathfrak{p}^2 \quad \wedge \quad \mathfrak{p} \text{ is nonprincipal} \overset{1}{\implies} \mathrm{Cl}(K) = \{1, [\mathfrak{p}]\} \cong \mathbb{Z}/2\mathbb{Z}.$$

The reduced forms are $[1,0,5]$ and $[2,2,3]$.

For $L = \mathbb{Q}(\sqrt{-23})$ and $\omega = (1+\sqrt{-23})/2$,

$$\mathrm{disc}(L/\mathbb{Q}) = -23, \qquad M_L = \frac{2}{\pi}\sqrt{23} < 4, \qquad \mathrm{N}_{L/\mathbb{Q}}(a + b\omega) = a^2 + ab + 6b^2.$$

$$\mathfrak{p}_2 = (2, \omega), \qquad \mathfrak{p}_3 = (3, \omega), \qquad 2\mathscr{O}_L = \mathfrak{p}_2\overline{\mathfrak{p}}_2, \qquad 3\mathscr{O}_L = \mathfrak{p}_3\overline{\mathfrak{p}}_3.$$

$$\omega\mathscr{O}_L = \mathfrak{p}_2\mathfrak{p}_3, \qquad [\mathfrak{p}_3] = [\overline{\mathfrak{p}}_2] = [\mathfrak{p}_2]^{-1}, \qquad [\overline{\mathfrak{p}}_3] = [\mathfrak{p}_2].$$

$$a^2 + ab + 6b^2 \notin \{2, 3\}, \qquad a^2 + ab + 6b^2 = 4 \iff (a,b) = (\pm 2, 0).$$

$$[\mathfrak{p}_2]^2 = 1 \implies \mathfrak{p}_2^2 = \alpha\mathscr{O}_L, \quad \mathrm{N}_{L/\mathbb{Q}}(\alpha) = 4 \implies \alpha = \pm 2 \implies \mathfrak{p}_2^2 = 2\mathscr{O}_L = \mathfrak{p}_2\overline{\mathfrak{p}}_2 \implies \mathfrak{p}_2 = \overline{\mathfrak{p}}_2,$$

which contradicts the splitting of 2. Thus

$$\mathrm{Cl}(L) \cong \mathbb{Z}/3\mathbb{Z}, \qquad \mathscr{F}(-23) = \{[1,1,6], [2,1,3], [2,-1,3]\}.$$

4. Localization and Relative Arithmetic

Let L/K be a finite extension of number fields. Localization isolates one prime of K, while the different and the decomposition groups measure its ramification and residue-field behavior in L.

4.1. **Valuations, the Different, and Relative Frobenius.**

Definition 4.1.1 (Localization)**.** Let A be an integral domain and let $S \subset A$ satisfy

$$1 \in S, \qquad 0 \notin S, \qquad s,t \in S \implies s \cdot t \in S.$$

The localization of A and an A-module M at S are

$$S^{-1}A := \left\{ \frac{a}{s} : a \in A,\ s \in S \right\} \subset \mathrm{Frac}(A), \qquad S^{-1}M := \left\{ \frac{m}{s} : m \in M,\ s \in S \right\}.$$

For a prime ideal $\mathfrak{p} \subset A$, put

$$A_{\mathfrak{p}} := (A \setminus \mathfrak{p})^{-1}A, \qquad M_{\mathfrak{p}} := (A \setminus \mathfrak{p})^{-1}M, \qquad \kappa(\mathfrak{p}) := \mathrm{Frac}(A/\mathfrak{p}).$$

Proposition 4.1.2 (Universal property and prime correspondence, [2])**.** *Let $f\colon A \to B$ satisfy $f(S) \subset B^{\times}$. There is a unique factorization*

$$\begin{tikzcd} A \arrow[r, "\lambda"] \arrow[dr, "f"'] & S^{-1}A \arrow[d, dashed, "\exists!\,\widetilde{f}"] \\ & B \end{tikzcd} \qquad \widetilde{f}\left(\frac{a}{s}\right) = f(a) \cdot f(s)^{-1}.$$

Extension and contraction give inverse order-preserving bijections

$$\begin{tikzcd} \mathrm{Spec}(S^{-1}A) \arrow[r, bend left, "\mathfrak{q} \mapsto \lambda^{-1}(\mathfrak{q})"] & \{\mathfrak{p} \in \mathrm{Spec}(A) : \mathfrak{p} \cap S = \varnothing\} \arrow[l, bend left, "\mathfrak{p} \mapsto S^{-1}\mathfrak{p}"] \end{tikzcd}$$

and, in particular,

$$\mathrm{Spec}(A_{\mathfrak{p}}) \longleftrightarrow \{\mathfrak{q} \in \mathrm{Spec}(A) : \mathfrak{q} \subseteq \mathfrak{p}\}, \qquad \mathfrak{p}A_{\mathfrak{p}} \text{ is the unique maximal ideal of } A_{\mathfrak{p}}.$$

Definition 4.1.3 (Discrete valuation ring)**.** A ring is local when it has a unique maximal ideal. A discrete valuation ring is a nonfield local Dedekind domain A, equivalently a nonfield local principal ideal domain whose nonzero ideals are the powers of its maximal ideal $\mathfrak{m}$. A uniformizer is an element $\pi \in A$ satisfying

$$\mathfrak{m} = \pi A, \qquad \{0 \neq \mathfrak{a} \subset A\} = \{\pi^n A : n \in \mathbb{Z}_{\geq 0}\}.$$

A normalized discrete valuation on a field F is a surjective map $v\colon F^{\times} \to \mathbb{Z}$ satisfying

$$v(xy) = v(x) + v(y), \qquad v(x+y) \geq \min(v(x), v(y)), \qquad v(0) := \infty.$$

Its valuation ring, maximal ideal, and unit group are

$$\mathcal{O}_v := \{x \in F : v(x) \geq 0\}, \qquad \mathfrak{m}_v := \{x \in F : v(x) > 0\}, \qquad \mathcal{O}_v^{\times} = \{x \in F : v(x) = 0\}.$$

Theorem 4.1.4 (Local Dedekind arithmetic, [2])**.** *Let A be a Dedekind domain, let $F = \mathrm{Frac}(A)$, and let $0 \neq \mathfrak{p} \subset A$ be prime. Then $A_{\mathfrak{p}}$ is a discrete valuation ring, and every $\pi \in \mathfrak{p} \setminus \mathfrak{p}^2$ is a uniformizer. The associated valuation satisfies*

$$A_{\mathfrak{p}} = \{x \in F : v_{\mathfrak{p}}(x) \geq 0\}, \qquad \mathfrak{p}A_{\mathfrak{p}} = \{x \in F : v_{\mathfrak{p}}(x) > 0\},$$
$$x = u\pi^{v_{\mathfrak{p}}(x)}, \qquad u \in A_{\mathfrak{p}}^{\times}, \qquad x \in F^{\times}.$$

Proof. Localize the unique prime-ideal factorization of A.

$$\mathfrak{q} \neq \mathfrak{p} \implies \exists s \in \mathfrak{q} \setminus \mathfrak{p} \implies s \in A_{\mathfrak{p}}^{\times} \cap \mathfrak{q}A_{\mathfrak{p}} \implies \mathfrak{q}A_{\mathfrak{p}} = A_{\mathfrak{p}}.$$

$$0 \neq \mathfrak{a} \subset A, \qquad \mathfrak{a} = \prod_{\mathfrak{q}} \mathfrak{q}^{v_{\mathfrak{q}}(\mathfrak{a})} \implies \mathfrak{a}A_{\mathfrak{p}} = \mathfrak{p}^{v_{\mathfrak{p}}(\mathfrak{a})}A_{\mathfrak{p}}.$$

$$0 \neq I \subset A_{\mathfrak{p}} \implies I = (I \cap A)A_{\mathfrak{p}} = \mathfrak{p}^n A_{\mathfrak{p}}, \qquad \pi \in \mathfrak{p} \setminus \mathfrak{p}^2 \implies \mathfrak{p}A_{\mathfrak{p}} = \pi A_{\mathfrak{p}}.$$

$$xA = \prod_{\mathfrak{q}} \mathfrak{q}^{v_{\mathfrak{q}}(x)} \implies xA_{\mathfrak{p}} = \pi^{v_{\mathfrak{p}}(x)}A_{\mathfrak{p}} \implies x = u\pi^{v_{\mathfrak{p}}(x)}, \quad u \in A_{\mathfrak{p}}^{\times}.$$

Thus localization at $\mathfrak{p}$ is the valuation ring of $v_{\mathfrak{p}}$. □

Corollary 4.1.5 (Local–global ideal dictionary)**.** *Let A be a Dedekind domain with fraction field F, and let $\mathfrak{a}, \mathfrak{b}$ be nonzero fractional ideals. Then*

$$\mathfrak{a}A_{\mathfrak{p}} = \mathfrak{p}^{v_{\mathfrak{p}}(\mathfrak{a})}A_{\mathfrak{p}}, \qquad \mathfrak{a} = \bigcap_{0 \neq \mathfrak{p} \in \operatorname{Spec}(A)} \mathfrak{a}A_{\mathfrak{p}} \subset F,$$

$$\mathfrak{a} \subseteq \mathfrak{b} \iff v_{\mathfrak{p}}(\mathfrak{a}) \geq v_{\mathfrak{p}}(\mathfrak{b}) \textit{ for every } \mathfrak{p} \iff \mathfrak{a}A_{\mathfrak{p}} \subseteq \mathfrak{b}A_{\mathfrak{p}} \textit{ for every } \mathfrak{p}.$$

Theorem 4.1.6 (Structure theorem over a principal ideal domain, [2])**.** *Let R be a principal ideal domain and let M be a finitely generated R-module. There are unique $r, t \in \mathbb{Z}_{\geq 0}$ and nonzero nonunits $d_1, \dots, d_t \in R$, unique up to associates, such that*

$$M \cong R^{\oplus r} \oplus \bigoplus_{i=1}^{t} R/(d_i), \qquad d_1 \mid d_2 \mid \cdots \mid d_t.$$

Consequently,

$$M \textit{ torsion-free} \iff M \cong R^{\oplus r}.$$

Remark 4.1.7. The principal-ideal hypothesis is essential,

$$(X,Y) \subset \mathbb{C}[X,Y], \qquad (X,Y) \text{ is finitely generated and torsion-free but not free.}$$

Definition 4.1.8 (Relative prime data)**.** Let L/K be finite of degree n, and let $0 \neq \mathfrak{p} \subset \mathscr{O}_K$ be prime. Write

$$\mathfrak{p}\mathscr{O}_L = \prod_{i=1}^{g} \mathfrak{q}_i^{e_i}, \qquad \mathscr{O}_{L,\mathfrak{p}} := (\mathscr{O}_K \setminus \mathfrak{p})^{-1}\mathscr{O}_L.$$

$$k_{\mathfrak{p}} := \mathscr{O}_K/\mathfrak{p}, \qquad k_{\mathfrak{q}_i} := \mathscr{O}_L/\mathfrak{q}_i, \qquad f_i := [k_{\mathfrak{q}_i} : k_{\mathfrak{p}}].$$

Here

$$e_i = e(\mathfrak{q}_i \mid \mathfrak{p}), \qquad f_i = f(\mathfrak{q}_i \mid \mathfrak{p}), \qquad \mathrm{N}_{L/K}(\mathfrak{q}_i) := \mathfrak{p}^{f_i}.$$

If $\mathfrak{p} \cap \mathbb{Z} = \ell\mathbb{Z}$ and $f_0 = [k_{\mathfrak{p}} : \mathbb{F}_\ell]$, then

$$k_{\mathfrak{p}} \cong \mathbb{F}_{\ell^{f_0}}, \qquad k_{\mathfrak{q}_i} \cong \mathbb{F}_{\ell^{f_0 f_i}}.$$

The prime $\mathfrak{p}$ is unramified, split completely, inert, or totally ramified according as

behavior	relative prime data
unramified	$e_i = 1$ for every i
split completely	$g = n, \quad e_i = f_i = 1$ for every i
inert	$g = 1, \quad e_1 = 1, \quad f_1 = n$
totally ramified	$g = 1, \quad e_1 = n, \quad f_1 = 1.$

Theorem 4.1.9 (Relative prime decomposition)**.** *Let L/K, $\mathfrak{p}$, and $\mathfrak{q}_i$ be as in Definition 4.1.8, and put*

$$A := \mathscr{O}_{K,\mathfrak{p}}, \qquad B := \mathscr{O}_{L,\mathfrak{p}}.$$

Then B is a free A-module of rank n, its maximal ideals are $\mathfrak{q}_i B$, and

$$\begin{tikzcd} \mathscr{O}_L \arrow[r, hook] \arrow[d, two heads] & B \arrow[d, two heads] \\ \mathscr{O}_L/\mathfrak{p}\mathscr{O}_L \arrow[r, "\sim"] \arrow[dr, "\sim"'] & B/\mathfrak{p}B \arrow[d, "\sim"] \\ & \prod_{i=1}^{g} \mathscr{O}_L/\mathfrak{q}_i^{e_i} \end{tikzcd} \qquad \sum_{i=1}^{g} e_i \cdot f_i = [L:K]. \tag{4.1.1}$$

Proof. Use local freeness and the primary decomposition of the special fiber.

$$B \text{ is finite and torsion-free over the DVR } A \overset{4.1.6}{\Longrightarrow} B \cong A^{\oplus n}, \qquad \dim_{k_\mathfrak{p}}(B/\mathfrak{p}B) = n.$$

$$\mathfrak{p}B = \prod_{i=1}^{g} (\mathfrak{q}_i B)^{e_i}, \qquad \mathfrak{q}_i B + \mathfrak{q}_j B = B \quad (i \neq j) \overset{2.1.4}{\Longrightarrow} B/\mathfrak{p}B \cong \prod_{i=1}^{g} B/(\mathfrak{q}_i B)^{e_i}.$$

$$(\mathfrak{q}_i B)^j/(\mathfrak{q}_i B)^{j+1} \cong k_{\mathfrak{q}_i}, \qquad 0 \le j < e_i \implies \dim_{k_\mathfrak{p}}(B/(\mathfrak{q}_i B)^{e_i}) = e_i \cdot f_i.$$

$$n = \dim_{k_\mathfrak{p}}(B/\mathfrak{p}B) = \sum_{i=1}^{g} \dim_{k_\mathfrak{p}}(B/(\mathfrak{q}_i B)^{e_i}) = \sum_{i=1}^{g} e_i \cdot f_i.$$

This proves the relative degree formula and the displayed fiber decomposition. □

Proposition 4.1.10 (Relative norm valuation, [2])**.** *Let $0 \neq x \in L$. For every prime $\mathfrak{p} \subset \mathscr{O}_K$,*

$$\mathrm{N}_{L/K}(x\mathscr{O}_L) = \mathrm{N}_{L/K}(x)\mathscr{O}_K, \qquad x\mathscr{O}_L = \prod_{\mathfrak{q}} \mathfrak{q}^{v_\mathfrak{q}(x)}.$$

$$v_\mathfrak{p}(\mathrm{N}_{L/K}(x)) = \sum_{\mathfrak{q}|\mathfrak{p}} f(\mathfrak{q} \mid \mathfrak{p}) \cdot v_\mathfrak{q}(x).$$

Theorem 4.1.11 (Relative Dedekind–Kummer, [2])**.** *Let $L = K(\alpha)$, where $\alpha \in \mathscr{O}_L$ has minimal polynomial $F(X) \in \mathscr{O}_K[X]$, and assume*

$$\mathscr{O}_{L,\mathfrak{p}} = \mathscr{O}_{K,\mathfrak{p}}[\alpha].$$

If

$$\overline{F}(X) = \prod_{i=1}^{g} \overline{G}_i(X)^{e_i} \in k_\mathfrak{p}[X]$$

is the factorization into distinct monic irreducibles, with lifts $G_i(X) \in \mathscr{O}_K[X]$, then

$$\begin{tikzcd} \mathscr{O}_{K,\mathfrak{p}}[X] \arrow[r, two heads, "X \mapsto \alpha"] \arrow[d, two heads] & \mathscr{O}_{L,\mathfrak{p}} \arrow[d, two heads] \\ k_\mathfrak{p}[X] \arrow[r, two heads] & k_\mathfrak{p}[X]/(\overline{F}) \end{tikzcd}$$

$$\mathfrak{p}\mathscr{O}_L = \prod_{i=1}^{g} \mathfrak{q}_i^{e_i}, \qquad \mathfrak{q}_i = \mathfrak{p}\mathscr{O}_L + G_i(\alpha)\mathscr{O}_L, \qquad k_{\mathfrak{q}_i} \cong k_\mathfrak{p}[X]/(\overline{G}_i), \qquad f_i = \deg(\overline{G}_i).$$

Theorem 4.1.12 (Tower laws, [2]). *Let $K \subset L \subset M$ be number fields and let $\mathfrak{r} \mid \mathfrak{q} \mid \mathfrak{p}$ be primes of $\mathscr{O}_M, \mathscr{O}_L, \mathscr{O}_K$, respectively. Then*

$$\begin{tikzcd} \mathscr{O}_{K,\mathfrak{p}} \arrow[r, hook] \arrow[d, two heads] & \mathscr{O}_{L,\mathfrak{q}} \arrow[r, hook] \arrow[d, two heads] & \mathscr{O}_{M,\mathfrak{r}} \arrow[d, two heads] \\ k_{\mathfrak{p}} \arrow[r, hook] & k_{\mathfrak{q}} \arrow[r, hook] & k_{\mathfrak{r}} \end{tikzcd}$$

$$e(\mathfrak{r} \mid \mathfrak{p}) = e(\mathfrak{r} \mid \mathfrak{q}) \cdot e(\mathfrak{q} \mid \mathfrak{p}), \qquad f(\mathfrak{r} \mid \mathfrak{p}) = f(\mathfrak{r} \mid \mathfrak{q}) \cdot f(\mathfrak{q} \mid \mathfrak{p}).$$

Proof. Compare prime exponents and residue-field dimensions through the tower.

$$\mathfrak{p}\mathscr{O}_M = (\mathfrak{p}\mathscr{O}_L)\mathscr{O}_M = \prod_{\mathfrak{q}'\mid\mathfrak{p}} \prod_{\mathfrak{r}'\mid\mathfrak{q}'} (\mathfrak{r}')^{e(\mathfrak{r}'\mid\mathfrak{q}')\cdot e(\mathfrak{q}'\mid\mathfrak{p})}.$$

$$v_{\mathfrak{r}}(\mathfrak{p}\mathscr{O}_M) = e(\mathfrak{r} \mid \mathfrak{p}) = e(\mathfrak{r} \mid \mathfrak{q}) \cdot e(\mathfrak{q} \mid \mathfrak{p}).$$

$$k_{\mathfrak{p}} \subset k_{\mathfrak{q}} \subset k_{\mathfrak{r}} \implies [k_{\mathfrak{r}} : k_{\mathfrak{p}}] = [k_{\mathfrak{r}} : k_{\mathfrak{q}}] \cdot [k_{\mathfrak{q}} : k_{\mathfrak{p}}].$$

$$f(\mathfrak{r} \mid \mathfrak{p}) = f(\mathfrak{r} \mid \mathfrak{q}) \cdot f(\mathfrak{q} \mid \mathfrak{p}).$$

This proves both tower laws. □

Definition 4.1.13 (Codifferent and different). The trace-dual lattice, codifferent, and different of L/K are

$$\mathscr{O}_L^\vee := \{x \in L : \mathrm{Tr}_{L/K}(x\mathscr{O}_L) \subseteq \mathscr{O}_K\}, \qquad \mathfrak{d}(L/K)^{-1} := \mathscr{O}_L^\vee, \qquad \mathfrak{d}(L/K) := (\mathscr{O}_L^\vee)^{-1}.$$

The relative discriminant ideal is

$$\mathrm{disc}(L/K) := \left(\det\left(\begin{bmatrix} \mathrm{Tr}_{L/K}(x_1 \cdot x_1) & \mathrm{Tr}_{L/K}(x_1 \cdot x_2) & \cdots & \mathrm{Tr}_{L/K}(x_1 \cdot x_n) \\ \mathrm{Tr}_{L/K}(x_2 \cdot x_1) & \mathrm{Tr}_{L/K}(x_2 \cdot x_2) & \cdots & \mathrm{Tr}_{L/K}(x_2 \cdot x_n) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Tr}_{L/K}(x_n \cdot x_1) & \mathrm{Tr}_{L/K}(x_n \cdot x_2) & \cdots & \mathrm{Tr}_{L/K}(x_n \cdot x_n) \end{bmatrix} \right) : x_1, \ldots, x_n \in \mathscr{O}_L \right) \subseteq \mathscr{O}_K.$$

Theorem 4.1.14 (Different identities, [2]). *Let $K \subset L \subset M$ be finite extensions of number fields. The trace maps and differents satisfy*

$$\begin{tikzcd} M \arrow[r, "\mathrm{Tr}_{M/L}"] \arrow[dr, "\mathrm{Tr}_{M/K}"'] & L \arrow[d, "\mathrm{Tr}_{L/K}"] \\ & K \end{tikzcd} \qquad \mathrm{Tr}_{M/K} = \mathrm{Tr}_{L/K} \circ \mathrm{Tr}_{M/L}, \qquad \mathfrak{d}(M/K) = \mathfrak{d}(M/L) \cdot \mathfrak{d}(L/K)\mathscr{O}_M. \tag{4.1.2}$$

For every multiplicative set $S \subset \mathscr{O}_K$,

$$S^{-1}(\mathscr{O}_L^\vee) = (S^{-1}\mathscr{O}_L)^\vee, \qquad S^{-1}\mathfrak{d}(L/K) = \mathfrak{d}(S^{-1}\mathscr{O}_L/S^{-1}\mathscr{O}_K).$$

If $\mathscr{O}_L = \mathscr{O}_K[\alpha]$ and $F(X)$ is the minimal polynomial of α, then

$$\mathscr{O}_L^\vee = \frac{1}{F'(\alpha)}\mathscr{O}_L, \qquad \mathfrak{d}(L/K) = F'(\alpha)\mathscr{O}_L.$$

Finally,

$$\mathrm{disc}(L/K) = \mathrm{N}_{L/K}(\mathfrak{d}(L/K)), \qquad \mathrm{disc}(M/K) = \mathrm{disc}(L/K)^{[M:L]} \cdot \mathrm{N}_{L/K}(\mathrm{disc}(M/L)).$$

Theorem 4.1.15 (Different exponent, [2]). *Let $\mathfrak{q} \mid \mathfrak{p}$ in L/K, and put*

$$d(\mathfrak{q} \mid \mathfrak{p}) := v_{\mathfrak{q}}(\mathfrak{d}(L/K)), \qquad e := e(\mathfrak{q} \mid \mathfrak{p}), \qquad p := \mathrm{char}(k_{\mathfrak{p}}).$$

Then

$$d(\mathfrak{q} \mid \mathfrak{p}) \geq e - 1, \qquad d(\mathfrak{q} \mid \mathfrak{p}) = e - 1 \iff p \nmid e,$$
$$d(\mathfrak{q} \mid \mathfrak{p}) = 0 \iff e(\mathfrak{q} \mid \mathfrak{p}) = 1.$$

Theorem 4.1.16 (Relative discriminant criterion)**.** *Let L/K be finite, let $\mathfrak{p} \subset \mathscr{O}_K$ be prime, and let $\mathfrak{q} \mid \mathfrak{p}$. Then*

$$\mathfrak{q} \mid \mathfrak{d}(L/K) \iff e(\mathfrak{q} \mid \mathfrak{p}) > 1, \qquad \mathfrak{p} \mid \operatorname{disc}(L/K) \iff \exists \mathfrak{q} \mid \mathfrak{p} : e(\mathfrak{q} \mid \mathfrak{p}) > 1.$$

Proof. Localize the trace pairing and reduce its discriminant modulo $\mathfrak{p}$.

$$B := (\mathscr{O}_K \setminus \mathfrak{p})^{-1}\mathscr{O}_L, \qquad \operatorname{disc}(L/K)\mathscr{O}_{K,\mathfrak{p}} = \operatorname{disc}(B/\mathscr{O}_{K,\mathfrak{p}}).$$

$$B/\mathfrak{p}B \cong \prod_{\mathfrak{q}\mid\mathfrak{p}} \mathscr{O}_L/\mathfrak{q}^{e(\mathfrak{q}\mid\mathfrak{p})}.$$

$$\mathfrak{p} \nmid \operatorname{disc}(L/K) \iff \operatorname{disc}((B/\mathfrak{p}B)/k_\mathfrak{p}) \neq 0 \iff B/\mathfrak{p}B \text{ is reduced.}$$

$$B/\mathfrak{p}B \text{ is reduced} \iff e(\mathfrak{q} \mid \mathfrak{p}) = 1 \text{ for every } \mathfrak{q} \mid \mathfrak{p} \overset{4.1.15}{\iff} \mathfrak{q} \nmid \mathfrak{d}(L/K) \text{ for every } \mathfrak{q} \mid \mathfrak{p}.$$

$$\mathfrak{p} \mid \operatorname{disc}(L/K) \overset{4.1.14}{\iff} \exists \mathfrak{q} \mid \mathfrak{p} : \mathfrak{q} \mid \mathfrak{d}(L/K) \iff \exists \mathfrak{q} \mid \mathfrak{p} : e(\mathfrak{q} \mid \mathfrak{p}) > 1.$$

This proves both ramification criteria. □

Corollary 4.1.17 (Finite ramification)**.** *Only finitely many primes of K ramify in L, namely*

$$\{\mathfrak{p} \subset \mathscr{O}_K : \mathfrak{p} \textit{ ramifies in } L\} = \{\mathfrak{p} \subset \mathscr{O}_K : \mathfrak{p} \mid \operatorname{disc}(L/K)\}.$$

Definition 4.1.18 (Relative decomposition and inertia)**.** Let L/K be Galois with $G := \operatorname{Gal}(L/K)$, and let $\mathfrak{q} \mid \mathfrak{p}$. The decomposition and inertia groups are

$$D(\mathfrak{q} \mid \mathfrak{p}) := \{\sigma \in G : \sigma(\mathfrak{q}) = \mathfrak{q}\},$$
$$I(\mathfrak{q} \mid \mathfrak{p}) := \{\sigma \in D(\mathfrak{q} \mid \mathfrak{p}) : \sigma(x) \equiv x \pmod{\mathfrak{q}} \text{ for every } x \in \mathscr{O}_L\}.$$

Theorem 4.1.19 (Relative decomposition–inertia sequence, [2])**.** *Let L/K be Galois with group G, and let $\mathfrak{q} \mid \mathfrak{p}$ have relative data (e, f, g). Then*

$$G/D(\mathfrak{q} \mid \mathfrak{p}) \xrightarrow{\sim} \{\mathfrak{q}' \subset \mathscr{O}_L : \mathfrak{q}' \mid \mathfrak{p}\}, \qquad \sigma D(\mathfrak{q} \mid \mathfrak{p}) \longmapsto \sigma(\mathfrak{q}),$$

$$1 \longrightarrow I(\mathfrak{q} \mid \mathfrak{p}) \hookrightarrow D(\mathfrak{q} \mid \mathfrak{p}) \xrightarrow{\sigma \mapsto (\bar{x} \mapsto \overline{\sigma(x)})} \operatorname{Gal}(k_\mathfrak{q}/k_\mathfrak{p}) \longrightarrow 1 \tag{4.1.3}$$

and

$$|D(\mathfrak{q} \mid \mathfrak{p})| = e \cdot f, \qquad |I(\mathfrak{q} \mid \mathfrak{p})| = e, \qquad D(\mathfrak{q} \mid \mathfrak{p})/I(\mathfrak{q} \mid \mathfrak{p}) \cong \operatorname{Gal}(k_\mathfrak{q}/k_\mathfrak{p}) \cong \mathbb{Z}/f\mathbb{Z}.$$

For every $\sigma \in G$,

$$D(\sigma(\mathfrak{q}) \mid \mathfrak{p}) = \sigma D(\mathfrak{q} \mid \mathfrak{p})\sigma^{-1}, \qquad I(\sigma(\mathfrak{q}) \mid \mathfrak{p}) = \sigma I(\mathfrak{q} \mid \mathfrak{p})\sigma^{-1}.$$

Theorem 4.1.20 (Decomposition groups in towers)**.** *Let L/K be Galois, let M/K be an intermediate Galois extension, and put*

$$G := \operatorname{Gal}(L/K), \qquad H := \operatorname{Gal}(L/M) \trianglelefteq G, \qquad \mathfrak{r} := \mathfrak{q} \cap \mathscr{O}_M, \qquad \mathfrak{p} := \mathfrak{q} \cap \mathscr{O}_K.$$

Restriction gives the commutative exact diagram

$$\begin{tikzcd}
1 \arrow[r] & I(\mathfrak{q} \mid \mathfrak{r}) \arrow[r, hook] \arrow[d, hook] & I(\mathfrak{q} \mid \mathfrak{p}) \arrow[r, two heads] \arrow[d, hook] & I(\mathfrak{r} \mid \mathfrak{p}) \arrow[r] \arrow[d, hook] & 1 \\
1 \arrow[r] & D(\mathfrak{q} \mid \mathfrak{r}) \arrow[r, hook] \arrow[d, two heads] & D(\mathfrak{q} \mid \mathfrak{p}) \arrow[r, two heads] \arrow[d, two heads] & D(\mathfrak{r} \mid \mathfrak{p}) \arrow[r] \arrow[d, two heads] & 1 \\
1 \arrow[r] & \operatorname{Gal}(k_\mathfrak{q}/k_\mathfrak{r}) \arrow[r, hook] & \operatorname{Gal}(k_\mathfrak{q}/k_\mathfrak{p}) \arrow[r, two heads] & \operatorname{Gal}(k_\mathfrak{r}/k_\mathfrak{p}) \arrow[r] & 1.
\end{tikzcd}$$

In particular,

$$D(\mathfrak{q} \mid \mathfrak{r}) = D(\mathfrak{q} \mid \mathfrak{p}) \cap H, \qquad I(\mathfrak{q} \mid \mathfrak{r}) = I(\mathfrak{q} \mid \mathfrak{p}) \cap H.$$

Definition 4.1.21 (Relative Frobenius)**.** Assume that $\mathfrak{p}$ is unramified in the Galois extension L/K, and let $q_{\mathfrak{p}} := \#(k_{\mathfrak{p}}) = \mathrm{N}(\mathfrak{p})$. The Frobenius element is the unique element

$$\mathrm{Frob}(\mathfrak{q} \mid \mathfrak{p}) \in D(\mathfrak{q} \mid \mathfrak{p}), \qquad \mathrm{Frob}(\mathfrak{q} \mid \mathfrak{p})(x) \equiv x^{q_{\mathfrak{p}}} \pmod{\mathfrak{q}} \quad (x \in \mathscr{O}_L). \tag{4.1.4}$$

Theorem 4.1.22 (Relative Frobenius laws, [2])**.** *Let L/K be Galois and let $\mathfrak{p}$ be unramified in L. For $\mathfrak{q} \mid \mathfrak{p}$,*

$$D(\mathfrak{q} \mid \mathfrak{p}) = \langle \mathrm{Frob}(\mathfrak{q} \mid \mathfrak{p}) \rangle, \qquad \mathrm{ord}(\mathrm{Frob}(\mathfrak{q} \mid \mathfrak{p})) = f(\mathfrak{q} \mid \mathfrak{p}),$$

$$\mathrm{Frob}(\sigma(\mathfrak{q}) \mid \mathfrak{p}) = \sigma \mathrm{Frob}(\mathfrak{q} \mid \mathfrak{p}) \sigma^{-1}, \qquad \mathrm{Frob}(\mathfrak{q} \mid \mathfrak{p}) = 1 \iff \mathfrak{p} \textit{ splits completely in } L.$$

If $K \subset M \subset L$, $\mathfrak{r} = \mathfrak{q} \cap \mathscr{O}_M$, and M/K is Galois, then

$$\begin{tikzcd} D(\mathfrak{q} \mid \mathfrak{p}) \arrow[r, "\sim"] \arrow[d, two heads, "\mathrm{res}"'] & \mathrm{Gal}(k_{\mathfrak{q}}/k_{\mathfrak{p}}) \arrow[d, two heads, "\mathrm{res}"] \\ D(\mathfrak{r} \mid \mathfrak{p}) \arrow[r, "\sim"] & \mathrm{Gal}(k_{\mathfrak{r}}/k_{\mathfrak{p}}) \end{tikzcd} \qquad \mathrm{Frob}(\mathfrak{q} \mid \mathfrak{p})|_M = \mathrm{Frob}(\mathfrak{r} \mid \mathfrak{p}),$$

$$\mathrm{Frob}(\mathfrak{q} \mid \mathfrak{r}) = \mathrm{Frob}(\mathfrak{q} \mid \mathfrak{p})^{f(\mathfrak{r} \mid \mathfrak{p})}.$$

5. Local Fields and Completions

Absolute values complete the localizations of § 4, and the resulting local fields detect global splitting, ramification, norms, and traces.

5.1. **Completions, Hensel Lifting, and Ramification.**

Definition 5.1.1 (Absolute values and places)**.** Let F be a field. An absolute value is a map $|\cdot| \colon F \to \mathbb{R}_{\geq 0}$ satisfying

$$|x| = 0 \iff x = 0, \qquad |xy| = |x||y|, \qquad |x+y| \leq |x| + |y|.$$

It is nonarchimedean when

$$|x+y| \leq \max(|x|, |y|), \qquad |x| \neq |y| \implies |x+y| = \max(|x|, |y|).$$

Two absolute values are equivalent when

$$|x|_1 = |x|_2^c \quad (x \in F) \qquad \text{for some } c > 0;$$

an equivalence class is a place of F. A normalized discrete valuation $v \colon F^\times \twoheadrightarrow \mathbb{Z}$, extended by $v(0) = \infty$, gives

$$|x|_v := c^{v(x)}, \qquad 0 < c < 1, \qquad v(x+y) \geq \min(v(x), v(y)).$$

Its valuation ring, maximal ideal, residue field, and unit filtration are

$$\mathscr{O}_v := \{x : v(x) \geq 0\}, \qquad \mathfrak{m}_v := \{x : v(x) > 0\}, \qquad k_v := \mathscr{O}_v/\mathfrak{m}_v, \qquad U_v^{(i)} := 1 + \mathfrak{m}_v^i.$$

Theorem 5.1.2 (Ostrowski and the product formula, [2])**.** *The nontrivial places of $\mathbb{Q}$ are represented by*

$$|x|_\infty := |x|, \qquad |x|_p := p^{-v_p(x)}, \qquad \prod_{p \leq \infty} |x|_p = 1 \quad (x \in \mathbb{Q}^\times).$$

For a number field K, the places are

$$\{\mathfrak{p} \subset \mathscr{O}_K : 0 \neq \mathfrak{p} \textit{ prime}\} \sqcup \{\sigma \colon K \hookrightarrow \mathbb{R}\} \sqcup \{\tau, \overline{\tau} \colon K \hookrightarrow \mathbb{C}\}/\textit{conjugation}.$$

The normalized place norms are

$$\|x\|_{\mathfrak{p}} := \mathrm{N}(\mathfrak{p})^{-v_{\mathfrak{p}}(x)}, \qquad \|x\|_\sigma := |\sigma(x)|, \qquad \|x\|_\tau := |\tau(x)|^2,$$

and satisfy

$$\prod_v \|x\|_v = 1 \quad (x \in K^\times), \qquad \prod_{\mathfrak{p} \nmid \infty} \|x\|_{\mathfrak{p}} = |\mathrm{N}_{K/\mathbb{Q}}(x)|^{-1}, \qquad \prod_{v \mid \infty} \|x\|_v = |\mathrm{N}_{K/\mathbb{Q}}(x)|.$$

Definition 5.1.3 (Completion)**.** Let $(F,|\cdot|)$ be valued. Its completion $\widehat{F}$ is the field of Cauchy sequences modulo null sequences, with

$$F \hookrightarrow \widehat{F}, \qquad x \longmapsto (x,x,\ldots), \qquad |(x_n)| := \lim_{n\to\infty}(|x_n|).$$

For every isometric embedding $j\colon F \hookrightarrow E$ into a complete valued field, there is a unique isometric extension $\widehat{j}$,

$$\begin{tikzcd} F \arrow[r, hook, "\iota"] \arrow[dr, hook, "j"'] & \widehat{F} \arrow[d, hook, dashed, "\exists!\,\widehat{j}"] \\ & E \end{tikzcd} \qquad j = \widehat{j}\circ\iota.$$

If A is a DVR with uniformizer π, its π-adic completion is

$$\widehat{A} := \varprojlim_m (A/\pi^m A) = \{(a_m)_{m\geq 1} : a_{m+1} \equiv a_m \pmod{\pi^m}\}.$$

For a number field K and $0 \neq \mathfrak{p} \subset \mathscr{O}_K$, put

$$\widehat{\mathscr{O}_{K,\mathfrak{p}}} := \varprojlim_m (\mathscr{O}_K/\mathfrak{p}^m), \qquad K_{\mathfrak{p}} := \mathrm{Frac}(\widehat{\mathscr{O}_{K,\mathfrak{p}}}).$$

Proposition 5.1.4 (Completed discrete valuation rings)**.** *Let A be a DVR with uniformizer π, fraction field F, and residue field k. Then*

$$\begin{tikzcd} A \arrow[r, hook, "\iota"] \arrow[d, two heads] & \widehat{A} \arrow[d, two heads] \\ A/\pi^n A \arrow[r, "\sim"'] & \widehat{A}/\pi^n\widehat{A} \end{tikzcd} \qquad \begin{tikzcd} \widehat{A} \arrow[r, "\sim"] \arrow[dr, "\rho_n"'] & \varprojlim_m (A/\pi^m A) \arrow[d, "\mathrm{pr}_n"] \\ & A/\pi^n A. \end{tikzcd}$$

Moreover,

$$\widehat{A} \textit{ is a complete DVR}, \qquad \mathfrak{m}_{\widehat{A}} = \pi\widehat{A}, \qquad \widehat{A}/\pi\widehat{A} \cong k, \qquad \widehat{F} := \mathrm{Frac}(\widehat{A}) \textit{ is complete}.$$

Proof. Use the compatible residue maps and separatedness.

$$\iota(a) := (a+\pi^m A)_{m\geq 1}, \qquad \ker(\iota) = \bigcap_{m\geq 1} \pi^m A = 0.$$

$$(a_j)_{j\geq 1} \in \widehat{A} \implies (a_j)_{j\geq 1} + \pi^m\widehat{A} = \iota(a_m) + \pi^m\widehat{A} \implies A/\pi^m A \xrightarrow{\sim} \widehat{A}/\pi^m\widehat{A}.$$
$$0 \neq a \in \widehat{A}, \qquad r := \sup(\{m \geq 0 : a \in \pi^m\widehat{A}\}) < \infty \implies a = \pi^r u, \quad u \notin \pi\widehat{A}.$$
$$u \notin \pi\widehat{A} \iff u + \pi\widehat{A} \neq 0 \iff u \in \widehat{A}^\times \implies \widehat{A} \text{ is a DVR}, \qquad \widehat{F} = \bigcup_{r\geq 0} \pi^{-r}\widehat{A}.$$

This proves the completed DVR structure. $\square$

Definition 5.1.5 (Local fields)**.** A local field is a complete discretely valued field F with finite residue field k_F. Writing $q := \#(k_F)$, one has

$$\mathscr{O}_F := \{x : v_F(x) \geq 0\}, \qquad \mathfrak{m}_F := \{x : v_F(x) > 0\} = \pi_F\mathscr{O}_F, \qquad \|x\|_F := q^{-v_F(x)}.$$

Every nondiscrete locally compact field belongs to exactly one of the families

$$\mathbb{R}, \qquad \mathbb{C}, \qquad F/\mathbb{Q}_p \text{ finite}, \qquad \mathbb{F}_q((T)).$$

For nonarchimedean F,

$$\mathrm{char}(F) = 0 \iff F/\mathbb{Q}_p \text{ is finite}, \qquad \mathrm{char}(F) = p \iff F \cong \mathbb{F}_q((T)).$$

Example 5.1.6 (The fields $\mathbb{Q}_p$ and $\mathbb{F}_q((T))$)**.** The standard expansions are

$$\mathbb{Z}_p = \left\{ \sum_{i=0}^{\infty} a_i p^i : 0 \le a_i < p \right\}, \qquad \mathbb{Q}_p = \left\{ \sum_{i=N}^{\infty} a_i p^i : N \in \mathbb{Z},\ 0 \le a_i < p \right\},$$
$$\mathbb{F}_q[[T]] = \left\{ \sum_{i=0}^{\infty} a_i T^i : a_i \in \mathbb{F}_q \right\}, \qquad \mathbb{F}_q((T)) = \left\{ \sum_{i=N}^{\infty} a_i T^i : N \in \mathbb{Z},\ a_i \in \mathbb{F}_q \right\}.$$

Thus

$$\mathbb{Z}_p/p^m\mathbb{Z}_p \cong \mathbb{Z}/p^m\mathbb{Z}, \qquad \mathbb{F}_q[[T]]/T^m\mathbb{F}_q[[T]] \cong \mathbb{F}_q[T]/(T^m).$$

Example 5.1.7 (A split completion of $\mathbb{Q}(i)$)**.** Let $K = \mathbb{Q}(i)$ and

$$\mathfrak{p}_5 := (5, 2+i), \qquad 5\mathbb{Z}[i] = \mathfrak{p}_5\overline{\mathfrak{p}}_5, \qquad e(\mathfrak{p}_5 \mid 5) = f(\mathfrak{p}_5 \mid 5) = 1.$$

Since $X^2+1 \equiv (X-2)(X+2) \pmod 5$,

$$K_{\mathfrak{p}_5} \cong \mathbb{Q}_5, \qquad K_{\overline{\mathfrak{p}}_5} \cong \mathbb{Q}_5, \qquad K \otimes_{\mathbb{Q}} \mathbb{Q}_5 \cong \mathbb{Q}_5 \times \mathbb{Q}_5.$$

Theorem 5.1.8 (Global–local decomposition, [2])**.** *Let L/K be finite, let $\mathfrak{p} \subset \mathscr{O}_K$ be nonzero, and write*

$$\mathfrak{p}\mathscr{O}_L = \prod_{i=1}^{g} \mathfrak{q}_i^{e_i}, \qquad f_i := f(\mathfrak{q}_i \mid \mathfrak{p}).$$

Then

$$\begin{tikzcd}
K \arrow[r, hook] \arrow[d, hook] & L \arrow[d, "x\mapsto x\otimes 1"] \arrow[dr, bend left, "x\mapsto (x)_{\mathfrak{q}}"] & \\
K_{\mathfrak{p}} \arrow[r, "a\mapsto 1\otimes a"'] \arrow[drr, bend right, "a\mapsto (a)_{\mathfrak{q}}"'] & L\otimes_K K_{\mathfrak{p}} \arrow[dr, "\sim"] & \\
& & \prod_{\mathfrak{q}\mid\mathfrak{p}} L_{\mathfrak{q}},
\end{tikzcd} \qquad x\otimes a \longmapsto (xa)_{\mathfrak{q}}, \qquad [L_{\mathfrak{q}_i} : K_{\mathfrak{p}}] = e_i f_i. \tag{5.1.1}$$

Trace and norm commute with completion,

$$\begin{tikzcd}
L\otimes_K K_{\mathfrak{p}} \arrow[r, "\sim"] \arrow[dr, "y\mapsto \mathrm{Tr}(y)"'] & \prod_{\mathfrak{q}\mid\mathfrak{p}} L_{\mathfrak{q}} \arrow[d, "(x_{\mathfrak{q}})_{\mathfrak{q}}\mapsto \sum_{\mathfrak{q}\mid\mathfrak{p}} \mathrm{Tr}_{L_{\mathfrak{q}}/K_{\mathfrak{p}}}(x_{\mathfrak{q}})"] \\
& K_{\mathfrak{p}}
\end{tikzcd}$$

$$\begin{tikzcd}
L\otimes_K K_{\mathfrak{p}} \arrow[r, "\sim"] \arrow[dr, "y\mapsto \mathrm{N}(y)"'] & \prod_{\mathfrak{q}\mid\mathfrak{p}} L_{\mathfrak{q}} \arrow[d, "(x_{\mathfrak{q}})_{\mathfrak{q}}\mapsto \prod_{\mathfrak{q}\mid\mathfrak{p}} \mathrm{N}_{L_{\mathfrak{q}}/K_{\mathfrak{p}}}(x_{\mathfrak{q}})"] \\
& K_{\mathfrak{p}}.
\end{tikzcd}$$

$$v_{\mathfrak{p}}(\mathrm{N}_{L/K}(x)) = \sum_{\mathfrak{q}\mid\mathfrak{p}} f(\mathfrak{q}\mid\mathfrak{p})\, v_{\mathfrak{q}}(x) \qquad (x \in L^{\times}).$$

If L/K is Galois, restriction extends continuously and gives

$$\begin{tikzcd} D(\mathfrak{q}_i \mid \mathfrak{p}) \arrow[r, "\sim"] \arrow[d, two heads] & \mathrm{Gal}(L_{\mathfrak{q}_i}/K_{\mathfrak{p}}) \arrow[d, two heads] \\ \mathrm{Gal}(k_{\mathfrak{q}_i}/k_{\mathfrak{p}}) \arrow[r, equal] & \mathrm{Gal}(k_{\mathfrak{q}_i}/k_{\mathfrak{p}}) \end{tikzcd}$$

$$\ker(D(\mathfrak{q}_i \mid \mathfrak{p}) \to \mathrm{Gal}(k_{\mathfrak{q}_i}/k_{\mathfrak{p}})) = I(\mathfrak{q}_i \mid \mathfrak{p}), \qquad \ker(\mathrm{Gal}(L_{\mathfrak{q}_i}/K_{\mathfrak{p}}) \to \mathrm{Gal}(k_{\mathfrak{q}_i}/k_{\mathfrak{p}})) = I(L_{\mathfrak{q}_i}/K_{\mathfrak{p}}).$$

Proof. Apply polynomial Chinese remaindering after writing $L = K(\alpha)$ and $f_\alpha(X) = \prod_{i=1}^g F_i(X)$ over $K_{\mathfrak{p}}$, with $(F_i, F_j) = 1$ for $i \neq j$.

$$L \otimes_K K_{\mathfrak{p}} \cong K_{\mathfrak{p}}[X]/(f_\alpha) \cong \prod_{i=1}^{g} K_{\mathfrak{p}}[X]/(F_i) \cong \prod_{i=1}^{g} L_{\mathfrak{q}_i}.$$

$$\deg(F_i) = [L_{\mathfrak{q}_i} : K_{\mathfrak{p}}] = e_i f_i, \qquad \sum_i e_i f_i = [L : K].$$

$$y \longleftrightarrow (y_{\mathfrak{q}})_{\mathfrak{q}|\mathfrak{p}}, \qquad m_y = \bigoplus_{\mathfrak{q}|\mathfrak{p}} m_{y_{\mathfrak{q}}} \implies \mathrm{Tr}(m_y) = \sum_{\mathfrak{q}|\mathfrak{p}} \mathrm{Tr}(m_{y_{\mathfrak{q}}}), \qquad \det(m_y) = \prod_{\mathfrak{q}|\mathfrak{p}} \det(m_{y_{\mathfrak{q}}}).$$

$$\sigma(\mathfrak{q}_i) = \mathfrak{q}_i \iff v_{\mathfrak{q}_i} \circ \sigma = v_{\mathfrak{q}_i} \iff v_{\mathfrak{q}_i}(\sigma(x)) = v_{\mathfrak{q}_i}(x) \quad (\forall x \in L^\times) \implies D(\mathfrak{q}_i \mid \mathfrak{p}) \hookrightarrow \mathrm{Aut}_{K_{\mathfrak{p}}}(L_{\mathfrak{q}_i}).$$

$$|D(\mathfrak{q}_i \mid \mathfrak{p})| = e_i f_i = [L_{\mathfrak{q}_i} : K_{\mathfrak{p}}] \implies L_{\mathfrak{q}_i}/K_{\mathfrak{p}} \text{ is Galois}, \quad D(\mathfrak{q}_i \mid \mathfrak{p}) \cong \mathrm{Gal}(L_{\mathfrak{q}_i}/K_{\mathfrak{p}}).$$

This proves the decomposition and its Galois compatibility. ☐

Theorem 5.1.9 (Unique extension of absolute values, [2])**.** *Let L/K be a finite extension of local fields of degree n, and equip K with $|a|_K := q_K^{-v_K(a)}$. Its unique extension to L is*

$$|x|_{L/K} := |\mathrm{N}_{L/K}(x)|_K^{1/n}, \qquad x \in L.$$

Equivalently,

$$\begin{tikzcd} L^\times \arrow[r, "\mathrm{N}_{L/K}"] \arrow[d, "|\cdot|_{L/K}"'] & K^\times \arrow[d, "|\cdot|_K"] \\ \mathbb{R}_{>0} \arrow[r, "t \mapsto t^n"'] & \mathbb{R}_{>0}. \end{tikzcd}$$

For the normalized valuations, ramification index e, residue degree f, and $q_K = \#(k_K)$, one has

$$v_L|_K = e\, v_K, \qquad v_K(\mathrm{N}_{L/K}(x)) = f\, v_L(x), \qquad |x|_{L/K} = q_K^{-v_L(x)/e}.$$

The intrinsic residue-normalized modulus on L is

$$\|x\|_L := q_L^{-v_L(x)} = |x|_{L/K}^n, \qquad q_L = q_K^f, \qquad \|a\|_L = \|a\|_K^n \quad (a \in K).$$

Moreover,

$$L \text{ is complete}, \qquad [L : K] = ef, \qquad \pi_K = u\pi_L^e \quad \text{for some } u \in \mathscr{O}_L^\times.$$

Theorem 5.1.10 (Hensel root lifting, [2])**.** *Let F be complete discretely valued, let $f(X) \in \mathscr{O}_F[X]$, and let $a_0 \in \mathscr{O}_F$. If*

$$v_F(f(a_0)) > 2v_F(f'(a_0)), \tag{5.1.2}$$

then there is a unique $a \in \mathscr{O}_F$ satisfying

$$f(a) = 0, \qquad v_F(a - a_0) > v_F(f'(a_0)).$$

In particular, reduction gives a bijection on simple-zero loci,

$$Z_f^{\mathrm{sm}}(\mathscr{O}_F) := \{a \in \mathscr{O}_F : f(a) = 0,\ f'(a) \in \mathscr{O}_F^\times\}, \qquad Z_{\overline{f}}^{\mathrm{sm}}(k_F) := \{\overline{a} \in k_F : \overline{f}(\overline{a}) = 0,\ \overline{f}'(\overline{a}) \neq 0\},$$

$$\begin{tikzcd} Z_f^{\mathrm{sm}}(\mathscr{O}_F) \arrow[r, hook] \arrow[d, shift right, "\mathrm{red}"'] & \mathscr{O}_F \arrow[d, two heads] \\ Z_{\overline{f}}^{\mathrm{sm}}(k_F) \arrow[r, hook] \arrow[u, dashed, shift right, "\exists!\,\mathrm{lift}"'] & k_F \end{tikzcd} \qquad Z_f^{\mathrm{sm}}(\mathscr{O}_F) \xrightarrow{\sim} Z_{\overline{f}}^{\mathrm{sm}}(k_F).$$

Proof. Apply Newton iteration with $s = v_F(f'(a_0))$.

$$a_{m+1} := a_m - \frac{f(a_m)}{f'(a_m)}, \qquad t_m := v_F(f(a_m)), \qquad t_0 > 2s.$$

$$v_F(f'(a_m)) = s, \qquad v_F(a_{m+1} - a_m) = t_m - s > s, \qquad t_{m+1} \geq 2(t_m - s).$$

$$t_m - s \geq 2^m(t_0 - 2s) + s \longrightarrow \infty \implies (a_m)_{m\geq 0} \text{ is Cauchy} \implies a_m \longrightarrow a \in \mathscr{O}_F, \quad f(a) = 0.$$

$$f(a) = f(b) = 0, \quad a \neq b, \quad v_F(a - a_0), v_F(b - a_0) > s \implies \frac{f(a) - f(b)}{a - b} \equiv f'(a_0) \not\equiv 0 \pmod{\mathfrak{m}_F^{s+1}}.$$

$$0 = f(a) - f(b) = (a - b) \cdot \frac{f(a) - f(b)}{a - b}, \qquad a - b \neq 0, \qquad \frac{f(a) - f(b)}{a - b} \neq 0 \implies \perp.$$

This proves existence and uniqueness. □

Theorem 5.1.11 (Hensel factorization, [2]). *Let F be a local field, let $f(X) \in \mathscr{O}_F[X]$ be monic, and suppose*

$$\overline{f}(X) = \overline{g}(X)\overline{h}(X), \qquad (\overline{g}, \overline{h}) = 1, \qquad \overline{g}, \overline{h} \in k_F[X] \text{ monic}.$$

There are unique monic lifts $g, h \in \mathscr{O}_F[X]$ of the same degrees satisfying

$$f(X) = g(X)h(X).$$

The lift is encoded by

$$\begin{tikzcd} (g,h) \arrow[r, mapsto, "(g,h)\mapsto gh"] \arrow[d, shift right, "\mathrm{red}"'] & f \arrow[d, "\mathrm{red}"] \\ (\overline{g}, \overline{h}) \arrow[r, mapsto, "(\overline{g},\overline{h})\mapsto \overline{g}\overline{h}"'] \arrow[u, shift right, "\exists!"'] & \overline{f}. \end{tikzcd}$$

Example 5.1.12 (Roots of $X^2 + 1$ in $\mathbb{Z}_5$). The factorization

$$X^2 + 1 \equiv (X - 2)(X + 2) \pmod 5, \qquad 2 \cdot 2 \not\equiv 0 \pmod 5, \qquad 2 \cdot (-2) \not\equiv 0 \pmod 5$$

lifts uniquely to

$$X^2 + 1 = (X - \alpha_+)(X - \alpha_-) \in \mathbb{Z}_5[X], \qquad \alpha_+ \equiv 2 \pmod 5, \qquad \alpha_- \equiv -2 \pmod 5.$$

For the positive lift,

$$2 \pmod 5 \longleftarrow 7 \pmod{25} \longleftarrow 57 \pmod{125} \longleftarrow 182 \pmod{625}, \qquad \alpha_- = -\alpha_+.$$

Corollary 5.1.13 (Teichmüller representatives)**.** *Let F have residue field $k_F \cong \mathbb{F}_q$. Every $\overline{a} \in k_F^\times$ has a unique lift $[\overline{a}] \in \mathscr{O}_F^\times$ satisfying*

$$[\overline{a}]^{q-1} = 1, \qquad [\overline{a}] \equiv \overline{a} \pmod{\mathfrak{m}_F}, \qquad [\overline{a}\overline{b}] = [\overline{a}][\overline{b}].$$

Thus

$$\begin{tikzcd} 1 \arrow[r] & U_F^{(1)} \arrow[r, hook] & \mathscr{O}_F^\times \arrow[r, two heads] & k_F^\times \arrow[r] \arrow[l, bend left, "\overline{a}\mapsto[\overline{a}]"] & 1 \end{tikzcd}$$

and

$$F^\times \cong \pi_F^{\mathbb{Z}} \times \mu_{q-1} \times U_F^{(1)}, \qquad U_F^{(i)}/U_F^{(i+1)} \cong (k_F, +) \quad (i \geq 1).$$

If $F/\mathbb{Q}_p$ is finite, then

$$U_F^{(1)} \cong \mu_{p^\infty}(F) \times \mathbb{Z}_p^{[F:\mathbb{Q}_p]} \quad \textit{as topological groups}, \qquad U_F^{(m)} \cong \mathbb{Z}_p^{[F:\mathbb{Q}_p]} \quad \textit{for all sufficiently large } m.$$

Definition 5.1.14 (Newton polygon)**.** Let F be complete discretely valued and

$$f(X) = \sum_{i=0}^{n} a_i X^i \in F[X], \qquad a_0 a_n \neq 0.$$

The Newton polygon $\mathrm{NP}(f)$ is the lower convex hull of

$$\{(i, v_F(a_i)) : a_i \neq 0\} \subset \mathbb{R}^2.$$

The Newton slope of a side is the negative of its geometric slope, and its multiplicity is its horizontal length.

Theorem 5.1.15 (Newton polygon theorem, [2])**.** *Let the Newton slopes of $f \in F[X]$ be $s_1 > \cdots > s_r$ with multiplicities $m_1, \ldots, m_r$, and write v for the extension of v_F to a splitting field E. If $Z_s(f)$ denotes the multiset of roots α satisfying $v(\alpha) = s$, then*

$$\#(Z_{s_j}(f)) = m_j.$$

Moreover,

$$\begin{tikzcd} \{(s_j, m_j)\}_{j=1}^r \arrow[r, mapsto, "(s_j,m_j)\mapsto f_j"] \arrow[dr, bend right, "(s_j,m_j)\mapsto Z_{s_j}(f)"'] & \{f_j\}_{j=1}^r \arrow[d, mapsto, "f_j\mapsto Z(f_j)"] \\ & \{Z_{s_j}(f)\}_{j=1}^r \end{tikzcd} \qquad f = a_n \cdot \prod_{j=1}^{r} f_j, \quad \deg(f_j) = m_j, \quad Z(f_j) = Z_{s_j}(f),$$

where each $f_j(X) \in F[X]$ is monic and $\mathrm{NP}(f_j)$ has the single slope s_j.

Example 5.1.16 (A polygon over $\mathbb{Q}_2$)**.** For

$$f(X) = \frac{X^6}{2} - X^5 + \frac{X^4}{4} - \frac{X^3}{111} + \frac{X^2}{10} + X - 3,$$

the coefficient points are

$$(0,0), (1,0), (2,-1), (3,0), (4,-2), (5,0), (6,-1).$$

The vertices are $(0,0), (4,-2), (6,-1)$, hence

$$\mathrm{NP}(f): \quad s_1 = \frac{1}{2},\ m_1 = 4, \qquad s_2 = -\frac{1}{2},\ m_2 = 2, \qquad f = \frac{1}{2} \cdot f_1 f_2, \quad \deg(f_1) = 4, \quad \deg(f_2) = 2.$$

Theorem 5.1.17 (Eisenstein and total ramification, [2])**.** *Let F be a local field and*

$$E(X)=X^n+a_{n-1}X^{n-1}+\cdots+a_0\in\mathscr{O}_F[X],\qquad v_F(a_i)\geq 1,\qquad v_F(a_0)=1.$$

If π_L is a root and $L=F(\pi_L)$, then

$$E\text{ is irreducible},\qquad v_L(\pi_L)=1,\qquad \mathscr{O}_L=\mathscr{O}_F[\pi_L],\qquad e(L/F)=n,\qquad f(L/F)=1.$$

Conversely, if L/F is totally ramified of degree n, every uniformizer π_L generates L/F, its minimal polynomial $P(X)=X^n+a_{n-1}X^{n-1}+\cdots+a_0$ is Eisenstein, and

$$\mathscr{O}_L=\mathscr{O}_F[\pi_L].$$

Proof. Use the Newton polygon and the value-group index.

$$\mathrm{NP}(E)=[(0,1),(n,0)],\qquad v|_F=v_F\overset{5.1.15}{\Longrightarrow}v(\alpha)=\frac{1}{n}\quad\text{for every root }\alpha.$$

$$v(L^\times)=\frac{1}{e(L/F)}\mathbb{Z}\ni\frac{1}{n}\implies n\mid e(L/F)\leq[L:F]\leq n\implies e(L/F)=[L:F]=n,\quad f(L/F)=1.$$

$$(b_i)_{i=0}^{n-1}\in F^n,\qquad v\left(\sum_{i=0}^{n-1}b_i\pi_L^i\right)=\min_{0\leq i<n}\left(v_F(b_i)+\frac{i}{n}\right)\geq 0\iff(b_i)_{i=0}^{n-1}\in\mathscr{O}_F^n\implies\mathscr{O}_L=\mathscr{O}_F[\pi_L].$$

$$M:=F(\pi_L),\qquad n\mid e(M/F)\leq[M:F]\leq[L:F]=n\implies M=L,\quad \mathscr{O}_L=\mathscr{O}_F[\pi_L].$$

$$v(\alpha)=\frac{1}{n}\quad(\forall\,\alpha:P(\alpha)=0)\implies v_F(a_i)\geq 1\quad(i<n),\quad v_F(a_0)=v_F(\mathrm{N}_{L/F}(\pi_L))=1\implies P\text{ Eisenstein.}$$

This proves both directions. □

Theorem 5.1.18 (Unramified extensions, [2])**.** *Let F be a local field with residue field $k_F\cong\mathbb{F}_q$, and let L/F be finite and unramified.*

(1) *There is $\alpha\in\mathscr{O}_L$ such that*

$$\mathscr{O}_L=\mathscr{O}_F[\alpha],\qquad k_L=k_F(\overline{\alpha}),\qquad [L:F]=[k_L:k_F].$$

Proof. Lift a primitive residue-field element by Hensel's lemma.

$$k_L=k_F(\overline{\alpha}),\qquad \overline{g}(X):=f_{\overline{\alpha},k_F}(X),\qquad \overline{g}'(\overline{\alpha})\neq 0.$$

$$g\in\mathscr{O}_F[X]\text{ monic},\quad g\equiv\overline{g}\pmod{\mathfrak{m}_F}\overset{5.1.10}{\Longrightarrow}\exists\,\alpha\in\mathscr{O}_L:g(\alpha)=0,\quad\alpha\equiv\overline{\alpha}\pmod{\mathfrak{m}_L}.$$

$$\mathscr{O}_L=\mathscr{O}_F[\alpha]+\mathfrak{m}_F\mathscr{O}_L\implies\mathscr{O}_L=\mathscr{O}_F[\alpha],\qquad k_L=k_F(\overline{\alpha}).$$

$$[L:F]=e(L/F)f(L/F)=f(L/F)=[k_L:k_F].$$

This proves the monogenic description. □

(2) *For every finite extension E/F, reduction induces*

$$\begin{tikzcd} \mathscr{O}_L \arrow[r, "\varphi|_{\mathscr{O}_L}"] \arrow[d, two heads] & \mathscr{O}_E \arrow[d, two heads] \\ k_L \arrow[r, "\overline{\varphi}"'] & k_E \end{tikzcd}\qquad \mathrm{Hom}_F(L,E)\xrightarrow{\sim}\mathrm{Hom}_{k_F}(k_L,k_E),\quad\varphi\longmapsto\overline{\varphi}.$$

Proof. Use the generator from Item 1.

$$\varphi\in\mathrm{Hom}_F(L,E)\implies\varphi(\mathscr{O}_L)\subseteq\mathscr{O}_E\implies\overline{\varphi}\in\mathrm{Hom}_{k_F}(k_L,k_E).$$

$$L=F(\alpha),\qquad\varphi\longmapsto\varphi(\alpha),\qquad\overline{\varphi}\longmapsto\overline{\varphi}(\overline{\alpha}).$$

$$g(\alpha)=0,\quad\overline{g}'(\overline{\alpha})\neq 0\overset{5.1.10}{\Longrightarrow}\{\text{roots of }g\text{ in }E\}\xrightarrow{\sim}\{\text{roots of }\overline{g}\text{ in }k_E\}.$$

This proves the Hom-set bijection. □

(3) *For every $m \geq 1$, there is a unique unramified extension F_m/F of degree m, and*
$$k_{F_m} \cong \mathbb{F}_{q^m}, \qquad \mathrm{Gal}(F_m/F) \cong \mathrm{Gal}(\mathbb{F}_{q^m}/\mathbb{F}_q) = \langle \mathrm{Fr}_q \rangle, \qquad \mathrm{Fr}_q(x) = x^q.$$

Proof. Lift an irreducible polynomial over the residue field.
$$\overline{g} \in \mathbb{F}_q[X] \text{ irreducible}, \quad \deg(\overline{g}) = m, \quad g \in \mathscr{O}_F[X] \text{ monic lift}, \qquad g \equiv \overline{g} \pmod{\mathfrak{m}_F}.$$
$$A := \mathscr{O}_F[X]/(g), \qquad F_m := \mathrm{Frac}(A) = F[X]/(g).$$
$$A/\pi_F A \cong \mathbb{F}_q[X]/(\overline{g}) \cong \mathbb{F}_{q^m} \implies \mathfrak{m}_A = \pi_F A, \quad A \text{ is a DVR}, \quad A = \mathscr{O}_{F_m}, \quad e(F_m/F) = 1, \quad f(F_m/F) = m.$$
$$E/F \text{ unramified of degree } m \overset{2}{\implies} \mathrm{Hom}_F(F_m, E) \cong \mathrm{Hom}_{\mathbb{F}_q}(\mathbb{F}_{q^m}, k_E) \neq \varnothing \implies E \cong F_m.$$
$$\mathrm{Aut}_F(F_m) \cong \mathrm{Aut}_{\mathbb{F}_q}(\mathbb{F}_{q^m}) = \langle x \mapsto x^q \rangle \implies F_m/F \text{ is cyclic Galois of degree } m.$$
This proves existence, uniqueness, and the Frobenius description. □

(4) *Every finite extension E/F has a unique maximal unramified subextension $E/E_0/F$, and*
$$\begin{tikzcd} \mathscr{O}_F \arrow[r, hook, "\text{unramified}"] \arrow[d, two heads] & \mathscr{O}_{E_0} \arrow[r, hook, "\text{totally ramified}"] \arrow[d, two heads] & \mathscr{O}_E \arrow[d, two heads] \\ k_F \arrow[r, hook, "{[k_E:k_F]=f(E/F)}"'] & k_{E_0} \arrow[r, "\sim"'] & k_E \end{tikzcd} \qquad [E_0 : F] = f(E/F), \quad [E : E_0] = e(E/F).$$

Proof. Lift the full residue extension inside E.
$$k_E/k_F \text{ has degree } f(E/F) \overset{3}{\implies} \exists!\, E_0/F \text{ unramified} : k_{E_0} = k_E.$$
$$\mathrm{id}_{k_E} \in \mathrm{Hom}_{k_F}(k_{E_0}, k_E) \overset{2}{\implies} E_0 \hookrightarrow E.$$
$$F \subseteq M \subseteq E, \quad M/F \text{ unramified} \implies k_M \subseteq k_E = k_{E_0} \overset{2}{\implies} M \subseteq E_0.$$
$$[E_0 : F] = f(E/F), \qquad [E : E_0] = \frac{e(E/F)f(E/F)}{f(E/F)} = e(E/F) \implies E/E_0 \text{ is totally ramified.}$$
This proves maximality and the degree formulas. □

Corollary 5.1.19 (Structure over $\mathbb{Q}_p$)**.** *Let $F/\mathbb{Q}_p$ be finite, let $k_F \cong \mathbb{F}_{p^f}$, and put $e = e(F/\mathbb{Q}_p)$. Its maximal unramified subfield F_0 and every uniformizer π_F satisfy*
$$\begin{tikzcd} F \arrow[d, dash, "e"'] & \mathscr{O}_F = \mathscr{O}_{F_0}[\pi_F] \\ F_0 \arrow[d, dash, "f"'] & \mathscr{O}_{F_0} \arrow[u, hook] \\ \mathbb{Q}_p & \mathbb{Z}_p, \arrow[u, hook] \end{tikzcd} \qquad [F : \mathbb{Q}_p] = ef.$$

Moreover,
$$F_0/\mathbb{Q}_p \textit{ is cyclic unramified of degree } f, \qquad F/F_0 \textit{ is totally ramified of degree } e,$$
$$f_{\pi_F, F_0}(X) \in \mathscr{O}_{F_0}[X] \textit{ is Eisenstein}, \qquad \mathscr{O}_F = \mathscr{O}_{F_0}[\pi_F].$$

Example 5.1.20 (Cyclotomic unramified extensions)**.** For every $m \geq 1$, a primitive (p^m-1)-st root of unity gives

$$\mathbb{Q}_p^{(m)} := \mathbb{Q}_p(\zeta_{p^m-1}), \qquad [\mathbb{Q}_p^{(m)} : \mathbb{Q}_p] = m, \qquad k_{\mathbb{Q}_p^{(m)}} \cong \mathbb{F}_{p^m}.$$

Thus

$$\mathrm{Gal}(\mathbb{Q}_p^{(m)}/\mathbb{Q}_p) \cong \mathrm{Gal}(\mathbb{F}_{p^m}/\mathbb{F}_p) = \langle \mathrm{Fr}_p \rangle, \qquad \mathrm{Fr}_p(\bar{x}) = \bar{x}^p.$$

Definition 5.1.21 (Higher ramification groups)**.** Let L/F be finite Galois with group G, residue characteristic p, and uniformizer π_L. The lower ramification filtration is

$$G_{-1} := G, \qquad G_i := \{\sigma \in G : v_L(\sigma(x) - x) \geq i+1 \text{ for every } x \in \mathscr{O}_L\} \quad (i \geq 0). \tag{5.1.3}$$

$$G_{-1} \supseteq G_0 \supseteq G_1 \supseteq \cdots \supseteq 1.$$

The extension is tamely ramified when $p \nmid e(L/F)$, wildly ramified when $p \mid e(L/F)$, and G_1 is its wild inertia group.

Proposition 5.1.22 (Ramification filtration)**.** *Let L/F be as in Definition 5.1.21.*

(1) *The initial terms and fixed fields satisfy*

$$G_0 = I(L/F), \qquad L^{G_0}/F \text{ is maximal unramified}, \qquad L/L^{G_0} \text{ is totally ramified}.$$

Proof. Compare the action on the residue field.

$$G \longrightarrow \mathrm{Gal}(k_L/k_F), \qquad \sigma \longmapsto (\bar{x} \mapsto \overline{\sigma(x)}), \qquad \ker(G \to \mathrm{Gal}(k_L/k_F)) = I(L/F).$$

$$\sigma \in I(L/F) \iff \sigma(x) \equiv x \pmod{\mathfrak{m}_L} \quad (x \in \mathscr{O}_L) \iff \sigma \in G_0.$$

$$G/G_0 \cong \mathrm{Gal}(k_L/k_F), \qquad [L^{G_0} : F] = f(L/F), \qquad [L : L^{G_0}] = e(L/F).$$

This proves the inertia and fixed-field assertions. □

(2) *There is a canonical injection*

$$G_0/G_1 \hookrightarrow k_L^\times, \qquad \sigma G_1 \longmapsto \frac{\sigma(\pi_L)}{\pi_L} \pmod{\mathfrak{m}_L},$$

so G_0/G_1 is cyclic of order prime to p.

Proof. Use the leading coefficient of a uniformizer displacement.

$$\sigma \in G_0 \implies v_L(\sigma(\pi_L)) = v_L(\pi_L) = 1 \implies \sigma(\pi_L)/\pi_L \in \mathscr{O}_L^\times.$$

$$\frac{\sigma\tau(\pi_L)}{\pi_L} \equiv \frac{\sigma(\pi_L)}{\pi_L} \cdot \frac{\tau(\pi_L)}{\pi_L} \pmod{\mathfrak{m}_L}.$$

$$\sigma(\pi_L)/\pi_L \equiv 1 \pmod{\mathfrak{m}_L} \iff \sigma \in G_1 \implies G_0/G_1 \hookrightarrow k_L^\times.$$

$$\#(k_L^\times) = \#(k_L) - 1 \not\equiv 0 \pmod{p} \implies p \nmid |G_0/G_1|.$$

This proves the tame-quotient assertion. □

(3) *For every $i \geq 1$,*

$$G_i/G_{i+1} \hookrightarrow (k_L, +), \qquad \sigma G_{i+1} \longmapsto \frac{\sigma(\pi_L) - \pi_L}{\pi_L^{i+1}} \pmod{\mathfrak{m}_L}.$$

Hence G_i/G_{i+1} is elementary abelian of exponent p, and G_1 is the unique Sylow p-subgroup of G_0.

Proof. Use the first nonzero additive displacement.

$$\sigma \in G_i \implies \frac{\sigma(\pi_L)-\pi_L}{\pi_L^{i+1}} \in \mathscr{O}_L.$$

$$\frac{\sigma\tau(\pi_L)-\pi_L}{\pi_L^{i+1}} \equiv \frac{\sigma(\pi_L)-\pi_L}{\pi_L^{i+1}} + \frac{\tau(\pi_L)-\pi_L}{\pi_L^{i+1}} \pmod{\mathfrak{m}_L}.$$

$$\frac{\sigma(\pi_L)-\pi_L}{\pi_L^{i+1}} \equiv 0 \pmod{\mathfrak{m}_L} \iff \sigma \in G_{i+1} \implies G_i/G_{i+1} \hookrightarrow (k_L,+).$$

$$G_j = 1 \quad (j \gg 0), \qquad G_i/G_{i+1} \text{ is a } p\text{-group} \quad (i \geq 1) \implies G_1 \text{ is a } p\text{-group}.$$

$$G_1 \trianglelefteq G_0, \qquad p \nmid [G_0 : G_1] \overset{2}{\implies} G_1 \text{ is the unique Sylow } p\text{-subgroup of } G_0.$$

This proves the wild-quotient assertions. □

(4) *The fixed-field tower is*

$$\begin{tikzcd}
L \arrow[d, dash, "G_1\ (\text{wild})"'] & 1 \arrow[d, hook] \\
L^{G_1} \arrow[d, dash, "G_0/G_1\ (\text{tame})"'] & G_1 \arrow[d, hook] \\
L^{G_0} \arrow[d, dash, "G/G_0\ (\text{unramified})"'] & G_0 \arrow[d, hook] \\
F = L^G & G,
\end{tikzcd}$$

where L^{G_0}/F is maximal unramified, L^{G_1}/F is maximal tamely ramified, and

$$L/F \textit{ is tame} \iff G_1 = 1.$$

Proof. Use the quotient structure from the preceding items.

$$G/G_0 \cong \mathrm{Gal}(k_L/k_F) \implies L^{G_0}/F \text{ is maximal unramified.}$$

$$G_0/G_1 \hookrightarrow k_L^\times \overset{2}{\implies} p \nmid [G_0 : G_1] \implies L^{G_1}/L^{G_0} \text{ is tamely ramified.}$$

$$G_1 \in \mathrm{Syl}_p(G_0) \overset{3}{\implies} M/F \text{ tame}, \quad M \subseteq L \implies G_1 \subseteq \mathrm{Gal}(L/M) \implies M \subseteq L^{G_1}.$$

$$p \nmid e(L/F) = |G_0| \iff G_1 = 1.$$

This proves maximality and the tame criterion. □

Theorem 5.1.23 (Hilbert different formula, [2])**.** *Let L/F be a finite Galois extension of local fields with lower ramification groups G_i, and set*

$$\mathscr{O}_L^\vee := \{x \in L : \mathrm{Tr}_{L/F}(x\mathscr{O}_L) \subseteq \mathscr{O}_F\}, \qquad \mathfrak{d}(L/F) := (\mathscr{O}_L^\vee)^{-1}.$$

Then

$$v_L(\mathfrak{d}(L/F)) = \sum_{i=0}^{\infty}(|G_i|-1), \qquad \mathrm{disc}(L/F) = \mathrm{N}_{L/F}(\mathfrak{d}(L/F)).$$

In particular,

$$L/F \textit{ unramified} \iff \mathfrak{d}(L/F) = \mathscr{O}_L, \qquad L/F \textit{ tame} \implies v_L(\mathfrak{d}(L/F)) = e(L/F) - 1.$$

6. LOCAL CLASS FIELD THEORY

Local reciprocity converts finite abelian extensions of a local field into finite quotients of its multiplicative group. We retain the arithmetic Frobenius normalization of § 4.

6.1. **Reciprocity, Norm Groups, and Lubin–Tate Theory.**

Definition 6.1.1 (Krull topology and profinite completion)**.** Let E/K be Galois, put $G := \mathrm{Gal}(E/K)$, and let $\mathscr{F}(E/K)$ denote the finite Galois intermediate extensions. The subgroups $\mathrm{Gal}(E/M)$, $M \in \mathscr{F}(E/K)$, form a neighborhood basis of 1, and restriction gives

$$\begin{tikzcd} G \arrow[r, "\sim"] \arrow[dr, "\mathrm{res}_M"'] & \varprojlim_{N \in \mathscr{F}(E/K)} \mathrm{Gal}(N/K) \arrow[d, "\mathrm{pr}_M"] \\ & \mathrm{Gal}(M/K). \end{tikzcd}$$

For a topological group Γ, its profinite completion is

$$\widehat{\Gamma} := \varprojlim_{N \trianglelefteq_{\mathrm{open}} \Gamma,\, [\Gamma:N]<\infty} \Gamma/N, \qquad \widehat{\mathbb{Z}} = \varprojlim_{m \geq 1} \mathbb{Z}/m\mathbb{Z}.$$

Thus a topological group is profinite precisely when it is an inverse limit of finite discrete groups.

Proposition 6.1.2 (Maximal abelian subextension)**.** *Let E/K be Galois with group G, and put $G' := \overline{[G,G]}$. The field $E^{\mathrm{ab}} := E^{G'}$ satisfies*

$$G\left(\begin{array}{c} E \\ \big| {\scriptstyle G'} \\ E^{\mathrm{ab}} \\ \big| {\scriptstyle G/G'} \\ K \end{array}\right. \qquad \mathrm{Gal}(E^{\mathrm{ab}}/K) \cong G^{\mathrm{ab}} := G/G', \qquad M/K \textit{ abelian} \iff M \subseteq E^{\mathrm{ab}}.$$

Proof. Apply the Galois correspondence to $H := \mathrm{Gal}(E/M)$.

$$M/K \text{ Galois} \iff H \trianglelefteq G, \qquad \mathrm{Gal}(M/K) \cong G/H.$$
$$G/H \text{ abelian} \iff [G,G] \subseteq H \iff \overline{[G,G]} \subseteq H.$$
$$G' \subseteq H \iff E^H \subseteq E^{G'} \iff M \subseteq E^{\mathrm{ab}}.$$

This proves the universal property and the quotient formula. □

Definition 6.1.3 (Local Weil group)**.** Let K^{s}/K be a separable closure of a nonarchimedean local field with residue field $k_K \cong \mathbb{F}_q$, absolute Galois group $G_K := \mathrm{Gal}(K^{\mathrm{s}}/K)$, and inertia group I_K. Writing $\mathrm{Fr}_q(\bar{x}) = \bar{x}^q$, the Weil group is the inverse image of $\langle \mathrm{Fr}_q \rangle \cong \mathbb{Z}$ under $G_K \twoheadrightarrow \mathrm{Gal}(\bar{k}_K/k_K) \cong \widehat{\mathbb{Z}}$:

$$\begin{tikzcd} 1 \arrow[r] & I_K \arrow[r, hook] \arrow[d, equal] & W_K \arrow[r, two heads, "v_K"] \arrow[d, hook] & \mathbb{Z} \arrow[r] \arrow[d, hook] & 0 \\ 1 \arrow[r] & I_K \arrow[r, hook] & G_K \arrow[r, two heads] & \widehat{\mathbb{Z}} \arrow[r] & 0. \end{tikzcd}$$

Set $W_K^{\mathrm{ab}} := W_K/\overline{[W_K, W_K]}$ and $K^{\mathrm{ab}} := (K^{\mathrm{s}})^{\overline{[G_K, G_K]}}$.

Theorem 6.1.4 (Local Artin reciprocity, [2]). *Let K be a nonarchimedean local field. There is a unique topological isomorphism*

$$\mathrm{Art}_K\colon K^\times \xrightarrow{\sim} W_K^{\mathrm{ab}}, \qquad \widehat{K^\times} \xrightarrow{\sim} \mathrm{Gal}(K^{\mathrm{ab}}/K), \qquad \mathrm{Art}_K(\pi_K)|_{K^{\mathrm{ur}}} = \mathrm{Fr}_q. \tag{6.1.1}$$

For every finite abelian L/K, restriction gives

$$\begin{tikzcd} K^\times \arrow[r, "\mathrm{Art}_K"] \arrow[d, two heads] & W_K^{\mathrm{ab}} \arrow[d, two heads, "\mathrm{res}_L"] \\ K^\times/\mathrm{N}_{L/K}(L^\times) \arrow[r, "\sim", "\overline{\mathrm{Art}}_{L/K}"'] & \mathrm{Gal}(L/K), \end{tikzcd}$$

where

$$\mathrm{Art}_{L/K} := \mathrm{res}_L \circ \mathrm{Art}_K, \qquad \ker(\mathrm{Art}_{L/K}) = \mathrm{N}_{L/K}(L^\times), \qquad \overline{\mathrm{Art}}_{L/K}(a\mathrm{N}_{L/K}(L^\times)) = \mathrm{Art}_{L/K}(a).$$

If L_0/K is the maximal unramified subextension and $I(L/K) := \mathrm{Gal}(L/L_0)$, then

$$\mathrm{Art}_{L/K}(\mathscr{O}_K^\times) = I(L/K), \qquad \mathrm{Art}_{L/K}(\pi_K)|_{L_0} = \mathrm{Frob}(L_0/K).$$

Theorem 6.1.5 (Local class field correspondence, [2]). *Let $\mathscr{E}_K$ be the finite abelian extensions in K^{ab}, and let $\mathscr{U}_K$ be the open finite-index subgroups of $K^\times$. Local reciprocity induces the order-reversing bijection*

$$\mathscr{E}_K \underset{U \mapsto L_U}{\overset{L \mapsto \mathrm{N}_{L/K}(L^\times)}{\rightleftarrows}} \mathscr{U}_K \qquad [L:K] = [K^\times : \mathrm{N}_{L/K}(L^\times)]. \tag{6.1.2}$$

For $L_1, L_2 \in \mathscr{E}_K$, writing $N_i := \mathrm{N}_{L_i/K}(L_i^\times)$, one has

$$L_1 \subseteq L_2 \iff N_2 \subseteq N_1, \qquad \mathrm{N}_{L_1L_2/K}((L_1L_2)^\times) = N_1 \cap N_2, \qquad \mathrm{N}_{(L_1\cap L_2)/K}((L_1\cap L_2)^\times) = N_1N_2.$$

$$\begin{tikzcd} & L_1L_2 & & & & N_1\cap N_2 \arrow[dl, hook] \arrow[dr, hook] & \\ L_1 \arrow[ur, hook] & & L_2 \arrow[ul, hook'] & \overset{\mathrm{op}}{\longleftrightarrow} & N_1 \arrow[dr, hook] & & N_2 \arrow[dl, hook'] \\ & L_1\cap L_2 \arrow[ul, hook'] \arrow[ur, hook] & & & & N_1N_2 & \end{tikzcd}$$

Definition 6.1.6 (Local norm group and conductor). Let K be nonarchimedean, and put

$$U_K^{(0)} := \mathscr{O}_K^\times, \qquad U_K^{(m)} := 1 + \mathfrak{m}_K^m \quad (m \geq 1).$$

For finite abelian L/K, define

$$\mathscr{N}_{L/K} := \mathrm{N}_{L/K}(L^\times), \qquad a(L/K) := \min\{m \geq 0 : U_K^{(m)} \subseteq \mathscr{N}_{L/K}\}.$$

An arbitrary subgroup $U \subseteq K^\times$ is open of finite index precisely when, for some $d \geq 1$ and $m \geq 0$,

$$\pi_K^{d\mathbb{Z}} U_K^{(m)} \subseteq U \subseteq K^\times.$$

Proposition 6.1.7 (Local norm indices). *Let L/K be finite abelian with ramification index e and residue degree f. Then*

$$v_K(\mathscr{N}_{L/K}) = f\mathbb{Z}, \qquad \mathscr{N}_{L/K} \cap \mathscr{O}_K^\times = \mathrm{N}_{L/K}(\mathscr{O}_L^\times),$$
$$[K^\times : \mathscr{N}_{L/K}] = ef = [L:K], \qquad [\mathscr{O}_K^\times : \mathrm{N}_{L/K}(\mathscr{O}_L^\times)] = e.$$

Proof. Use valuation compatibility and local reciprocity.

$$v_K(\mathrm{N}_{L/K}(x)) = f \cdot v_L(x) \overset{4.1.10}{\Longrightarrow} v_K(\mathscr{N}_{L/K}) = f\mathbb{Z}.$$
$$v_K(\mathrm{N}_{L/K}(x)) = 0 \iff v_L(x) = 0 \iff x \in \mathscr{O}_L^\times \implies \mathscr{N}_{L/K} \cap \mathscr{O}_K^\times = \mathrm{N}_{L/K}(\mathscr{O}_L^\times).$$
$$K^\times \cong \pi_K^{\mathbb{Z}} \times \mathscr{O}_K^\times \overset{5.1.13}{\Longrightarrow} [K^\times : \mathscr{N}_{L/K}] = f \cdot [\mathscr{O}_K^\times : \mathrm{N}_{L/K}(\mathscr{O}_L^\times)].$$
$$K^\times/\mathscr{N}_{L/K} \cong \mathrm{Gal}(L/K) \overset{6.1.4}{\Longrightarrow} [K^\times : \mathscr{N}_{L/K}] = |\mathrm{Gal}(L/K)| = [L:K] = ef \implies [\mathscr{O}_K^\times : \mathrm{N}_{L/K}(\mathscr{O}_L^\times)] = e.$$

This proves both norm-index formulas. $\square$

Theorem 6.1.8 (Ramification reciprocity, [2])**.** *Let L/K be finite abelian with group G. Extend the lower filtration of Definition 5.1.21 by $G_t := G_{\lceil t \rceil}$ for $t > 0$, and put*

$$\varphi_{L/K}(u) := \int_0^u \frac{\mathrm{d}t}{[G_0 : G_t]}, \qquad \psi_{L/K} := \varphi_{L/K}^{-1}, \qquad G^u := G_{\psi_{L/K}(u)} \quad (u \geq 0).$$

The Artin map identifies the unit and upper ramification filtrations:

$$\begin{tikzcd}
U_K^{(0)} \arrow[d, two heads, "\mathrm{Art}_{L/K}"'] & U_K^{(1)} \arrow[l, hook'] \arrow[d, two heads, "\mathrm{Art}_{L/K}"'] & \cdots \arrow[l, hook'] & U_K^{(m)} \arrow[l, hook'] \arrow[d, two heads, "\mathrm{Art}_{L/K}"'] & U_K^{(m+1)} \arrow[l, hook'] \arrow[d, two heads, "\mathrm{Art}_{L/K}"'] \\
G^0 & G^1 \arrow[l, hook'] & \cdots \arrow[l, hook'] & G^m \arrow[l, hook'] & G^{m+1}. \arrow[l, hook']
\end{tikzcd} \tag{6.1.3}$$

Consequently,

$$\mathrm{Art}_{L/K}(U_K^{(m)}) = G^m, \qquad \frac{U_K^{(m)}\mathscr{N}_{L/K}}{U_K^{(m+1)}\mathscr{N}_{L/K}} \cong \frac{G^m}{G^{m+1}},$$
$$a(L/K) = \min\{m : G^m = 1\}, \qquad a(L/K) = 0 \iff L/K \textit{ unramified},$$
$$a(L/K) = 1 \iff L/K \textit{ tame and ramified}.$$

Proposition 6.1.9 (Unramified Artin map)**.** *Let L/K be the unramified extension of degree n. Then*

$$\mathscr{N}_{L/K} = \pi_K^{n\mathbb{Z}} \times \mathscr{O}_K^\times, \qquad \begin{tikzcd}
K^\times \arrow[r, two heads, "\mathrm{Art}_{L/K}"] \arrow[d, "v_K \bmod n"'] & \mathrm{Gal}(L/K) \arrow[d, "\mathrm{Frob}(L/K)^j \mapsto j"] \\
\mathbb{Z}/n\mathbb{Z} \arrow[r, equal] & \mathbb{Z}/n\mathbb{Z}.
\end{tikzcd}$$

Proof. Use the unramified classification and finite-field norm.

$$L/K \text{ unramified of degree } n \overset{3}{\Longrightarrow} e(L/K) = 1, \qquad f(L/K) = n, \qquad \pi_L = \pi_K.$$
$$\mathrm{N}_{L/K}(\pi_K) = \pi_K^n, \qquad v_K(\mathscr{N}_{L/K}) = n\mathbb{Z}.$$
$$e(L/K) = 1 \overset{6.1.7}{\Longrightarrow} \mathrm{N}_{L/K}(\mathscr{O}_L^\times) = \mathscr{O}_K^\times \implies \mathscr{N}_{L/K} = \pi_K^{n\mathbb{Z}} \times \mathscr{O}_K^\times.$$
$$K^\times/\mathscr{N}_{L/K} \cong \mathbb{Z}/n\mathbb{Z} \overset{6.1.4}{\Longrightarrow} \mathrm{Art}_{L/K}(\pi_K) = \mathrm{Frob}(L/K).$$

This proves the norm group and the Frobenius square. $\square$

Definition 6.1.10 (Lubin–Tate formal group)**.** Let K be nonarchimedean with residue cardinality q, and fix a uniformizer π_K. A Lubin–Tate series is

$$f(X) \in \mathscr{O}_K[[X]], \qquad f(X) \equiv \pi_K X \pmod{X^2}, \qquad f(X) \equiv X^q \pmod{\pi_K}.$$

Its one-dimensional commutative formal $\mathscr{O}_K$-module $\mathscr{F}$ satisfies

$$\mathscr{F}(X,Y) = X + Y + \text{terms of degree at least 2}, \qquad [\pi_K]_{\mathscr{F}} = f,$$
$$[a+b]_{\mathscr{F}} = \mathscr{F}([a]_{\mathscr{F}}, [b]_{\mathscr{F}}), \qquad [ab]_{\mathscr{F}} = [a]_{\mathscr{F}} \circ [b]_{\mathscr{F}},$$
$$[a]_{\mathscr{F}}(X) \equiv aX \pmod{X^2} \quad (a \in \mathscr{O}_K).$$

For $m \geq 1$, set

$$\mathscr{F}[\pi_K^m] := \{x \in \mathfrak{m}_{\overline{K}} : [\pi_K^m]_{\mathscr{F}}(x) = 0\}, \qquad K_m := K(\mathscr{F}[\pi_K^m]).$$

Theorem 6.1.11 (Lubin–Tate extensions, [2])**.** *Let $m \geq 1$ and let K_m be as in Definition 6.1.10. Then*

$$K \subset K_1 \subset K_2 \subset \cdots, \qquad K_m/K \text{ totally ramified abelian}, \qquad [K_m : K] = (q-1)q^{m-1},$$
$$\mathrm{Gal}(K_m/K) \cong (\mathscr{O}_K/\pi_K^m \mathscr{O}_K)^\times, \qquad \mathscr{N}_{K_m/K} = \pi_K^{\mathbb{Z}} U_K^{(m)}.$$

$$\begin{tikzcd}
\mathrm{Gal}(K_{m+1}/K) \arrow[r, two heads, "\mathrm{res}"] \arrow[d, "\sim"] & \mathrm{Gal}(K_m/K) \arrow[d, "\sim"] \\
(\mathscr{O}_K/\pi_K^{m+1}\mathscr{O}_K)^\times \arrow[r, two heads] & (\mathscr{O}_K/\pi_K^m \mathscr{O}_K)^\times.
\end{tikzcd}$$

If K_n^{ur}/K is unramified of degree n, then

$$\begin{tikzcd}
 & K_m K_n^{\mathrm{ur}} \arrow[dl, no head, "n"'] \arrow[dr, no head, "(q-1)q^{m-1}"] & \\
K_m \arrow[dr, no head, "(q-1)q^{m-1}"'] & & K_n^{\mathrm{ur}} \arrow[dl, no head, "n"] \\
 & K &
\end{tikzcd}
\qquad
\begin{tikzcd}
K^\times/\pi_K^{n\mathbb{Z}} U_K^{(m)} \arrow[r, "\sim"] & \mathrm{Gal}(K_m K_n^{\mathrm{ur}}/K) \\
\mathbb{Z}/n\mathbb{Z} \times (\mathscr{O}_K/\pi_K^m \mathscr{O}_K)^\times \arrow[u, "\sim"] &
\end{tikzcd}$$

and

$$\mathrm{N}_{K_m K_n^{\mathrm{ur}}/K}\big((K_m K_n^{\mathrm{ur}})^\times\big) = \pi_K^{n\mathbb{Z}} U_K^{(m)}.$$

Corollary 6.1.12 (Local Kronecker–Weber)**.** *For every prime p,*

$$\mathbb{Q}_p^{\mathrm{ab}} = \mathbb{Q}_p^{\mathrm{ur}} \mathbb{Q}_p(\mu_{p^\infty}) = \bigcup_{m \geq 1} \mathbb{Q}_p(\zeta_m),$$

$$\begin{tikzcd}
 & \mathbb{Q}_p^{\mathrm{ab}} \arrow[dl, no head] \arrow[dr, no head] & \\
\mathbb{Q}_p^{\mathrm{ur}} \arrow[dr, no head] & & \mathbb{Q}_p(\mu_{p^\infty}) \arrow[dl, no head] \\
 & \mathbb{Q}_p &
\end{tikzcd}
\qquad \mathrm{Gal}(\mathbb{Q}_p^{\mathrm{ab}}/\mathbb{Q}_p) \cong \widehat{\mathbb{Z}} \times \mathbb{Z}_p^\times \cong \widehat{\mathbb{Q}_p^\times}.$$

Theorem 6.1.13 (Functoriality of local reciprocity, [2])**.** *Let L/K be finite, fix $\overline{L} = \overline{K}$, and let $i\colon W_L \hookrightarrow W_K$. The reciprocity squares are*

$$\begin{tikzcd}
L^\times \arrow[r, "\mathrm{Art}_L"] \arrow[d, "\mathrm{N}_{L/K}"'] & W_L^{\mathrm{ab}} \arrow[d, "i_*"] \\
K^\times \arrow[r, "\mathrm{Art}_K"'] & W_K^{\mathrm{ab}}
\end{tikzcd}
\qquad
\begin{tikzcd}
K^\times \arrow[r, "\mathrm{Art}_K"] \arrow[d, hook, "\iota"'] & W_K^{\mathrm{ab}} \arrow[d, "\mathrm{Ver}_{L/K}"] \\
L^\times \arrow[r, "\mathrm{Art}_L"'] & W_L^{\mathrm{ab}}.
\end{tikzcd}$$

For finite abelian M/K and $K \subseteq L \subseteq M$, restriction gives

$$\begin{tikzcd} K^\times \arrow[r, "\mathrm{Art}_{M/K}"] \arrow[dr, "\mathrm{Art}_{L/K}"'] & \mathrm{Gal}(M/K) \arrow[d, two heads, "\mathrm{res}"] \\ & \mathrm{Gal}(L/K). \end{tikzcd}$$

Definition 6.1.14 (Hilbert symbol)**.** Let K be nonarchimedean, let $n \geq 1$, $\mathrm{char}(K) \nmid n$, and let $\mu_n \subset K$. For $a, b \in K^\times$, choose $\beta^n = b$, put $K_b := K(\beta)$, and define

$$\begin{tikzcd} K^\times \arrow[r, "\mathrm{Art}_{K_b/K}"] \arrow[dr, "a \mapsto (a,b)_{K,n}"'] & \mathrm{Gal}(K_b/K) \arrow[d, "\sigma \mapsto \sigma(\beta)/\beta"] \\ & \mu_n, \end{tikzcd} \qquad (a,b)_{K,n} := \frac{\mathrm{Art}_{K_b/K}(a)(\beta)}{\beta}.$$

Proposition 6.1.15 (Hilbert-symbol identities, [2])**.** *Under Definition 6.1.14, the Hilbert symbol induces the perfect pairing*

$$K^\times/K^{\times n} \times K^\times/K^{\times n} \longrightarrow \mu_n, \qquad (a,b)_{K,n} = (b,a)_{K,n}^{-1},$$
$$(aa',b)_{K,n} = (a,b)_{K,n}(a',b)_{K,n}, \qquad (a,bb')_{K,n} = (a,b)_{K,n}(a,b')_{K,n},$$
$$(a,1-a)_{K,n} = 1 \quad (a \neq 1), \qquad (a,-a)_{K,n} = 1.$$

Lemma 6.1.16 (Hilbert norm criterion)**.** *Let $K_b = K(\sqrt[n]{b})$ under Definition 6.1.14. Then*

$$(a,b)_{K,n} = 1 \iff a \in \mathrm{N}_{K_b/K}(K_b^\times).$$

Proof. The norm criterion is the reciprocity kernel.

$$(a,b)_{K,n} = 1 \iff \mathrm{Art}_{K_b/K}(a)(\beta) = \beta.$$
$$K_b = K(\beta) \implies \mathrm{Art}_{K_b/K}(a)(\beta) = \beta \iff \mathrm{Art}_{K_b/K}(a) = 1.$$
$$\mathrm{Art}_{K_b/K}(a) = 1 \overset{6.1.4}{\iff} a \in \mathrm{N}_{K_b/K}(K_b^\times).$$

This proves the norm–residue criterion. □

Theorem 6.1.17 (Tame Hilbert symbol, [2])**.** *Let K have residue characteristic p, residue field $\mathbb{F}_q$, and let $(n,p) = 1$. Then*

$$\mu_n \subset K \iff n \mid (q-1).$$

Assume these conditions, let $\omega \colon \mathbb{F}_q^\times \to \mu_{q-1} \subset \mathscr{O}_K^\times$ be the Teichmüller lift, and write $a = \pi_K^\alpha u$, $b = \pi_K^\beta v$ with $u, v \in \mathscr{O}_K^\times$. Then

$$(a,b)_{K,n} = \omega\left(\overline{(-1)^{\alpha\beta} v^\alpha u^{-\beta}}^{(q-1)/n}\right). \tag{6.1.4}$$

In particular,

$$(u,v)_{K,n} = 1, \qquad (\pi_K,u)_{K,n} = \omega(\overline{u}^{(q-1)/n}), \qquad (u,\pi_K)_{K,n} = (\pi_K,u)_{K,n}^{-1}.$$

Proof. Reduce the symbol to units and a uniformizer.

$$(n,p) = 1 \overset{5.1.10}{\implies} \mu_n(K) \xrightarrow{\sim} \mu_n(\mathbb{F}_q) \implies \mu_n \subset K \iff n \mid (q-1).$$
$$u, v \in \mathscr{O}_K^\times \implies K(\sqrt[n]{v})/K \text{ unramified} \overset{6.1.4}{\implies} (u,v)_{K,n} = 1.$$
$$\mathrm{Art}_K(\pi_K)|_{K^{\mathrm{ur}}} = \mathrm{Fr}_q \overset{6.1.1}{\implies} (\pi_K,u)_{K,n} = \omega(\overline{u}^{(q-1)/n}), \quad (u,\pi_K)_{K,n} = (\pi_K,u)_{K,n}^{-1}.$$

$$1 = (\pi_K, -\pi_K)_{K,n} \overset{6.1.15}{\Longrightarrow} (\pi_K, \pi_K)_{K,n} = (\pi_K, -1)_{K,n}^{-1} = \omega(\overline{-1}^{(q-1)/n}).$$

$$(a,b)_{K,n} \overset{6.1.15}{=} \big(\pi_K, \pi_K\big)_{K,n}^{\alpha\beta} \big(\pi_K, v\big)_{K,n}^{\alpha} \big(u, \pi_K\big)_{K,n}^{\beta} \big(u, v\big)_{K,n} = \omega\left(\overline{(-1)^{\alpha\beta} v^{\alpha} u^{-\beta}}^{(q-1)/n}\right).$$

This proves Equation (6.1.4). $\square$

7. Global Class Field Theory

Global reciprocity assembles § 6 through idèle classes and ray classes.

7.1. **Idèles, Ray Class Fields, and Reciprocity.**

Definition 7.1.1 (Idèles and idèle classes)**.** Let K be a number field with places v. Its idèle and idèle class groups are

$$\mathbb{A}_K^\times := \prod_{v\mid\infty} K_v^\times \times \prod_{\mathfrak{p}\nmid\infty}{}' K_{\mathfrak{p}}^\times, \qquad C_K := \mathbb{A}_K^\times / \Delta(K^\times), \tag{7.1.1}$$

where $x_{\mathfrak{p}} \in \mathscr{O}_{K_{\mathfrak{p}}}^\times$ for almost all $\mathfrak{p}$, and $\Delta(a) := (a)_v$. Thus

$$\begin{tikzcd} 1 \arrow[r] & K^\times \arrow[r, hook, "\Delta"] & \mathbb{A}_K^\times \arrow[r, two heads, "q_K"] & C_K \arrow[r] & 1 \\ & & K_v^\times \arrow[u, hook, "\iota_v"] \arrow[ur, bend right, "q_K\circ\iota_v"'] & & \end{tikzcd}$$

with $\iota_v(x)$ equal to x at v and 1 elsewhere. For finite L/K, idèlic norm is

$$\mathrm{N}_{L/K}((y_w)_w) := \left(\prod_{w\mid v} \mathrm{N}_{L_w/K_v}(y_w)\right)_v, \qquad \mathrm{N}_{L/K}(\Delta_L(a)) = \Delta_K(\mathrm{N}_{L/K}(a)),$$

and descends through

$$\begin{tikzcd} L^\times \arrow[r, hook, "\Delta_L"] \arrow[d, "\mathrm{N}_{L/K}"'] & \mathbb{A}_L^\times \arrow[r, two heads, "q_L"] \arrow[d, "\mathrm{N}_{L/K}"'] & C_L \arrow[d, "\mathrm{N}_{L/K}"] \\ K^\times \arrow[r, hook, "\Delta_K"'] & \mathbb{A}_K^\times \arrow[r, two heads, "q_K"'] & C_K. \end{tikzcd}$$

Theorem 7.1.2 (Global Artin reciprocity, [2])**.** *Let L/K be finite abelian with group G. For each place v, choose $w \mid v$ and identify* $\mathrm{Gal}(L_w/K_v) \cong D(w \mid v) \subseteq G$. *At infinity use*

$$\mathrm{Art}_{\mathbb{C}/\mathbb{C}} = \mathrm{Art}_{\mathbb{R}/\mathbb{R}} = 1, \qquad \mathrm{Art}_{\mathbb{C}/\mathbb{R}}(x) = \begin{cases} 1, & x > 0, \\ \textit{complex conjugation}, & x < 0. \end{cases}$$

The product of local Artin maps is a continuous surjection satisfying

$$\mathrm{Art}_{L/K} \colon \mathbb{A}_K^\times \twoheadrightarrow G, \qquad (x_v)_v \longmapsto \prod_v \mathrm{Art}_{L_w/K_v}(x_v), \qquad \ker(\mathrm{Art}_{L/K}) = \Delta(K^\times) \cdot \mathrm{N}_{L/K}(\mathbb{A}_L^\times). \tag{7.1.2}$$

It induces

$$\begin{tikzcd} \mathbb{A}_K^\times \arrow[r, two heads, "\mathrm{Art}_{L/K}"] \arrow[d, two heads, "q_K"'] & G \arrow[d, equal] \\ C_K \arrow[r, two heads] \arrow[dr, two heads] & G \\ & C_K/\mathrm{N}_{L/K}(C_L) \arrow[u, "\sim"'] \end{tikzcd} \qquad C_K/\mathrm{N}_{L/K}(C_L) \cong \mathrm{Gal}(L/K).$$

Taking inverse limits defines

$$\mathrm{Art}_K \colon C_K \twoheadrightarrow \mathrm{Gal}(K^{\mathrm{ab}}/K).$$

If C_K^0 is the identity component, then

$$\ker(\mathrm{Art}_K) = C_K^0, \qquad C_K/C_K^0 \xrightarrow{\sim} \mathrm{Gal}(K^{\mathrm{ab}}/K), \qquad \widehat{C_K} \xrightarrow{\sim} \mathrm{Gal}(K^{\mathrm{ab}}/K).$$

Moreover,

$$\{L/K \subset K^{\mathrm{ab}} : [L:K] < \infty\} \underset{U \mapsto L_U}{\overset{L \mapsto \mathrm{N}_{L/K}(C_L)}{\rightleftarrows}} \{U \subset C_K : U \text{ open},\ [C_K : U] < \infty\}$$

is order reversing.

Theorem 7.1.3 (Local–global compatibility)**.** *Let L/K be finite abelian, let $\mathfrak{q} \mid \mathfrak{p}$, and embed $K_{\mathfrak{p}}^\times$ as the $\mathfrak{p}$-component of $\mathbb{A}_K^\times$. Let*

$$\iota_{\mathfrak{q}} \colon \mathrm{Gal}(L_{\mathfrak{q}}/K_{\mathfrak{p}}) \xrightarrow{\sim} D(\mathfrak{q} \mid \mathfrak{p}) \hookrightarrow \mathrm{Gal}(L/K).$$

Then

$$\begin{tikzcd} K_{\mathfrak{p}}^\times \arrow[r, "\mathrm{Art}_{L_{\mathfrak{q}}/K_{\mathfrak{p}}}"] \arrow[d, hook, "\iota_{\mathfrak{p}}"'] & \mathrm{Gal}(L_{\mathfrak{q}}/K_{\mathfrak{p}}) \arrow[d, hook, "\iota_{\mathfrak{q}}"] \\ \mathbb{A}_K^\times \arrow[r, "\mathrm{Art}_{L/K}"'] & \mathrm{Gal}(L/K). \end{tikzcd}$$

If $\mathfrak{p}$ is unramified, then

$$\mathrm{Art}_{L/K}(\iota_{\mathfrak{p}}(\pi_{\mathfrak{p}})) = \mathrm{Frob}(\mathfrak{q} \mid \mathfrak{p}), \qquad \mathrm{Art}_{L/K}(\Delta(a)) = \prod_v \mathrm{Art}_{L_w/K_v}(a) = 1 \quad (a \in K^\times).$$

Definition 7.1.4 (Modulus and ray class group)**.** A modulus of K is

$$\mathfrak{m} = \mathfrak{m}_0 \mathfrak{m}_\infty, \qquad \mathfrak{m}_0 = \prod_{\mathfrak{p}} \mathfrak{p}^{m_{\mathfrak{p}}}, \qquad \mathfrak{m}_\infty = \prod_{v \in S_\infty(\mathfrak{m})} v,$$

where

$$m_{\mathfrak{p}} \in \mathbb{Z}_{\geq 0}, \qquad m_{\mathfrak{p}} = 0 \text{ for almost all } \mathfrak{p}, \qquad S_\infty(\mathfrak{m}) \subseteq \{v : v \text{ real}\}.$$

Write $\mathfrak{m} \mid \mathfrak{n}$ when

$$\mathfrak{m}_0 \mid \mathfrak{n}_0, \qquad S_\infty(\mathfrak{m}) \subseteq S_\infty(\mathfrak{n}).$$

The ray groups are

$$I_K(\mathfrak{m}_0) := \left\{ \prod_{\mathfrak{p}} \mathfrak{p}^{a_{\mathfrak{p}}} : a_{\mathfrak{p}} \in \mathbb{Z},\ a_{\mathfrak{p}} = 0 \text{ for almost all } \mathfrak{p},\ a_{\mathfrak{p}} = 0 \text{ if } \mathfrak{p} \mid \mathfrak{m}_0 \right\},$$

$$K_{\mathfrak{m},1} := \left\{ \alpha \in K^\times : v_{\mathfrak{p}}(\alpha - 1) \geq m_{\mathfrak{p}} \text{ for } \mathfrak{p} \mid \mathfrak{m}_0,\ \sigma_v(\alpha) > 0 \text{ for } v \in S_\infty(\mathfrak{m}) \right\},$$

$$P_{K,1}(\mathfrak{m}) := \{(\alpha) : \alpha \in K_{\mathfrak{m},1}\}, \qquad \mathrm{Cl}_K(\mathfrak{m}) := I_K(\mathfrak{m}_0)/P_{K,1}(\mathfrak{m}). \tag{7.1.3}$$

Proposition 7.1.5 (Ray-class exact sequence)**.** *Let*

$$\mathscr{O}_{K,\mathfrak{m},1}^{\times} := \mathscr{O}_K^{\times} \cap K_{\mathfrak{m},1}, \qquad \Sigma_{\mathfrak{m}} := (\mathscr{O}_K/\mathfrak{m}_0)^{\times} \times \{\pm 1\}^{S_\infty(\mathfrak{m})}.$$

Reduction and real signs give

$$1 \longrightarrow \mathscr{O}_{K,\mathfrak{m},1}^{\times} \hookrightarrow \mathscr{O}_K^{\times} \xrightarrow{\rho_{\mathfrak{m}}} \Sigma_{\mathfrak{m}} \xrightarrow{\delta_{\mathfrak{m}}} \mathrm{Cl}_K(\mathfrak{m}) \twoheadrightarrow \mathrm{Cl}(K) \longrightarrow 1,$$

$$\rho_{\mathfrak{m}}(u) = \big(u \bmod \mathfrak{m}_0, (\mathrm{sgn}(\sigma_v(u)))_{v \in S_\infty(\mathfrak{m})}\big), \qquad \delta_{\mathfrak{m}}\big(\overline{a}, (\varepsilon_v)_v\big) = [(a)]_{\mathfrak{m}},$$
$$a \bmod \mathfrak{m}_0 = \overline{a}, \qquad \mathrm{sgn}(\sigma_v(a)) = \varepsilon_v \quad (v \in S_\infty(\mathfrak{m})), \qquad \#(\mathrm{Cl}_K(\mathfrak{m})) < \infty.$$

Definition 7.1.6 (Global conductor)**.** Let L/K be finite abelian. For each finite $\mathfrak{p}$, choose $\mathfrak{q} \mid \mathfrak{p}$ and use the exponent of Definition 6.1.6; for a real place v, put $\varepsilon_v = 1$ precisely when v becomes complex in L. The conductor is

$$\mathfrak{f}(L/K) := \prod_{\mathfrak{p} \nmid \infty} \mathfrak{p}^{a(L_{\mathfrak{q}}/K_{\mathfrak{p}})} \cdot \prod_{\substack{v \mid \infty \\ v \text{ real}}} v^{\varepsilon_v}.$$

The exponents are independent of $\mathfrak{q} \mid \mathfrak{p}$, and $\mathfrak{f}(L/K)$ is the least modulus for which the prime-to- $\mathfrak{m}_0$ Artin map factors through $\mathrm{Cl}_K(\mathfrak{m})$.

Theorem 7.1.7 (Ray class fields, [2])**.** *For every modulus* $\mathfrak{m}$*, there is a unique ray class field* $K(\mathfrak{m})/K$ *such that*

$$\begin{tikzcd} I_K(\mathfrak{m}_0) \arrow[r, two heads, "\mathrm{Art}_{\mathfrak{m}}"] \arrow[d, two heads] & \mathrm{Gal}(K(\mathfrak{m})/K) \arrow[d, equal] \\ \mathrm{Cl}_K(\mathfrak{m}) \arrow[r, "\sim"'] & \mathrm{Gal}(K(\mathfrak{m})/K), \end{tikzcd} \qquad [\mathfrak{p}] \longmapsto \mathrm{Frob}(\mathfrak{p}) \quad (\mathfrak{p} \nmid \mathfrak{m}_0). \tag{7.1.4}$$

For every finite abelian L/K*,*

$$\mathfrak{f}(L/K) \mid \mathfrak{m} \iff L \subseteq K(\mathfrak{m}).$$

Under these conditions,

$$\begin{tikzcd} I_K(\mathfrak{m}_0) \arrow[r, two heads, "\mathrm{Art}_{L/K}^{\mathfrak{m}}"] \arrow[d, two heads] & \mathrm{Gal}(L/K) \arrow[d, equal] \\ \mathrm{Cl}_K(\mathfrak{m}) \arrow[r, two heads, "\overline{\mathrm{Art}}_{L/K}^{\mathfrak{m}}"'] & \mathrm{Gal}(L/K) \end{tikzcd}$$

$$\ker(\mathrm{Art}_{L/K}^{\mathfrak{m}}) = P_{K,1}(\mathfrak{m}) \cdot \mathrm{N}_{L/K}\big(I_L(\mathfrak{m}_0\mathscr{O}_L)\big), \qquad \frac{\mathrm{Cl}_K(\mathfrak{m})}{\ker(\overline{\mathrm{Art}}_{L/K}^{\mathfrak{m}})} \cong \mathrm{Gal}(L/K).$$

If $\mathfrak{m} \mid \mathfrak{n}$*, then*

$$\begin{tikzcd} \mathrm{Cl}_K(\mathfrak{n}) \arrow[r, two heads] \arrow[d, "\sim"] & \mathrm{Cl}_K(\mathfrak{m}) \arrow[d, "\sim"] \\ \mathrm{Gal}(K(\mathfrak{n})/K) \arrow[r, two heads, "\mathrm{res}"'] & \mathrm{Gal}(K(\mathfrak{m})/K) \end{tikzcd} \qquad \begin{tikzcd} K(\mathfrak{n}) \arrow[d, dash] \\ K(\mathfrak{m}) \arrow[d, dash] \\ K. \end{tikzcd}$$

Corollary 7.1.8 (Hilbert and narrow class fields)**.** *Let* $\mathbf{1}$ *be the empty modulus, let* $\mathfrak{m}_\infty^{\mathrm{all}}$ *be the product of all real places, and put*

$$H_K := K(\mathbf{1}), \qquad H_K^+ := K(\mathfrak{m}_\infty^{\mathrm{all}}), \qquad \mathrm{Cl}^+(K) := \mathrm{Cl}_K(\mathfrak{m}_\infty^{\mathrm{all}}).$$

Then

$$\mathrm{Gal}(H_K/K) \cong \mathrm{Cl}(K), \qquad [H_K : K] = h_K, \qquad \mathrm{Gal}(H_K^+/K) \cong \mathrm{Cl}^+(K),$$

$$\begin{array}{ccc} H_K^+ & & \mathrm{Cl}^+(K) \\ | & & \twoheaddownarrow \\ H_K & & \mathrm{Cl}(K) \\ | & & \\ K & & \end{array}$$

$$H_K/K \textit{ is maximal abelian unramified at every place.}$$
$$H_K^+/K \textit{ is maximal abelian unramified at every finite place.}$$

Corollary 7.1.9 (Principal primes and complete splitting)**.** *For every nonzero prime* $\mathfrak{p} \subset \mathscr{O}_K$*,*

$$\mathfrak{p} \textit{ is principal} \iff \mathfrak{p} \textit{ splits completely in } H_K.$$

Proof. Pass from ideal classes to Frobenius.

$$\mathfrak{p} \text{ principal} \overset{3.1.1}{\iff} [\mathfrak{p}] = 1 \text{ in } \mathrm{Cl}(K).$$
$$[\mathfrak{p}] = 1 \overset{7.1.4}{\iff} \mathrm{Frob}(\mathfrak{p}) = 1 \text{ in } \mathrm{Gal}(H_K/K).$$
$$\mathrm{Frob}(\mathfrak{p}) = 1 \overset{4.1.22}{\iff} \mathfrak{p} \text{ splits completely in } H_K.$$

This proves the splitting criterion. □

Theorem 7.1.10 (Hilbert reciprocity, [2])**.** *Let* $\mu_n \subset K$*, and let* $a, b \in K^\times$*. At infinity,*

$$(a,b)_{\mathbb{C},n} = 1, \qquad (a,b)_{\mathbb{R},1} = 1, \qquad (a,b)_{\mathbb{R},2} = \begin{cases} -1, & a < 0 \wedge b < 0, \\ 1, & a > 0 \vee b > 0. \end{cases}$$

All local symbols satisfy

$$(a,b)_{K_v,n} = 1 \quad \textit{for almost all } v, \qquad \prod_v (a,b)_{K_v,n} = 1.$$

Proof. Apply global reciprocity to $L := K(\sqrt[n]{b})$.

$$\mathfrak{p} \nmid n(a)(b) \overset{6.1.4}{\Longrightarrow} (a,b)_{K_\mathfrak{p},n} = 1.$$

$$\Delta(a) \in \Delta(K^\times) \cdot \mathrm{N}_{L/K}(\mathbb{A}_L^\times) = \ker(\mathrm{Art}_{L/K}) \overset{7.1.2}{\Longrightarrow} 1 = \mathrm{Art}_{L/K}(\Delta(a)) = \prod_v \mathrm{Art}_{L_w/K_v}(a).$$

$$\frac{\mathrm{Art}_{L_w/K_v}(a)(\sqrt[n]{b})}{\sqrt[n]{b}} = (a,b)_{K_v,n} \overset{6.1.14}{\Longrightarrow} \prod_v (a,b)_{K_v,n} = \frac{\left(\prod_v \mathrm{Art}_{L_w/K_v}(a)\right)(\sqrt[n]{b})}{\sqrt[n]{b}} = 1.$$

This proves the product formula. □

Definition 7.1.11 (Power-residue symbol)**.** Let $\mu_n \subset K$, let $a \in \mathscr{O}_K \setminus \{0\}$, and let $\mathfrak{p} \nmid n(a)$. Put $L_a := K(\sqrt[n]{a})$, and let $\mathrm{Frob}_a(\mathfrak{p}) \in \mathrm{Gal}(L_a/K)$ be its arithmetic Frobenius. Define

$$\left(\frac{a}{\mathfrak{p}}\right)_n := \frac{\mathrm{Frob}_a(\mathfrak{p})(\sqrt[n]{a})}{\sqrt[n]{a}} \in \mu_n.$$

For an integral ideal $\mathfrak{b}$ coprime to $n(a)$, put

$$\left(\frac{a}{\mathfrak{b}}\right)_n := \prod_{\mathfrak{p}\mid\mathfrak{b}} \left(\frac{a}{\mathfrak{p}}\right)_n^{v_\mathfrak{p}(\mathfrak{b})}.$$

Theorem 7.1.12 (Power reciprocity, [2]). *Let $\mu_n \subset K$, and let $a,b \in \mathcal{O}_K \setminus \{0\}$ have coprime principal ideals, each prime to n. Then*

$$\left(\frac{a}{(b)}\right)_n \left(\frac{b}{(a)}\right)_n^{-1} = \prod_{v\mid n\infty} (a,b)_{K_v,n}.$$

Proof. Separate the tame finite places in Hilbert reciprocity.

$$\mathfrak{p}\nmid n \quad \wedge \quad ((a),(b)) = \mathcal{O}_K \overset{6.1.4}{\Longrightarrow} (a,b)_{K_\mathfrak{p},n} = \begin{cases} \left(\dfrac{b}{\mathfrak{p}}\right)_n^{v_\mathfrak{p}(a)}, & \mathfrak{p}\mid(a), \\ \left(\dfrac{a}{\mathfrak{p}}\right)_n^{-v_\mathfrak{p}(b)}, & \mathfrak{p}\mid(b), \\ 1, & \mathfrak{p}\nmid(ab). \end{cases}$$

$$\prod_{\mathfrak{p}\nmid n} (a,b)_{K_\mathfrak{p},n} \overset{7.1.11}{=} \left(\frac{b}{(a)}\right)_n \left(\frac{a}{(b)}\right)_n^{-1}.$$

$$\prod_v (a,b)_{K_v,n} = 1 \overset{7.1.10}{\Longrightarrow} 1 = \left(\frac{b}{(a)}\right)_n \left(\frac{a}{(b)}\right)_n^{-1} \prod_{v\mid n\infty} (a,b)_{K_v,n}$$

$$\Longleftrightarrow \left(\frac{a}{(b)}\right)_n \left(\frac{b}{(a)}\right)_n^{-1} = \prod_{v\mid n\infty} (a,b)_{K_v,n}.$$

This proves the reciprocity law. □

Corollary 7.1.13 (Kronecker–Weber, [2]). *Every finite abelian extension $L/\mathbb{Q}$ satisfies*

$$L \subseteq \mathbb{Q}(\zeta_m) = \mathbb{Q}((m)\infty) \quad \text{for some } m \geq 1, \qquad \mathbb{Q}^{\mathrm{ab}} = \bigcup_{m\geq 1} \mathbb{Q}(\zeta_m).$$

8. Applications of Global Class Field Theory

The unit and continued-fraction applications occur in § 3; here reciprocity is applied to characters, L-functions, class numbers, and complex multiplication.

8.1. **Characters, Zeta Functions, and Complex Multiplication.**

Definition 8.1.1 (Dirichlet and ray class characters). A Dirichlet character modulo m is a homomorphism

$$\chi\colon (\mathbb{Z}/m\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times, \qquad \chi(a) := 0 \quad \text{when } (a,m) > 1.$$

Its conductor $f_\chi \mid m$ is the least divisor through which χ factors; χ is primitive when $f_\chi = m$. Thus a unique primitive χ^* modulo f_χ satisfies

$$\begin{tikzcd} (\mathbb{Z}/m\mathbb{Z})^\times \arrow[r, two heads, "\rho_{m,f_\chi}"] \arrow[dr, "\chi"'] & (\mathbb{Z}/f_\chi\mathbb{Z})^\times \arrow[d, "\chi^*"] \\ & \mathbb{C}^\times. \end{tikzcd}$$

For a finite abelian group G, put

$$G^\vee := \mathrm{Hom}(G,\mathbb{C}^\times), \qquad G \xrightarrow{\sim} (G^\vee)^\vee, \qquad g \longmapsto (\chi \mapsto \chi(g)), \qquad |G^\vee| = |G|.$$

A finite-order Hecke character of K is a continuous character $\chi\colon C_K\to\mathbb{C}^\times$ with finite image. Its conductor $\mathfrak{f}(\chi)$ is the least modulus for which

$$\begin{tikzcd} C_K \arrow[r, two heads] \arrow[dr, "\chi"'] & \mathrm{Cl}_K(\mathfrak{f}(\chi)) \arrow[d, "\chi_{\mathfrak{f}}"] \\ & \mathbb{C}^\times \end{tikzcd}$$

commutes. Global reciprocity gives, for finite abelian L/K,

$$\begin{tikzcd} C_K \arrow[r, two heads, "\mathrm{Art}_{L/K}"] \arrow[d, two heads] & \mathrm{Gal}(L/K) \arrow[r, "\psi"] & \mathbb{C}^\times \\ \mathrm{Cl}_K(\mathfrak{m}) \arrow[urr, bend right, "\psi\circ\overline{\mathrm{Art}}^{\mathfrak{m}}_{L/K}"'] & & \end{tikzcd} \qquad \mathfrak{f}(L/K)\mid\mathfrak{m}, \qquad \psi\in\mathrm{Gal}(L/K)^\vee.$$

For $K=\mathbb{Q}$, Theorem 2.1.21 identifies these with Dirichlet characters.

Proposition 8.1.2 (Primitive induction)**.** *Let χ modulo m be induced by the primitive character χ^* modulo f_χ. For* $\mathrm{Re}(s)>1$*,*

$$L(s,\chi)=L(s,\chi^*)\prod_{\substack{p\mid m\\ p\nmid f_\chi}}\left(1-\frac{\chi^*(p)}{p^s}\right).$$

Consequently every Dirichlet L-function reduces to one primitive L-function and finitely many Euler factors.

Definition 8.1.3 (Dirichlet L-function and Gauss sum)**.** For a Dirichlet character χ modulo m, set

$$L(s,\chi):=\sum_{n\ge1}\frac{\chi(n)}{n^s}=\prod_p\left(1-\frac{\chi(p)}{p^s}\right)^{-1},\qquad \mathrm{Re}(s)>1. \tag{8.1.1}$$

If χ is primitive modulo f, put

$$a_\chi:=\frac{1-\chi(-1)}{2}\in\{0,1\},\qquad \tau(\chi):=\sum_{a=1}^{f}\chi(a)e^{2\pi ia/f},$$

$$\Lambda(s,\chi):=\left(\frac{f}{\pi}\right)^{(s+a_\chi)/2}\Gamma\left(\frac{s+a_\chi}{2}\right)L(s,\chi),\qquad \varepsilon(\chi):=\frac{\tau(\chi)}{i^{a_\chi}\sqrt{f}}.$$

Theorem 8.1.4 (Dirichlet functional equation, [2])**.** *Let χ be primitive modulo f. Then*

$$|\tau(\chi)|=\sqrt{f},\qquad \tau(\chi)\tau(\overline{\chi})=\chi(-1)f,\qquad |\varepsilon(\chi)|=1,$$

and $L(s,\chi)$ extends to $\mathbb{C}$, holomorphically when $\chi\neq1$ and meromorphically with a simple pole at $s=1$ when $\chi=1$. Moreover,

$$\Lambda(s,\chi)=\varepsilon(\chi)\Lambda(1-s,\overline{\chi}). \tag{8.1.2}$$

Thus

$$\chi(-1)=(-1)^{a_\chi},\qquad L(-2r,\chi)=0\quad(a_\chi=0,\ r\ge1),\qquad L(-(2r+1),\chi)=0\quad(a_\chi=1,\ r\ge0).$$

Theorem 8.1.5 (Algebraic special values, [2])**.** *For a primitive character χ of conductor f, define*

$$\sum_{a=1}^{f}\chi(a)\frac{te^{at}}{e^{ft}-1}=\sum_{r\ge0}B_{r,\chi}\frac{t^r}{r!}.$$

For every $r \geq 1$,

$$L(1-r,\chi) = -\frac{B_{r,\chi}}{r}, \qquad \chi \neq 1 \wedge \chi(-1) \neq (-1)^r \implies B_{r,\chi} = L(1-r,\chi) = 0.$$

If $\chi(-1) = (-1)^r$, then

$$\frac{f^r L(r,\chi)}{(2\pi i)^r \tau(\chi)} \in \mathbb{Q}(\chi),$$

with the explicit value obtained from Theorem 8.1.4 and $B_{r,\overline{\chi}}$.

Definition 8.1.6 (Dedekind zeta and Hecke L-functions)**.** For a number field K and a finite-order Hecke character χ, set

$$\zeta_K(s) := \sum_{0 \neq \mathfrak{a} \subseteq \mathscr{O}_K} \frac{1}{\mathrm{N}(\mathfrak{a})^s} = \prod_{\mathfrak{p}} \left(1 - \frac{1}{\mathrm{N}(\mathfrak{p})^s}\right)^{-1}, \qquad L_K(s,\chi) := \prod_{\mathfrak{p} \nmid \mathfrak{f}(\chi)} \left(1 - \frac{\chi(\mathfrak{p})}{\mathrm{N}(\mathfrak{p})^s}\right)^{-1}, \tag{8.1.3}$$

for $\mathrm{Re}(s) > 1$, where

$$\chi(\mathfrak{p}) := \chi_{\mathfrak{f}}([\mathfrak{p}]).$$

Theorem 8.1.7 (Abelian factorization and conductor–discriminant, [2])**.** *Let L/K be finite abelian with group G. Reciprocity identifies each $\chi \in G^\vee$ with a finite-order Hecke character whose unique primitive associate has conductor $\mathfrak{f}(\chi)$, and*

$$\zeta_L(s) = \prod_{\chi \in G^\vee} L_K(s,\chi), \qquad \mathfrak{d}_{L/K} = \prod_{\chi \in G^\vee} \mathfrak{f}_0(\chi),$$

where $\mathfrak{f}_0(\chi)$ is the finite part of $\mathfrak{f}(\chi)$. Equivalently,

$$\begin{tikzcd}
G^\vee \arrow[r, "\chi \mapsto \chi \circ \mathrm{Art}_{L/K}"] \arrow[d, "\chi \mapsto L_K(s,\chi)"'] & \left(C_K/\mathrm{N}_{L/K}(C_L)\right)^\vee \arrow[d, "\eta \mapsto L_K(s,\eta)"] \\
\{\textit{Euler products}\} \arrow[r, equal] & \{\textit{Euler products}\}.
\end{tikzcd}$$

Proof. Compare the local factors at every prime.

$$\mathfrak{q} \mid \mathfrak{p}, \qquad I := I(\mathfrak{q} \mid \mathfrak{p}), \qquad e := |I|.$$

$$\overline{g} := \mathrm{Frob}_{\mathfrak{q}/\mathfrak{p}} \in D(\mathfrak{q} \mid \mathfrak{p})/I, \qquad \overline{g}(x) = x^{\mathrm{N}(\mathfrak{p})} \quad (x \in k_{\mathfrak{q}}), \qquad f := \mathrm{ord}(\overline{g}).$$

$$\zeta_{L,\mathfrak{p}}(s) = \left(1 - \mathrm{N}(\mathfrak{p})^{-fs}\right)^{-r}, \qquad r := \frac{|G|}{ef}, \qquad \{\chi(\overline{g}) : \chi \in G^\vee,\ \chi|_I = 1\} = \mu_f \text{ with multiplicity } r.$$

$$\prod_{\chi \in G^\vee} L_{K,\mathfrak{p}}(s,\chi) = \prod_{\substack{\chi \in G^\vee \\ \chi|_I = 1}} \left(1 - \chi(\overline{g})\mathrm{N}(\mathfrak{p})^{-s}\right)^{-1} = \left(1 - \mathrm{N}(\mathfrak{p})^{-fs}\right)^{-r} = \zeta_{L,\mathfrak{p}}(s).$$

$$\prod_{\mathfrak{p}} \zeta_{L,\mathfrak{p}}(s) = \prod_{\chi \in G^\vee} L_K(s,\chi), \qquad \mathfrak{d}_{L/K} = \prod_{\chi \in G^\vee} \mathfrak{f}_0(\chi).$$

This proves the factorization; the last identity is the local conductor formula. □

Corollary 8.1.8 (Quadratic zeta factorization)**.** *Let D be a fundamental discriminant, put $K = \mathbb{Q}(\sqrt{D})$, and let*

$$\chi_D(n) := \left(\frac{D}{n}\right).$$

Then

$$f_{\chi_D} = |D|, \qquad \zeta_K(s) = \zeta(s) L(s,\chi_D),$$

$$\begin{tikzcd} \mathrm{Gal}(K/\mathbb{Q})^\vee \arrow[r, "\sim"] & \{1,\chi_D\} \arrow[d, "\chi\mapsto L(s,\chi)"] \\ & \{\zeta(s), L(s,\chi_D)\} \end{tikzcd} \qquad \chi_D(p) = \begin{cases} 1, & p \text{ splits in } K, \\ 0, & p \text{ ramifies in } K, \\ -1, & p \text{ is inert in } K. \end{cases}$$

Theorem 8.1.9 (Analytic class number formula, [2]). *Let K have signature (r_1, r_2), class number $h_K := |\mathrm{Cl}(K)|$, regulator $\mathrm{Reg}(K)$ from Definition 3.1.9, and $w_K := |\mu_K|$. Then $\zeta_K(s)$ extends meromorphically to $\mathbb{C}$, has a unique simple pole at $s = 1$, and*

$$\mathrm{Res}_{s=1}(\zeta_K(s)) = \frac{2^{r_1}(2\pi)^{r_2} h_K \mathrm{Reg}(K)}{w_K\sqrt{|\mathrm{disc}(K/\mathbb{Q})|}}. \tag{8.1.4}$$

Writing

$$\Gamma_{\mathbb{R}}(s) := \pi^{-s/2}\Gamma(s/2), \qquad \Gamma_{\mathbb{C}}(s) := 2(2\pi)^{-s}\Gamma(s),$$

the completed function

$$\Lambda_K(s) := |\mathrm{disc}(K/\mathbb{Q})|^{s/2}\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\zeta_K(s)$$

satisfies

$$\Lambda_K(s) = \Lambda_K(1-s).$$

For fundamental D, Corollary 8.1.8 gives

$$L(1,\chi_D) = \begin{cases} \dfrac{2\pi h_K}{w_K\sqrt{|D|}}, & D < 0, \\ \dfrac{2h_K \log(\varepsilon_D)}{\sqrt{D}}, & D > 0, \end{cases}$$

where $\varepsilon_D > 1$ generates $\mathscr{O}_K^\times/\mu_K$ as in Theorem 3.1.14.

Theorem 8.1.10 (Complex multiplication and Kronecker's Jugendtraum, [2]). *Let $K \subset \mathbb{C}$ be imaginary quadratic, let $\mathfrak{a}$ be a fractional ideal, and put $E_{\mathfrak{a}} := \mathbb{C}/\mathfrak{a}$. Then*

$$j(\mathfrak{a}) := j(E_{\mathfrak{a}}) \in \overline{\mathbb{Q}}, \qquad H_K = K(j(\mathfrak{a})), \qquad \mathrm{Gal}(H_K/K) \cong \mathrm{Cl}(K).$$

For every ideal $\mathfrak{b}$ prime to the relevant conductor,

$$\mathrm{Art}_{H_K/K}([\mathfrak{b}])\big(j(\mathfrak{a})\big) = j(\mathfrak{b}^{-1}\mathfrak{a}).$$

Let

$$\mathfrak{h}_{\mathfrak{a}} \colon E_{\mathfrak{a}} \longrightarrow \mathbb{P}^1$$

be a Weber function with the standard CM normalization, and

$$E_{\mathfrak{a}}[\mathfrak{m}] := \{P \in E_{\mathfrak{a}} : [\alpha]P = 0 \text{ for every } \alpha \in \mathfrak{m}\}.$$

For every nonzero integral ideal $\mathfrak{m}$,

$$K(\mathfrak{m}) = K\big(j(\mathfrak{a}), \mathfrak{h}_{\mathfrak{a}}(P) : P \in E_{\mathfrak{a}}[\mathfrak{m}]\big),$$
$$K^{\mathrm{ab}} = K\big(j(\mathfrak{a}), \mathfrak{h}_{\mathfrak{a}}(P) : P \in E_{\mathfrak{a},\mathrm{tors}}\big).$$

Thus

$$\begin{tikzcd} K^{\mathrm{ab}} \arrow[d, dash] \\ K(\mathfrak{m}) \arrow[d, dash] \\ H_K = K(j(\mathfrak{a})) \arrow[d, dash] \\ K \end{tikzcd} \qquad \begin{tikzcd} \mathrm{Gal}(K^{\mathrm{ab}}/K) \arrow[d, two heads] & \\ \mathrm{Gal}(K(\mathfrak{m})/K) \arrow[d, two heads] & \mathrm{Cl}_K(\mathfrak{m}) \arrow[l, "\sim"'] \arrow[d, two heads] \\ \mathrm{Gal}(H_K/K) & \mathrm{Cl}(K) \arrow[l, "\sim"'] \end{tikzcd}$$

which realizes every finite abelian extension of K by special values of modular functions at CM points.

Part 2. Seminar Talks[4]

9. Sieve Problems

Sieve theory converts local divisibility data into global estimates for primes, almost primes, and additive functions.

9.1. Möbius Randomness and Prime Patterns.

Definition 9.1.1 (Arithmetic weights)**.** For $n=\prod_p p^{v_p(n)}\ge 1$, set

$$\omega(n):=\#\{p:p\mid n\},\qquad \Omega(n):=\sum_p v_p(n),\qquad \lambda(n):=(-1)^{\Omega(n)},$$

$$\mu(n):=\begin{cases}1, & n=1,\\ (-1)^{\omega(n)}, & n\text{ squarefree},\\ 0, & p^2\mid n\text{ for some prime }p,\end{cases}\qquad \Lambda(n):=\begin{cases}\log(p), & n=p^k,\quad k\ge 1,\\ 0, & \text{otherwise.}\end{cases}$$

For arithmetic functions $f,g\colon\mathbb{N}\to\mathbb{C}$, their Dirichlet convolution is

$$(f*g)(n):=\sum_{d\mid n}f(d)g(n/d),\qquad \mathbf{1}(n):=1,\qquad \varepsilon(n):=\begin{cases}1, & n=1,\\ 0, & n>1.\end{cases}$$

Proposition 9.1.2 (Möbius inversion)**.** *For* $n\ge 1$*,*

$$\sum_{d\mid n}\mu(d)=\varepsilon(n),\qquad \mu*\mathbf{1}=\varepsilon,\qquad g(n)=\sum_{d\mid n}f(d)\iff f(n)=\sum_{d\mid n}\mu(d)g(n/d).\tag{9.1.1}$$

For $\mathrm{Re}(s)>1$*,*

$$\frac{1}{\zeta(s)}=\sum_{n\ge1}\frac{\mu(n)}{n^s}=\prod_p\left(1-\frac{1}{p^s}\right),\qquad -\frac{\zeta'(s)}{\zeta(s)}=\sum_{n\ge1}\frac{\Lambda(n)}{n^s}.$$

Proof. Multiplicativity reduces the divisor sum to prime powers.

$$n=\prod_{j=1}^r p_j^{a_j}\implies\sum_{d\mid n}\mu(d)=\prod_{j=1}^r\big(\mu(1)+\mu(p_j)\big)=\begin{cases}1, & n=1,\\ 0, & n>1.\end{cases}$$

$$g=f*\mathbf{1}\overset{9.1.1}{\iff}\mu*g=(\mu*\mathbf{1})*f=\varepsilon*f=f.$$

$$\zeta(s)=\prod_p(1-p^{-s})^{-1}\implies\zeta(s)^{-1}=\prod_p(1-p^{-s})=\sum_{n\ge1}\frac{\mu(n)}{n^s}.$$

$$\log(\zeta(s))=\sum_p\sum_{k\ge1}\frac{p^{-ks}}{k}\implies-\frac{\zeta'(s)}{\zeta(s)}=\sum_p\sum_{k\ge1}\frac{\log(p)}{p^{ks}}=\sum_{n\ge1}\frac{\Lambda(n)}{n^s}.$$

This proves the inversion and Euler-product formulas. □

Definition 9.1.3 (Summatory functions and correlations)**.** For $x\ge1$ and distinct $h_1,\dots,h_k\in\mathbb{Z}$, set

$$M(x):=\sum_{n\le x}\mu(n),\qquad \psi(x):=\sum_{n\le x}\Lambda(n),$$

$$C_\mu(x;h_1,\dots,h_k):=\sum_{n\le x}\prod_{j=1}^k\mu(n+h_j),\qquad C_\Lambda(x;h):=\sum_{n\le x}\Lambda(n)\Lambda(n+h).$$

Terms with $n+h_j\le 0$ are omitted.

[4] Based on lectures by Austin Lei, others [1].

Theorem 9.1.4 (Mertens criteria). *Write* RH *for*

$$\zeta(\rho)=0 \quad \wedge \quad 0<\mathrm{Re}(\rho)<1 \implies \mathrm{Re}(\rho)=\frac{1}{2}.$$

For $\mathrm{Re}(s)>1$*, partial summation gives*

$$\frac{1}{\zeta(s)}=s\int_1^\infty \frac{M(x)}{x^{s+1}}\,\mathrm{d}x. \tag{9.1.2}$$

Consequently,

$$M(x)=o(x) \iff \psi(x)\sim x \iff \zeta(s)\neq 0 \quad \big(\mathrm{Re}(s)=1,\ s\neq 1\big),$$

$$\mathrm{RH} \iff M(x)=O_\varepsilon\Big(x^{1/2+\varepsilon}\Big) \quad \textit{for every } \varepsilon>0,$$

and the Riemann hypothesis implies the Lindelöf bound

$$\zeta\left(\frac{1}{2}+it\right)=O_\varepsilon((1+|t|)^\varepsilon) \quad \textit{for every } \varepsilon>0.$$

The dependence is summarized by

$$M(x)=o(x) \rightleftarrows \psi(x)\sim x \rightleftarrows \zeta(s)\neq 0 \textit{ on } \mathrm{Re}(s)=1,\ s\neq 1$$

$$M(x)=O_\varepsilon(x^{1/2+\varepsilon}) \rightleftarrows \mathrm{RH} \Longrightarrow \textit{Lindelöf}.$$

Warning 9.1.5 (Mertens conjecture). The historical estimate

$$|M(x)|\le \sqrt{x} \quad (x\ge 1)$$

is false; neither its failure nor Theorem 9.1.4 contradicts the Riemann hypothesis.

Conjecture 9.1.6 (Chowla). *For* $k\ge 2$ *and distinct* $h_1,\dots,h_k\in\mathbb{Z}$*,*

$$C_\mu(x;h_1,\dots,h_k)=o(x).$$

In particular,

$$\sum_{n\le x}\mu(n)\mu(n+1)=o(x).$$

Definition 9.1.7 (Sieve datum). Let $\mathscr{A}$ be a finite multiset of integers, let $\mathscr{P}$ be a set of primes, and put

$$P(z):=\prod_{\substack{p<z\\ p\in\mathscr{P}}}p, \qquad \mathscr{A}_d:=\{a\in\mathscr{A}:d\mid a\}, \qquad S(\mathscr{A},\mathscr{P},z):=\#\{a\in\mathscr{A}:(a,P(z))=1\},$$

$$\mathbf{1}_{(a,P(z))=1}=\sum_{d\mid(a,P(z))}\mu(d), \qquad S(\mathscr{A},\mathscr{P},z)=\sum_{d\mid P(z)}\mu(d)\#(\mathscr{A}_d).$$

For squarefree $d<D$ with $d\mid P(z)$, let g be multiplicative and write

$$\#(\mathscr{A}_d)=Xg(d)+r_d, \qquad 0\le g(p)<1.$$

The datum has sieve dimension κ when

$$V(z):=\prod_{\substack{p<z\\ p\in\mathscr{P}}}(1-g(p))\asymp\big(\log(z)\big)^{-\kappa}.$$

Its local-to-global structure is

$$\begin{tikzcd}
\big(\#(\mathscr{A}_d)\big)_{d\mid P(z)} \arrow[r, "\sum_{d\mid P(z)}\mu(d)\cdot(\ )"] \arrow[d, "\#(\mathscr{A}_d)=Xg(d)+r_d"'] & S(\mathscr{A},\mathscr{P},z) \arrow[d, dashed, "9.1.8"] \\
\big(Xg(d)+r_d\big)_{d<D} \arrow[r, dashed, "s=\log(D)/\log(z)"'] & XV(z)+O\big(R(D,z)\big).
\end{tikzcd}$$

Theorem 9.1.8 (Fundamental lemma of the combinatorial sieve)**.** *Assume Definition 9.1.7 and, for some $A>0$ and every $2\le w<z$,*

$$\prod_{\substack{w\le p<z\\ p\in\mathscr{P}}}\big(1-g(p)\big)^{-1}\le\left(\frac{\log(z)}{\log(w)}\right)^{\kappa}\left(1+\frac{A}{\log(w)}\right).$$

Put

$$D:=z^s,\qquad R(D,z):=\sum_{\substack{d<D\\ d\mid P(z)}}|r_d|.$$

For sufficiently large s, the fundamental sieve weights give

$$S(\mathscr{A},\mathscr{P},z)=XV(z)\left(1+O_{\kappa,A}\left(e^{-s\log(s)+O_\kappa(s)}\right)\right)+O\big(R(D,z)\big).$$

Consequently,

$$s\longrightarrow\infty\quad\wedge\quad R(D,z)=o\big(XV(z)\big)\implies S(\mathscr{A},\mathscr{P},z)\sim XV(z).$$

Definition 9.1.9 (Admissible tuples and singular series)**.** Let $\mathscr{H}=\{h_1,\dots,h_k\}\subset\mathbb{Z}$, and put

$$\nu_p(\mathscr{H}):=\#\{h \bmod p : h\in\mathscr{H}\}.$$

The tuple is admissible when

$$\nu_p(\mathscr{H})<p\quad\text{for every prime } p,$$

and its singular series is

$$\mathfrak{S}(\mathscr{H}):=\prod_p\left(1-\frac{\nu_p(\mathscr{H})}{p}\right)\left(1-\frac{1}{p}\right)^{-k}.$$

Thus

$$\begin{tikzcd}
\mathscr{H} \arrow[r, "\bmod p"] \arrow[dr, bend right, "\text{all } p"'] & \nu_p(\mathscr{H}) \arrow[d, "\nu_p(\mathscr{H})<p"] & \\
& \mathscr{H}\ \text{admissible} \arrow[r] & \mathfrak{S}(\mathscr{H})>0.
\end{tikzcd}$$

Conjecture 9.1.10 (Hardy–Littlewood prime tuples)**.** *For every admissible $\mathscr{H}=\{h_1,\dots,h_k\}$,*

$$\#\{n\le x : n+h_1,\dots,n+h_k \text{ are prime}\}\sim\mathfrak{S}(\mathscr{H})\int_2^x\frac{\mathrm{d}t}{(\log(t))^k}\sim\mathfrak{S}(\mathscr{H})\frac{x}{(\log(x))^k}. \tag{9.1.3}$$

For $\mathscr{H}=\{0,2\}$, this is the twin-prime conjecture and predicts

$$\sum_{n\le x}\Lambda(n)\Lambda(n+2)\sim 2C_2x,\qquad C_2:=\prod_{p>2}\left(1-\frac{1}{(p-1)^2}\right).$$

Conjecture 9.1.11 (Goldbach)**.** *Every even $N \geq 4$ is the sum of two primes. More precisely, the Hardy–Littlewood prediction for the number $R_2(N)$ of ordered representations $N = p+q$ is*

$$R_2(N) \sim 2C_2 \frac{N}{(\log(N))^2} \prod_{\substack{p \mid N \\ p>2}} \frac{p-1}{p-2}.$$

Theorem 9.1.12 (Brun and Chen)**.** *Put*

$$\pi_2(x) := \#\{p \leq x : p+2 \text{ is prime}\},$$

and let P_2 denote an integer with at most two prime factors counted with multiplicity. Then

$$\pi_2(x) \ll \frac{x}{(\log(x))^2}, \qquad \sum_{\substack{p \\ p+2 \text{ prime}}} \frac{1}{p} < \infty,$$

$$\#\{p \leq x : p+2 = P_2\} \gg \frac{x}{(\log(x))^2}, \qquad N \gg 1 \wedge 2 \mid N \implies N = p + P_2.$$

Thus the sieve reaches the almost-prime analogues of Conjectures 9.1.10 and 9.1.11.

Definition 9.1.13 (Kubilius model)**.** For primes $p \leq x$, let X_p be independent Bernoulli variables with

$$\Pr(X_p = 1) = \frac{1}{p}, \qquad \Pr(X_p = 0) = 1 - \frac{1}{p}.$$

Let Y_p be independent geometric variables with

$$\Pr(Y_p = j) = \left(1 - \frac{1}{p}\right) p^{-j} \quad (j \geq 0).$$

The Kubilius correspondences are

$$\mathbf{1}_{p\mid n} \rightsquigarrow X_p, \qquad \omega(n) = \sum_{p \leq x} \mathbf{1}_{p\mid n} \rightsquigarrow \sum_{p \leq x} X_p, \qquad \Omega(n) \rightsquigarrow \sum_{p \leq x} Y_p,$$

$$\sum_{p \leq x} \mathbb{E}(X_p) = \log(\log(x)) + O(1), \qquad \sum_{p \leq x} \mathrm{Var}(X_p) = \log(\log(x)) + O(1),$$
$$\sum_{p \leq x} \mathbb{E}(Y_p) = \log(\log(x)) + O(1), \qquad \sum_{p \leq x} \mathrm{Var}(Y_p) = \log(\log(x)) + O(1).$$

$$\begin{tikzcd}
n \leq x \arrow[r, mapsto] & (\mathbf{1}_{p\mid n})_{p \leq x} \arrow[r, "\sum_{p \leq x}"] \arrow[d, dashed, "\mathbf{1}_{p\mid n} \rightsquigarrow X_p"'] & \omega(n) \arrow[d, dashed, "\text{law}"] \\
& (X_p)_{p \leq x} \arrow[r, "\frac{\sum_{p \leq x} X_p - \log(\log(x))}{\sqrt{\log(\log(x))}}"'] & \mathcal{N}(0,1).
\end{tikzcd}$$

Theorem 9.1.14 (Erdős–Kac)**.** *For every $\alpha < \beta$,*

$$\lim_{x \to \infty} \frac{1}{x} \# \left\{ n \leq x : \alpha \leq \frac{\omega(n) - \log(\log(x))}{\sqrt{\log(\log(x))}} \leq \beta \right\} = \frac{1}{\sqrt{2\pi}} \int_\alpha^\beta e^{-t^2/2} \, \mathrm{d}t. \tag{9.1.4}$$

The same limit holds with $\Omega(n)$ in place of $\omega(n)$.

Warning 9.1.15 (Parity barrier). Classical combinatorial sieves control the local data $\#(\mathscr{A}_d) = Xg(d) + r_d$ but generally do not distinguish

$$\lambda(n) = 1 \iff \Omega(n) \equiv 0 \pmod 2, \qquad \lambda(n) = -1 \iff \Omega(n) \equiv 1 \pmod 2.$$

Thus small-divisor data alone does not isolate primes:

$$\begin{tikzcd}
(\#(\mathscr{A}_d))_{d\mid P(z)} \arrow[r, "9.1.8"] \arrow[ddr, dashed, bend right, "\not\Rightarrow"'] & S(\mathscr{A},\mathscr{P},z) \arrow[dr, dashed, "\not\Rightarrow"] & \\
& & \mathbf{1}_{\Omega(n)=1} \\
& \lambda(n) = (-1)^{\Omega(n)}. \arrow[ur, dashed, "\text{parity}"'] &
\end{tikzcd}$$

This obstructs direct sieve proofs of Conjectures 9.1.10 and 9.1.11.

Theorem 9.1.16 (Sato–Tate)**.** *Let $E/\mathbb{Q}$ be a non-CM elliptic curve with minimal discriminant Δ_E. For every prime $p \nmid \Delta_E$, set*

$$a_p := p + 1 - \#E(\mathbb{F}_p) = 2\sqrt{p}\cos(\theta_p), \qquad \theta_p \in [0, \pi].$$

Then, for $0 \le \alpha < \beta \le \pi$,

$$\lim_{x\to\infty} \frac{\#\{p \le x : p \nmid \Delta_E,\ \alpha \le \theta_p \le \beta\}}{\#\{p \le x : p \nmid \Delta_E\}} = \frac{2}{\pi}\int_\alpha^\beta \sin^2(\theta)\,\mathrm{d}\theta.$$

Part 3. **Appendix**

APPENDIX A. FOUNDATIONAL TOOLS

A.1. **Standard Algebraic and Analytic Tools.**

Fact A.1.1 (Primitive elements and Galois correspondence, [2]). *Every finite separable extension L/K is simple; in particular,*

$$\operatorname{char}(K) = 0 \implies \exists\, \alpha \in L : L = K(\alpha).$$

If Ω/K is Galois with Krull group $G := \operatorname{Gal}(\Omega/K)$, then

$$\begin{tikzcd} \{K \subseteq E \subseteq \Omega\} \arrow[rr, bend left, "E\mapsto \operatorname{Gal}(\Omega/E)"] & & \{H \leq G : H \text{ closed}\} \arrow[ll, bend left, "H\mapsto \Omega^H"] \end{tikzcd}$$

is inclusion-reversing, and

$$[E:K] < \infty \iff \operatorname{Gal}(\Omega/E) \text{ is open}, \qquad E/K \text{ Galois} \iff \operatorname{Gal}(\Omega/E) \trianglelefteq G,$$
$$\operatorname{Gal}(E/K) \cong G/\operatorname{Gal}(\Omega/E) \qquad \big(E/K \text{ Galois}\big).$$

Fact A.1.2 (Module remaindering and finite fields, [2]). *Let M be an A-module and let $\mathfrak{a}_1, \ldots, \mathfrak{a}_r \subset A$ be pairwise comaximal. Then*

$$0 \longrightarrow \left(\textstyle\prod_{i=1}^{r} \mathfrak{a}_i\right) M \hookrightarrow M \xrightarrow{m\mapsto (m+\mathfrak{a}_i M)_i} \prod_{i=1}^{r} M/\mathfrak{a}_i M \longrightarrow 0.$$

For $q = p^a$ and $f \geq 1$,

$$\mathbb{F}_{q^f}^{\times} \cong \mathbb{Z}/(q^f - 1)\mathbb{Z}, \qquad \operatorname{Gal}(\mathbb{F}_{q^f}/\mathbb{F}_q) = \langle \operatorname{Fr}_q \rangle \cong \mathbb{Z}/f\mathbb{Z}, \qquad \operatorname{Fr}_q(x) = x^q,$$
$$\operatorname{Tr}_{\mathbb{F}_{q^f}/\mathbb{F}_q}(x) = \sum_{j=0}^{f-1} x^{q^j}, \qquad \operatorname{N}_{\mathbb{F}_{q^f}/\mathbb{F}_q}(x) = \prod_{j=0}^{f-1} x^{q^j} = x^{(q^f-1)/(q-1)}.$$

Fact A.1.3 (Kummer theory, characters, and Hasse, [2, 1]). *Assume* $\operatorname{char}(K) \nmid n$ *and $\mu_n \subset K$. For $\beta^n = b \in K^{\times}$,*

$$K(\beta)/K \text{ is cyclic Galois}, \qquad \operatorname{Gal}(K(\beta)/K) \hookrightarrow \mu_n, \quad \sigma \longmapsto \frac{\sigma(\beta)}{\beta}, \qquad [K(\beta):K] = \operatorname{ord}\big(bK^{\times n}\big).$$

For a finite abelian group G, its dual $G^{\vee} := \operatorname{Hom}(G, \mathbb{C}^{\times})$ satisfies

$$|G^{\vee}| = |G|, \qquad \sum_{\chi \in G^{\vee}} \chi(g) = \begin{cases} |G|, & g = 1, \\ 0, & g \neq 1, \end{cases} \qquad \prod_{\zeta^m = 1} (1 - \zeta T) = 1 - T^m.$$

If $E/\mathbb{F}_q$ is an elliptic curve and $a_q := q + 1 - \#E(\mathbb{F}_q)$, then

$$|a_q| \leq 2\sqrt{q}, \qquad a_q = 2\sqrt{q}\cos(\theta_q) \quad \text{for a unique } \theta_q \in [0, \pi].$$

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